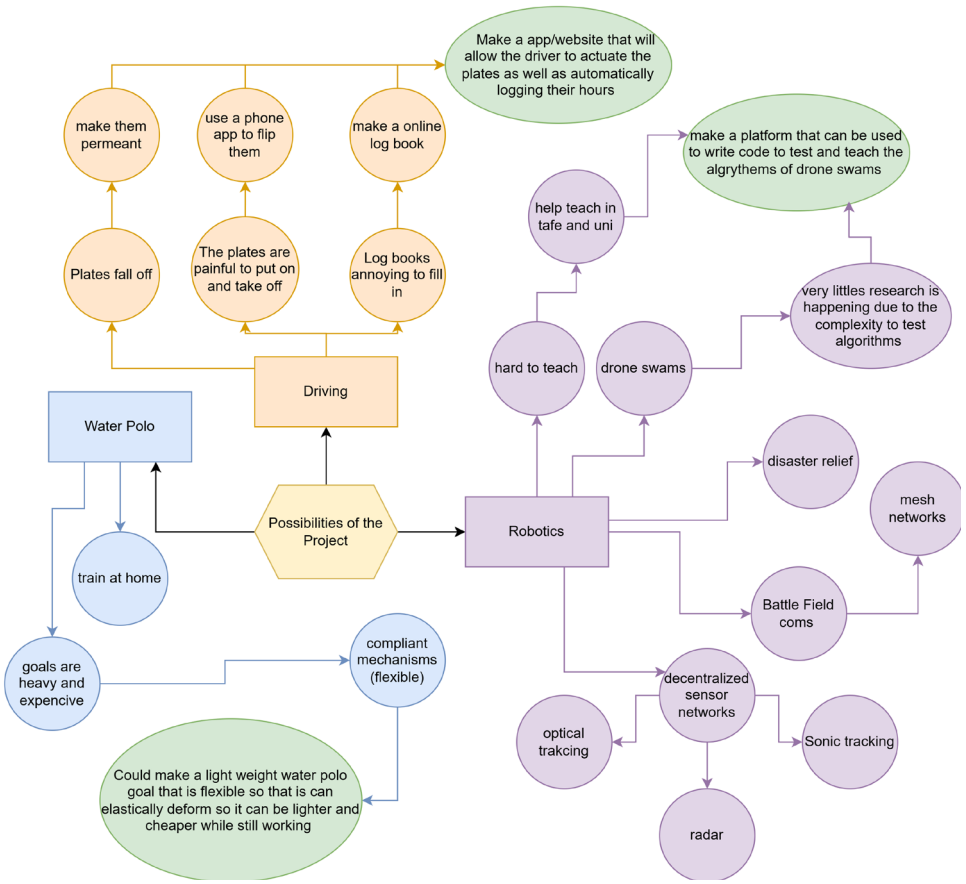
Lflip
It is all automatic
40200916

Project Proposal

Identification and Exploration of the Need: Exploration of Needs

Initial Mind Map: Several ideas were considered when selecting an option to pursue for my major design project.



Initial Ideas

Idea 1: The first idea was a drone swarm that can be used for teaching swarming technologies to students. The goal was to provide scientists and students with the hardware and software platform to develop new algorithms, as there is a lack of development in the field of swarms.

Ongoing Evaluation: The reason I did not choose this project was because it is incredibly complex and time consuming and hence I would not be able to make a polished final project.

Idea 2: The next idea was to create a water polo goal, as currently water polo goals are very heavy and expensive. They are designed this way so when people shoot at the goal with great force it can withstand it. To solve this problem, I was going to make a goal that is flexible whereby the soft material could move on impact and then recover to its original shape.

The reason I did not choose this project was because it would be difficult to iterate and test new designs as each version would require a new goal, which is a significant time investment.

Idea 3: The final option and the project I decided to go with was an automatic L-plate. This L-plate solves the problem that L-plates and P plates regularly fall off, leading to environmental pollution through the creation of litter and also great annoyance for the driver and public. In addition, the ongoing cost of replacement. My solution would also solve the tedious process of putting on L-plates and taking them off as well as filling in the log book through an app I would build with it. People often drive with the wrong plates due to laziness, which is both illegal and dangerous. I have lots of experience making electronics projects and designing 3D printed parts but I want to learn how to optimise mass production through 3D printing as this is a useful skill for the future. This project will also help me learn about software design, app development, microcontrollers, and GPS.

Despite this project being quite complicated, it is manageable if I follow my time management plan. I already have some skills and I will learn new skills over time to ensure that I can complete it.

Project Proposal

Existing Products

It is difficult to determine existing products for this type of proposal as there are none. The only functioning existing product that exists are physical L plates that need to be stuck on manually.

Why is it a need and what problem does it solve?

The project is needed because L-plates are annoying to take on and off and they often fall off, causing litter. The increased convenience and better mounting design will result in fewer people breaking the law by driving without the proper plates, and improve the accurate filling in of the logbook.

- Not filled out log books
- Less driving practice (lost opportunity to drive when can't be bothered to put plates on) → worse drivers
- Driving with incorrect plates
- Plates falling off → littering, illegal to drive without them, excess cost and plastic consumption to replace them
- Plastic Waste
- Inconvenience of removing and adding the plate to fit multiple drivers

Secondary Research

Plastic Waste

- Poor fixtures: L and P plates when displayed outside of the car often result in them getting damaged quicker. The clips/holders that are typically provided with the L and P plates are also generally made of poor quality plastic, adding to the waste.
- Non-Recycled Material: L plates create plastic waste due to the non-recyclable materials they use. So when they fall off on the street, they cause litter and pollution resulting in added issues to the environment especially if this happens in areas where councils don't do roadside litter collection.

This growing litter problem on our roads has an illegal solution

There is a growing litter problem on our roads in the form of P-plates, but the solution is actually illegal in some states.

Zane Dobie 16:00 06 September 2025 15 comments 9 shares

+ 3 Photos

Photos: Facebook

This article unpacked how widespread the issue of discarded L plates is. One of the key findings is that there were 101 L plates found on the side of the road along a 1 km stretch. This demonstrates that this is a systemic issue and licence plates are a large source of plastic pollution. Plastic pollution is harmful to the environment as it is a primary contributor to global warming, a prevalent issue in current society.
This research caused me to design an internal mounting system which is secure, reliable and can't fall off.

Website: https://www.drive.com.au/caradvice/this-growing-litter-problem-on-our-roads-has-an-illegal-solution/

This article unpacks how Australia has a growing pollution problem and how our recycling infrastructure cannot keep up with this. In 2005, we had a 14.7% recycling success rate, whereas now it has dropped to 14% in 2024. This is due to a lack of sufficient infrastructure and the fact that we are exceedingly reliant on plastic in Australia. The government intends to enact stricter laws regulating how packaging can be sold in products, for example banning the use of soft plastic bags. They are also investing more in the recycling sector and information campaigns to inform the public about recycling practices.
My project will reduce the amount of plastic waste therefore reducing the load on the recycling infrastructure of Australia.

Australia's amount of plastic waste surges as recycling rates fail to improve

Australians produced 3.2m tonnes of plastic waste in 2023-24, up from 3m tonnes the previous year

Get our breaking news email, free app or daily news podcast

The total amount of plastic waste has doubled since 2005 (from 1.57m tonnes) but recycling rates have largely shifted, despite investments in increased capacity. Photograph: Tracey

Website: https://www.theguardian.com/environment/2025/aug/24/australias-amount-of-plastic-waste-surges-as-recycling-rates-fail-to-improve

Project Proposal

Secondary Research

Plastic Waste

Evaluation: This article finds that there are large quantities of microplastics in Australian waterways. It conducted a survey in the Illawarra river and it found that the levels of microplastics were comparable to China. This means Australia is suffering from a severe environmental problem. You can see from my images below in my primary research that a lot of these L plates have started to break down into microplastics so my solution will reduce the prevalence of these microplastics. Also microplastics, while being an environmental issue are also a health issue for society as they can be consumed accidentally through food and water sources.

https://www.abc.net.au/news/2026-03-16/nsw-estuaries-microplastics-report/106458862

Evaluation: The NSW Government is investing $1.25 million in three research projects to find solutions for plastic waste, including CSIRO work measuring microplastics in soil, compost and sewage, and a UTS study of harmful chemicals in plastic fabrics. Environment Minister Penny Sharpe says NSW "is facing a landfill crisis" and that Greater Sydney's landfill capacity will be exhausted by 2030. This proves that there is a legislative desire for my product as my product will also reduce the amount of plastic waste each learner driver consumes. The faded plates I counted in my primary research below are the same plastic waste this funding is targeting.

https://www.nsw.gov.au/ministerial-releases/new-research-funded-to-find-plastic-waste-solutions

Boomerang Alliance

State/Territory	External only	Internal assumed	Not specified	Materials
Western Australia			x	Printable
South Australia		x	x	See-through plates for sale
Northern Territory			x	
Tasmania			x	
Victoria		x		
New South Wales	x			Plastic freebies
Australian Capital Territory			x	
Queensland			x	Downloadable template

This proves that P plate pollution is a large problem and that mounting plates inside is legal in most states. This organisation is advocating for government to allow internal mounting of plates to be nationwide.

https://boomerangalliance.org.au/pplate

Identification and Exploration of the Need: Exploration of Needs

Although this is a good proposed solution, the following issues are acknowledged:

- Visibility: Internal plate mounting in cars with dark tinted glass is unsuitable due to poor visibility.
- Thermal: Cars can get very hot, up to 40-50 degrees Celsius in the summer, which could result in the plastic on internal plates melting
- Internal plate mounting can block vision through rear window

How would my product improve user experience over existing or similar designs?

With fixed plates that can be electronically displayed and hidden my design removes the arduous and time consuming process of walking around to the front and back of the car to put L-plates on. Often learner drivers miss out on driving trips when passengers are rushed which reduces opportunities to develop their driving skills. The device will securely hold the plates to reduce likelihood of them falling off and littering the road.

Project Proposal

Point of Difference

Existing products are made of plastic and attached by a clip, suction or adhesion (blue tack or tape) or are a magnetic material that is put directly on the body of the car. These mounting options all have their strengths and weaknesses but the best option is the clip because it is easier to take on and off and it is quite reliable. My design solution will stay permanently attached to the car and will use an electronic system to toggle the plate on and off. This electronic system will move a flap or similar mechanical system to reveal and hide the L-plate, thus resulting in the effect of the plate becoming visible and not. This is quicker and easier than the manual systems. The app/website that I will build will allow the user to log hours easily and quickly. Note this app will be built in my HSC Software Design subject.

Primary Research

On a bike ride, when I rode out of suburbia I saw a lot of L and P plates littering the road and the shoulder. I collected data on the plates that I saw on the side of the road. I counted 67 plates on a 9 km ride. On this quantitative excursion, many of the plates were so old that they were faded to black or white (13 plates) and you could not recognise if they were P or L plates. I counted 33 learner plates and 21 provisional plates. To summaries, 19% of plates were faded, 49% were learner plates and 31% provisional plates.

This primary research makes me feel confident that my project is solving a real need. This helped to reaffirm that this project could benefit society in terms of both reducing litter and improving road safety for individual drivers. Because some of these are so incredibly faded it demonstrated as a long-term issue can't afford and councils can't afford to maintain the roads properly. It also suggests that there's lots of people driving on the road without their P plates on correctly which is a safety risk.

Project Proposal

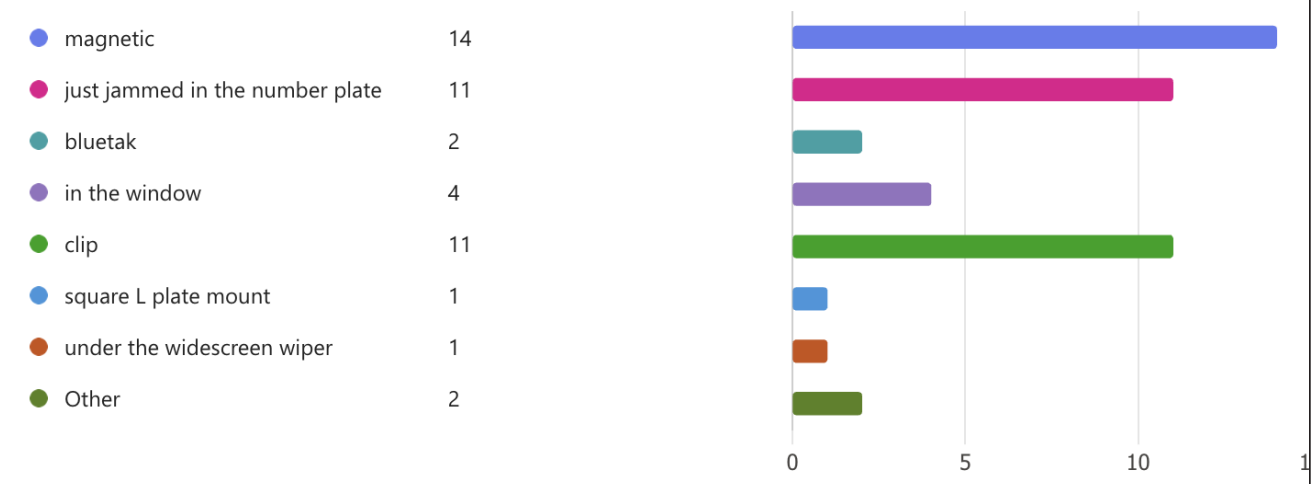
1. What type of driver are you?

This defines the scope of the problem by illustrating the reality that these plates are lost fairly regularly hence the current system need, improvement.

3. Have you ever taught someone to drive?

Demonstrates the log books are not filled in correctly. This shows that a solution is necessary to encourage people to fill in their logbook correctly complying with the legal requirements.

4. How do you attach the plates to the car?



shows that people falsify records out of laziness and other factors such as time constraints. This data provides evidence that people are not acting within the regulations of the laws, and then, consolidating the concept that users inherit laziness leading to danger.

Project Proposal
Primary Research

5. On average, how many days does a plate last with you (before it is lost)? More de

34
Responses

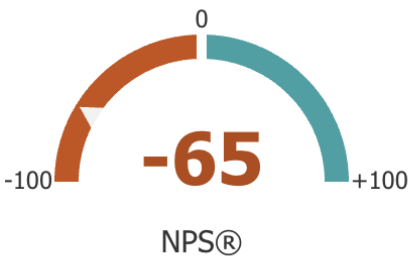
Latest Responses
"couple months"
"depends but maybe 50 days"
"I will be able to drive with it around 5 times before it's gone"
...

8 respondents (24%) answered month for this question.

This data demonstrates that uses typically lose their plates every month or more. This is very frequently! This proves that L plates falling off is a key issue that my target market faces.

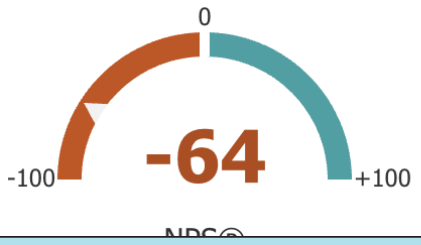
How frustrating is it to put the plates on?

Promoters	2
Passives	8
Detractors	24



How frustrating is it to take the plates off?

Promoters	4
Passives	4
Detractors	26



The response to this question indicate that it is very frustrating putting our plates on and off the car. This difficulty primarily leads people to not doing it or forgetting and being lazy about how they mount it as they want to just drive instead of dealing with their plates so they will use poor mounting methods. This results in them falling off and causing litter and more annoyance.

Project Proposal
Primary Research

8. How many times in the last year did you drive with incorrect plates (L driver no plates) or a (full license with plates). W hy did this occur?

More d

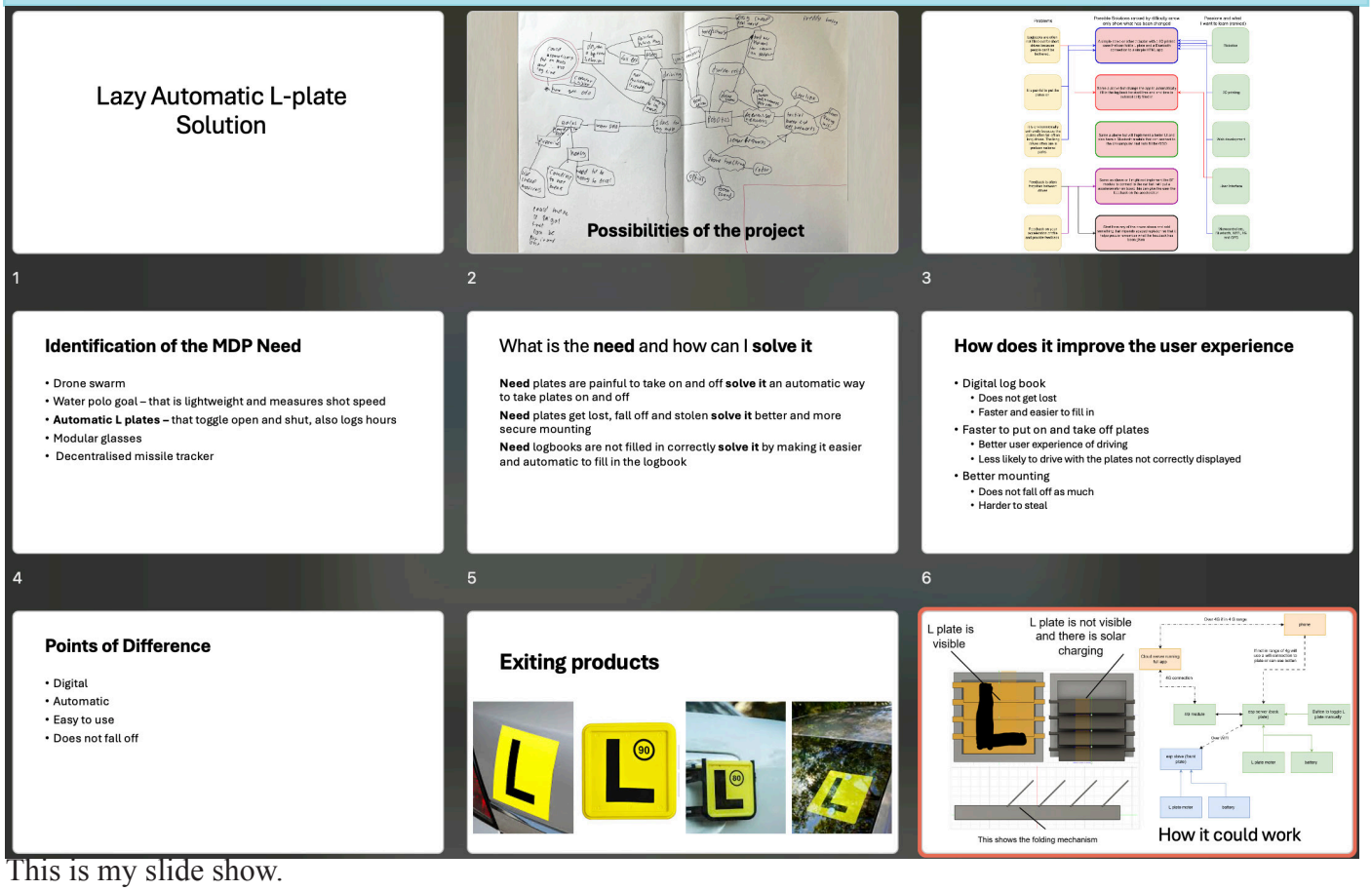
34
Responses

Latest Responses
"just a couple times, when my Ls would fall off and i didnt have another"
"none"
"Never occurred, i just didnt drive when i couldn't find it"
...

9 respondents (26%) answered times for this question.

This question shows that learner drivers will regularly drive without our plates or their parents will forget to take them off. This leads to the law getting broken regularly and unintentionally. This is bad because learner drivers need to have L plates for a reason so I can warn other drivers to be careful and if they drive without this it is dangerous to them.

This primary research I've conducted strongly supports my MDP, as it proves there is a need that I believe I can solve. This survey establishes that I can continue moving forward because I am actually addressing a real need.



This is my slide show.

Project Proposal

Design Brief

For my major design project, I plan to design an automatic L-plate. From my research and brainstorming, I have discovered a problem that I can solve which is the lack of usability of existing L and P plate systems. Using my experience of CAD for 3D printing as well as my experience in electronics design and microcontrollers gives me a good skill set for making this product. I intend to make an automatic L-plate which allows the user to go from displaying the plates to hiding them, without the need to take them on and off manually. This will be controlled from an app or a website on the user's phone, with an integrated log book that will be filled out automatically.

Design Situation

The project is needed because L-plates are annoying to take on and off and they often fall off, causing litter. The increased convenience and better mounting design will result in fewer people breaking the law by driving without the proper plates, and improve the accurate filling in of the logbook.

- Not filled out log books
- Less driving practice (lost opportunity to drive when can't be bothered to put plates on) → worse drivers
- Driving with incorrect plates
- Plates falling off → littering, illegal to drive without them, excess cost and plastic consumption to replace them
- Plastic waste
- Inconvenience of removing and adding the plate to fit multiple drivers

How does it improve the user experience over existing or similar design solutions?
With fixed plates that can be displayed and hidden, my design removes the arduous and time consuming process of walking around to the front and back of the car to put L-plates on. Often learner drivers miss out on driving trips when passengers are rushed which reduces opportunities to develop their driving skills. The device will securely hold the plates to reduce likelihood of them falling off and littering the road.

Target Market

Non-fully licensed drivers and anyone who shares a car with them. This includes the L-plate drivers learning in their parents' cars and P-plate drivers who share with their parents.

Where do they live? ->in every state of Australia.

Why will it appeal to the user? -> it will appeal to the user because it is better than the other options in the following ways

- Less litter from dropped plates and don't have to keep buying them
- Integrated logbook filling out
- Easier to switch from displaying plate to not (don't have to get out of car)
- Helps you become a better driver because of the AI teacher built into the app

It also saves the user time on every trip. Putting the plates on manually takes a typical learner driver about 40 to 50 seconds and has to be done at the start and the end of every drive, whereas my design switches them from inside the car and fills in the logbook automatically.

I'm very well suited to serving this target market because I am a member of the target market which means that I can effectively design this product as I know firsthand exactly what they want.
In addition, I have easy access to my target market as they are my peers and classmates. This lets me quickly take advantage of rapid iteration that the FDM process offers.

Project Proposal

Areas of Investigation

This chart allowed me to clearly unpack the things that I should investigate to build a good product. As there is a large amount of scope, this will take quite a significant amount of time to learn about. Once I have good general knowledge in these spaces I can then start doing specific research that is relevant to the project. I should first learn about general 3D printing, good aesthetic design and then learn specific skills for my project such as printing in place mechanisms.

Area of Investigation

Area of Investigation	Justification	What will be investigated	How it will be investigated
Where can I legally put an L plate?	It affects my mounting options. Legality is a highly important design parameter.	What the rules of mounting P and L plates is in each state.	I will ask an LLM to find me the relevant legislation. Asking an AI readthe legislation will be an addition to me reading the legislation myself as I believe it's important to not delegate these decisions to AI.
This entire project hinges on whether I am legally allowed to mount internally. If this product is illegal my target market and my entire solution would become irrelevant as you would be breaking the law.			
How to design print in place?	It will enable me to make a more reliable, stronger and easier to print the product	I need to build product design knowledge and inspiration.	I will use YouTube, places like thangs, my mentor and protyping.

Project Proposal
Area of Investigation

Area of Investigation	Justification	What will be investigated	How it will be investigated
Designing print in place is incredibly useful as it means that I can print mechanisms that normally are impossible to manufacture traditionally. So learning this skill would allow me to have a superior product that could be manufactured with something like injection moulding for example.			
The strengths and weaknesses of different 3D printed filaments	Different filaments are better and worse at different things for example, PLA is easy to print deforms in a hot car and also suffers from UV degradation.	I will do research on the YouTube channels "slant 3D", "CNC kitchen", "zach freedmen's" In addition, I will compare the published glass transition temperature of each filament against the temperature a parked car reaches in summer. I will test durability by driving with the plate mounted and checking that it does not fall off.	I will look online watch videos comparing them and ask on a forum I will cross check with an AI model. My mentor also has lots of experience in this area so I willto ask him also.
Every single filament is made up of different plastic polymers. They all have their own strengths and weaknesses for example PLA degrades under UV light whereas PETG does not but PETG does not print overhangs as well. Once I determined what filament to use it allows me to start dialling in the exact design but until then I can't optimise the 3D printing fully.			
What actuator is best suited for my design	There are many actuators on the market, from very large ones to really small ones. I need one with enough torque, waterproof, closed loop (this means that it knows where it is). I also want to be able to read the position of the actuator to verify that the plate actually moved.	I'll investigate different types of actuators such as: - **Servo motors** (you tell them an angle and they hold it) - **DC motors** (they just spin and do not know what position they are in) - **Linear motors** (they push and pull in a straight line instead of spinning) - **Lithium ion batteries** (the rechargeable type used in phones and laptops) - **Lithium polymer batteries** (same chemistry in a flat pouch so it fits thinner spaces) Investigating a wide range of actuators will allow me to choose the one that's best suited for my needs.	This will be investi-gated by researching online and also asking my mentor.
There are different actuators (some of them are louder some of them are stronger some of them are heavier some of them are bigger) that vary by strength, weight and size. Once I've established what actuator I need, I can then start designing the case for it. The actuators each have their own advantages and picking the right one will be key to the success of my project. If I pick the wrong one I'll be left with something underpowered, overpowered, too expensive or too low quality. There are so many variables that it is important to get this right as it makes up the core of the project.			

Project Proposal
Area of Investigation

Area of Investigation	Justification	What will be investigated	How it will be investigated
Learn to make an app how to program	Because I want to have an app to control the L plate	I am doing Software Design and Development as one of my HSC subjects and have learnt to build apps. I looked on Stack Overflow and Geeks for Geeks and asked an LLM when I got stuck, and the documentation for each language was my main source of information. I will investigate the different types of apps: - **Native** (written twice, once for iPhone and once for Android) - **Web apps** (nothing to install, you just go to a website) - **Progressive web apps** (a website you save onto the phone and open like an app) - **Cross platform such as React Native or Flutter** (written once, works on both)	High as I need to be able to program to make my app so then I can make a complete working product
Given that I want to build an app to work with the hardware I need to know the different types of apps as some of them have different levels of complexity and given that I'm a year 12 student some of them will not be in the scope of what I can attain so this means I have to have realistic expectations.			
How to design for 3D Printing	I intended to 3D to print my product as 3D printing can be mass produced as well as being accessible to a DT student. When designing for 3D printing you can make 'impossible' geometry.	There is lots of good information online and the 3D printing community is usually willing to answer questions. I will investigate things like: - **Overhangs** (anything with nothing underneath it while it prints) - **Bridging** (where the printer stretches plastic across a gap) - **Tolerances for print in place mechanisms** (the gap left between parts so they still move) - **Support material** (extra plastic that holds up overhangs) - **Elephant's foot and chamfers** (where the bottom layers squash out wider, and the angled cut that stops it) - **Print orientation** (parts are weaker along the layer lines so this changes the strength)	I will have discussions with my mentor and look at the plentiful knowledge online by watching videos and reading blogs.
Given that it is an absolute bottomless pit optimising for printing I expect I will be doing a large amount of prototyping and iterations. The way to get good at this skill is to practice it and learn after every single iteration.			

These areas of investigation let me effectively work out what were the gaps in my knowledge and how to cover them.
This enabled me to progress faster with my project and to make better researched and more informed decisions. Conducting the research also taught me lots about the other available actuators and parts and materials that would come in handy for future projects.
Investigating legality first was the most important decision, as the whole project depended on whether I could mount the plates inside the car, and there was no point designing a mechanism until I knew that. The filament research also directly changed what I built, because PLA deforms in a hot car, which is why I chose PETG for the body. Its limitation is that questions like the reliability of the suction cups could not be answered by reading and had to be tested physically.

Project Proposal

Criteria to Evaluate Success

Criteria	Relevance	Implementation
Function - Speed of operation: Easy to take on and off - Quick to put on and off	The L Plate must flip open, shut and log hours faster than it is to do it manually. The time it takes a typical learner driver to put plates is approximately 40-50 seconds depending on the size of the car. Displaying L-plates (and taking them off) must be significantly more convenient than the old way is because the user won't use it. Log hours must not take longer than manual logging and deliver additional benefits such as permanent record keeping (indelibility). The product will not be used, and therefore the need won't be solved.	The L-plate should take under 3 seconds to toggle and the log book component should take under 3 seconds to complete. I will be testing each prototype and the rate of which it takes to turn on and off the L plates and also the ease of it. If there are any obstructions I will be improving this from prototype to prototype. This also needs to be easy to use. So I will be evaluating the ease of the use of this product. I will also be surveying the demographic on
Aesthetics - Colour that does not stand out - Design should be sleek as it should go onto any car	The plate must blend into the car's aesthetic so that it is not an eyesore. If the plate looks ugly and does not blend into the car's aesthetics, some users won't want to use it. It should be a pleasing design to look at and should not stand out too much on the vehicle. The impact if not achieved is that the plate will have less adoption. This will result in people reverting back to the previous solution of the L and P plates. The L-plate should be visible from 10 metres away, and it should not have exposed wires. The colour should be neutral.	The L-plate should be visible from 10 metres away, and it should not have exposed wires. The frame colour should be neutral.
Legality - Follows the ACT guidelines	Because this is a legally mandated sign, it must display the L-Plate in a way that complies with the regulations If not followed, the entire MDP would fail because the plate would not comply with the legal requirements expected from the learner driver and their supervisor.	If the plates do not comply and are not legal, the entire project is a failure I will follow the ACT guidelines and plate usage on vehicles closely referring to their website.
Ergonomics -Must be easy and intuitive to use	Ergonomics is important because my product must be easy for the user to use without having to read complicated documentation or understand advanced concepts.	I will do this by applying user-centred design throughout the entire design process

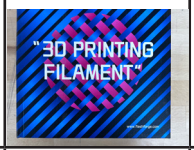
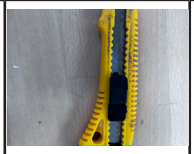
Project Proposal

Criteria to Evaluate Success continued...

Criteria	Relevance	Implementation
Quality - Life Cycle - Can sustain wind and other road conditions	The plate should feel high-quality and be high quality. Often 3D printed products are perceived as inferior quality, however this is not always true. It is easier to make premium products if taking full advantage of the 3D printing quality. If it does not feel high quality, people who care about their cars will not put it on their car. If the product is not high quality, it will not work well and break easily thus not solving the need of the plates falling off. The impact of an inferior product will result in a smaller portion of the target market adopting it.	There must be proper-looking packaging and no exposed wires. The product must work reliably and have longevity. Testing will occur with the final prototypes while driving with the device attached to the car to ensure it works. A longer test drive of two hours will be undertaken with various terrains and conditions.
Safety - Safe use for consumers - Should not damage the car	The product needs to be safe to both the consumer and the car. The consumers should not experience cuts while attaching the product on the car The car should not get damaged when using the product. This includes paint scratches or dents onto the vehicle due to the product	I will test each prototype to ensure that it does not damage the car. A stress test will also take place to determine if there are any impact on both the consumer or the vehicle.
Cost	The L plate must be a reasonable cost. It should not cost more than the current plates. A physical pack of L plates typically costs between $3 and $15 AUD, depending on the attachment style. Basic Plastic / Tie-on Plates: Around $3 – $5 per pack. Magnetic Plates: Around $7 – $10 per pack. Clip-on / Premium Holders: Around $12 – $15 per pack	I will closely evaluate the cost of my experimentation and resources used throughout the process to ensure the product does not exceed the current cost.

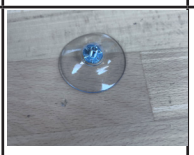
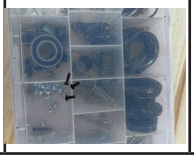
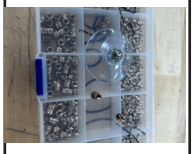
Writing out clearly what the requirements and criteria for this product to succeed were helped provide direction for me. When making decisions throughout the design process I was able to come back to this and work out which option would best maximise my criteria for success.
This was quite instrumental for helping choose to mount internally as this is one of the key decisions that I had to face.
It also helped me decide on the layout of having a master and a follower plate as this made it become more intuitive for the user to use.

Project Proposal
Finance Plan

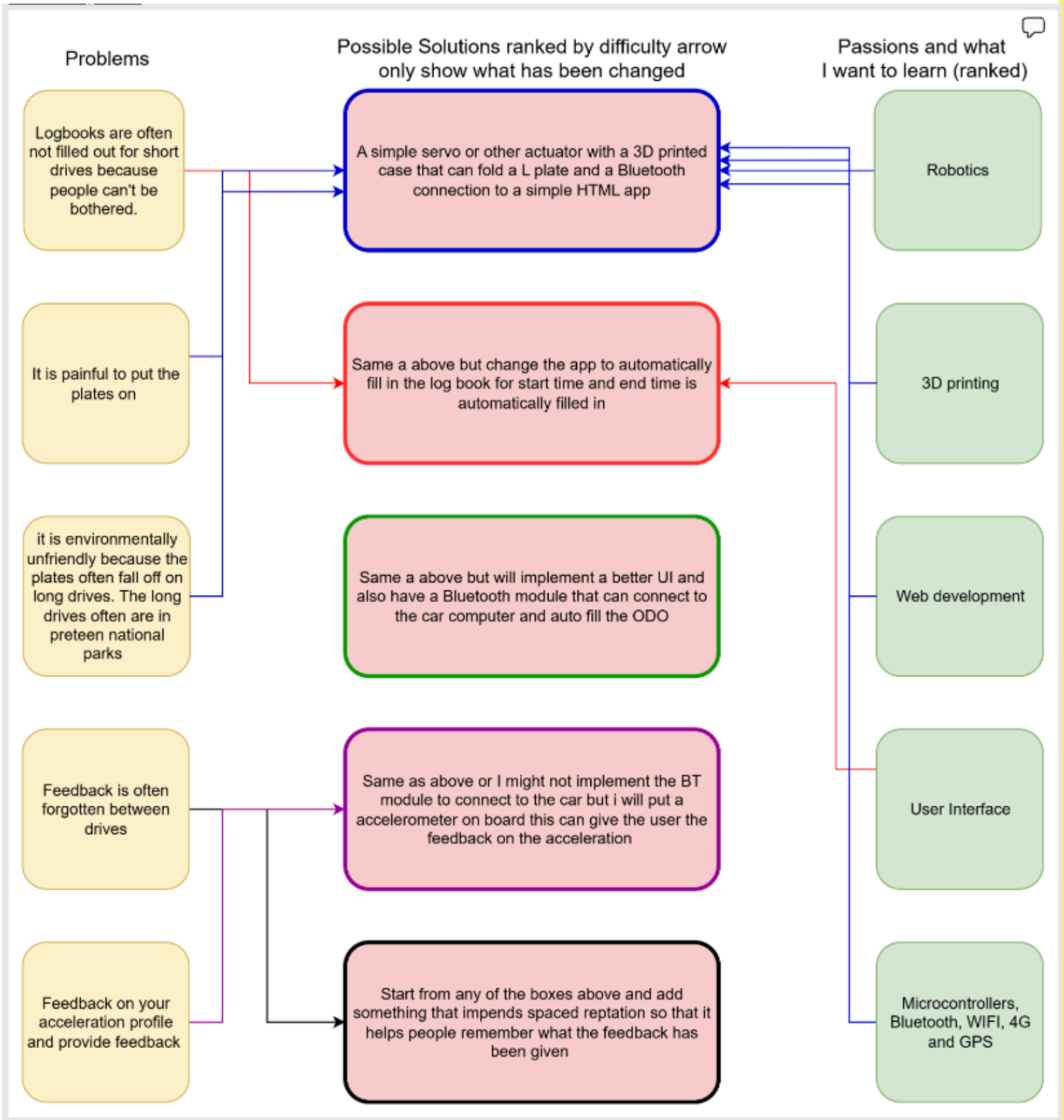
Item	Image	Proposed Cost	Actual Cost	Evaluation
Soldering wire with flux		$30	$0	This particular soldering flux is lead free. This is important as lead free solder is much safer to work with as lead is toxic. It is also of a small diameter which makes it ideal for microelectronics
Printing filament PETG PLA		$25x3	$0	This printing filament was provided by the school DT department. They had PLA and PETG in different colours
CA glue (Superglue)		$2	$0	This is superglue. I used it to adhere the acrylic to the 3D printed plastic although in the final product I did not need to use this as I had designed my solution to be reusable and repairable
Small cable ties		$5	$0	These are small cable ties. I used them to fix the servo horn to the 3D printed part. Initially, I used superglue then I used larger cable ties. Finally, I used small cable ties. I settled on this as it would suit my design better
Box cutter		$17	$0	This knife is really useful as it allows me to cut the support material away from my printed parts and also to cut wires as the school does not have wire cutters.
9g Servo		$6x10	$5.70x10 $57	These are very common in small electronics projects. They are a small servo motor. This is the actuator that I can control via a PWM signal (a way of commanding a position of the actuator), this then allows me to control the position of my louvre. The reason I had to buy so many of them is I broke a couple accidentally
Bambu Lab P1S		$1 200	$0	My mentor had this printer and it is a really good printer. It prints at very high speed with high precision and an excellent surface finish. This allowed me to rapidly iterate prototypes and I also got to learn how to use it
Spray booth		$12 00	$0	This is a spray booth which lets me safely spray paint. I needed it so that I could test whether spray painting was a valid option for applying the L to the plate
Spray paint		$20x3	$0	These are spray paint and primer. I needed them to spray paint. They also need to be the correct colours. The school didn't have all the colour so I tested without the colours and I established that it was not a viable approach.
Contact		$8x3	$8x3 $24	This is contact (the same thing you use to cover textbooks with just different colour). I needed them in the colours of white, red and black and it is surprisingly difficult to find. I had to go to 5 different stores to try and get it.

Project Proposal

Finance Plan

Item	Image	Proposed Cost (AUD)	Actual Cost (AUD)	Evaluation
Acrylic		$20	$0	This is acrylic. I initially used it for the base plate but my final design did not include it as it would make it more repairable and remove the need of planned obsolescence.
Band saw		$3840	$0	This is the band saw that I used to cut the acrylic. I did not end up needing to use it but it was used in prototyping.
Heat Gun		$600	$0	Due to inconsistencies in the manufacturing process of servo horns, they were of inconsistent size so I chose to use the heat gun to soften the 3D printed plastic so that they could fit
Wire		$15	$0	This is silicone shielded 30AWG wire. This is good as it is thin and easy to wrap and the silicone coating means it's flexible
Suction cup		$20	$10.99	These suction cups were a very good find as they let me successfully mount my L plate to the car at an economical price.
18650 Protected cell		$15x6	$15x6 90	This is an 18650 protective cell. If it was not a protected cell, my micro controller could drain the battery past where it is safe to potentially cause a fire.
ESP32-C3		$15x6	$10.65x6 $63.90	This ESP C3 is a common low power low cost micro controller. It has a good SDK meaning that it is easy to program on. It is very popular and common so that has lots of example code and forums where you can get help. It also has an integrated battery charger. This means I can reduce my part count thus reducing overall cost.
TS101 Soldering iron		$100-150	$0	This is a very good soldering iron. It is powered off a USB C port so that you do not need to buy a second power supply, meaning that it has high power capabilities while also not being bulky, heavy and at a low cost.
Self tapping screws		$5	$0	These allow me to fix the module's case in position. I chose self tapping screws as they were cheap and easy to use and as it was not a load-bearing part it was sufficiently strong.
Threaded inserts		$10	$0	These are threaded inserts. They are cheap while also being very easy to use. You need to heat up a soldering iron and then press them into a hole. This allows you to have a metal thread in plastic part. I have these at home as I have used them in personal projects before.

Project Proposal
Action Plan

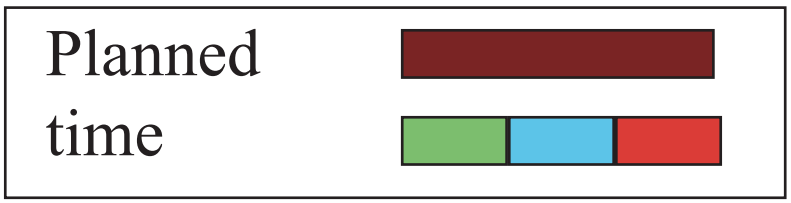


Because this major work is ambitious I have structured the project in such a way that depending on how my time management goes I can stop at various checkpoints along the development of the project and still have a finished prototype.
This plan also effectively outlines the different areas that I can pursue and each of them are mostly decoupled from each other meaning that I can work on one without negatively affecting others. This lets me modularise the time I spend so that I can become better at each of these areas separately and means that my overall skills can be lower.
An additional benefit is that I can find different mentors for each one of these areas and this means that I can get lots of help from experts in their areas.
The order of the plan is also deliberate. The CAD and 3D printing come before the electronics because the mechanism determines the size of the housing, and the housing determines how much room the electronics have, so doing it in the other order would have meant redesigning work I had already finished.

Project Proposal

Time plan

Task	term 4									holiday							
	W1	W2	W3	W4	W5	W6	W7	W8	W9	W1	W2	W3	W4	W5	W6	W7	W8
SECTION 1: PROJECT PROPOSAL & MANAGEMENT																	
1.1 Identification and Exploration of the Need																	
1.1.1 Possibilities of the project (Big Idea exploration)																	
1.1.2 Identification of the MDP Need (Design Brief & Situation)																	
1.1.3 Exploration of the Need: Initial Research																	
Target Market identification																	
1.2 Areas of Investigation																	
1.2.1 Areas to be considered (Cognitive organiser)																	
1.2.2 Research Plan: What is to be investigated and how																	
1.2.3 Limitations and parameters																	
1.3 Criteria for Evaluating Success																	
1.3.1 MDP criteria for evaluating success																	
1.3.2 Design factor consideration																	
1.3.3 Evaluation methods																	
1.4 Project Management																	
1.4.1 Proposed action plan																	
1.4.2 Time plan (Gantt chart)																	
1.4.3 Finance plan and budget																	



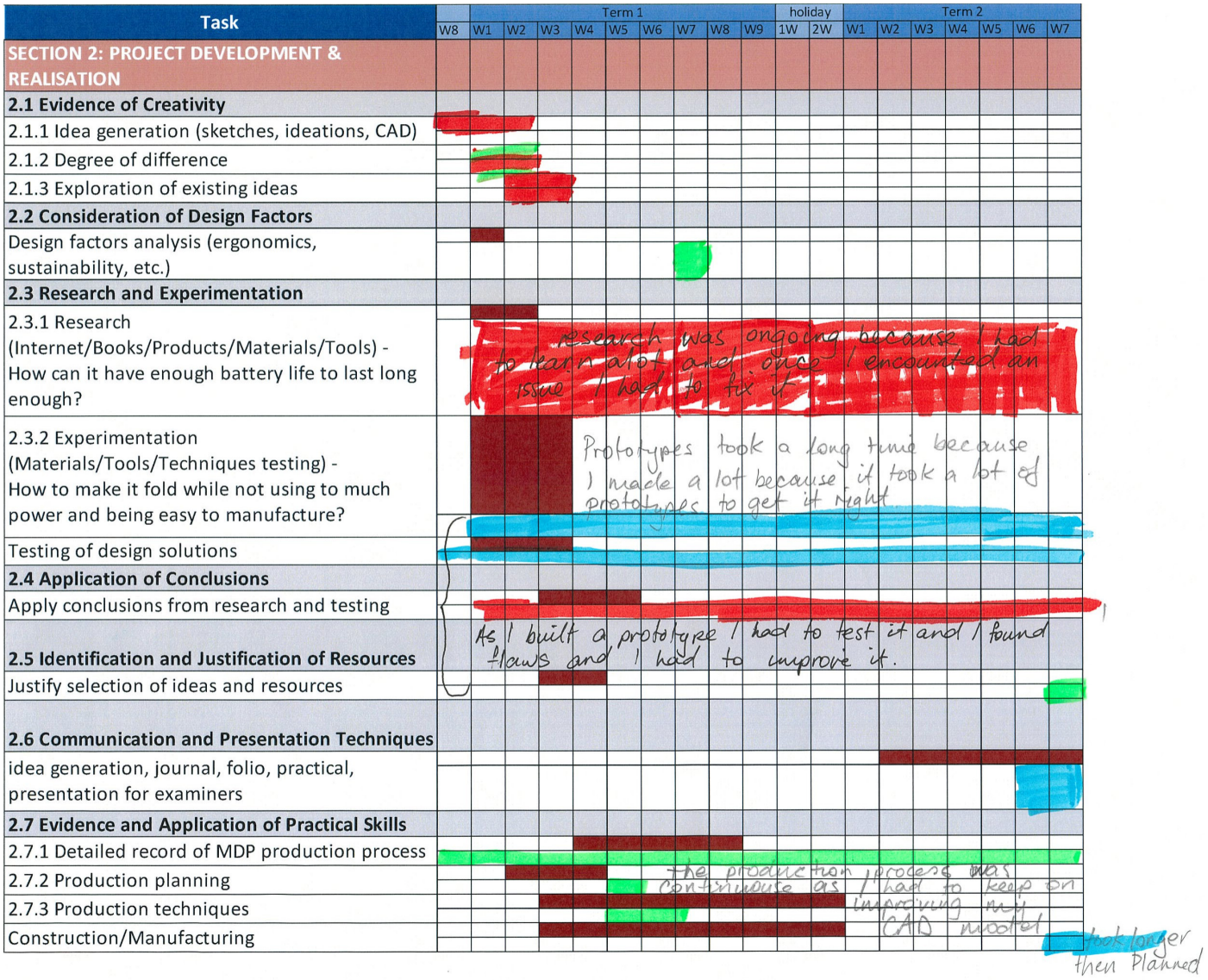
Structuring the plan around checkpoints also manages the risk of the project. If one area takes much longer than I have allowed for, I still have a working L plate from the previous checkpoint rather than a half finished product, so the worst case is a less advanced prototype and not a failed major work. This matters most for the electronics and the app, because these are the two areas I have the least experience in and therefore the areas I am least able to estimate accurately.

Decoupling the areas also lets me use waiting time productively. A single 3D print can take several hours, so instead of losing that time I can work on the app or solder the wiring while the printer runs, which means a delay in one area does not stop progress in the others.

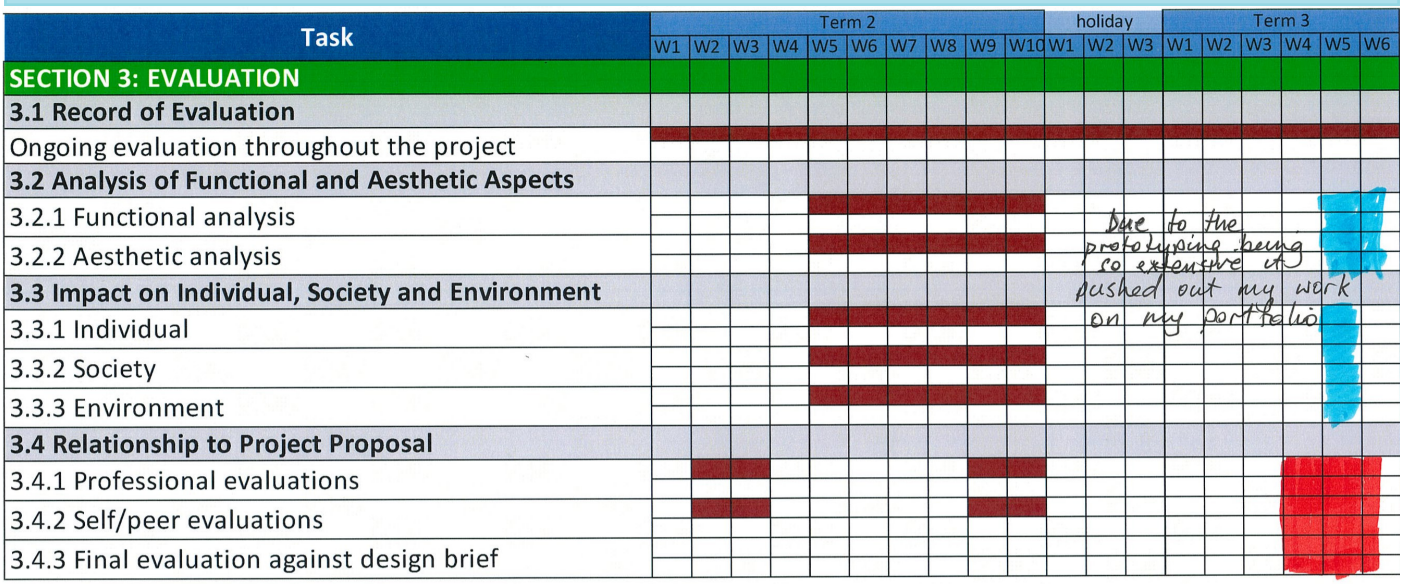
A limitation of this plan is that it assumes each area is equally predictable. In reality the areas I had done before, such as CAD, were easy to estimate, while the areas that were new to me were not, so the later checkpoints are far less reliable than the early ones.

Project Proposal

Time plan



This stage two also went quite well save for the prototyping research and experimentation as this was an iterative loop that extended well beyond into term three. Because after every prototype, I would have to address various issues and solve them in the next version that is why I had to iterate and continue researching.



Because my prototyping from section 2 took so long the time I had for my evaluation was reduced. This was poor time management on my behalf. This led to me spending many many many hours after school working on my evaluation and folio.

Project Development & Realisation

Existing Designs & Solutions

Magnetic nail plate

Plus:	Minus	Interesting:
- Magnetic L plates are successful because they are easy to put on and takeoff - Magnetic L Plates are also cheap and commonly available; through service stations and supermarkets	- When the magnets that fix the L plates are too weak or are applied to the wrong area of the car the L plates may fall off causing drivers to unintentionally disobey driving laws - Removing highly magnetic L plates can cause superficial damage to the exterior of the car.	- Magnetic L plates are inconsistent when it comes to attaching to cars; in some cases they affix to the exterior of the car extremely well while in other cases they continuously fall off. -Some vehicles, especially newer models which have plastic bumpers, aluminium panels, carbon fibre boots or flush glass tailgates, reject magnetic L plates

Although magnetic L plates are cheap and plentiful that does not result in a successful product as shown in the fact that magnetic L plates regularly fall and pollute the roadside environment. This lacking ability to affix to cars is the reason I have chosen not to pursue a magnetic attachment solution in my design. Furthermore, as my design is more expensive than a standard plate, I have made sure to use an attachment mechanism that prevents the plates being lost.

Competitor app analysis

- This app is successful because it has been utilised and tested, and has a fully functional user interface (UI) - The choice to categorise the data entry makes the app successful as it makes inputting data user friendly and easy to analyse	- This product is unsuccessful because logging data is time consuming as data must be manually put into the odometer - It does not support Dark Mode, an important safety feature for night driving	- The user interface of this mobile app is highly refined - This application also works for learner drivers in states all over Australia, however, use of this app has not been approved in certain states.

Analysing this application has helped me understand what the user interface of my design should look like, inspired me to include a categorised data entry for the weather (a variable that learner drivers must include in their logbook entries) and night or day driving. I will also implement a blank record recording screen that appears while driving and says "Don't look at your phone" as I believe that is a good safety feature. I also believe that this is an efficient solution to the challenge of learner drivers and fully licensed drivers consistently being caught on their phones while driving. In my design I will remove the start and suburb input variables as this is data that can be automatically collected with GPS data.

Project Development & Realisation

Existing Designs & Solutions

Clip on P plate

- A successful aspect of the clip-on P plate is that it is versatile, as it can be held in many orientations

According to my primary research, I did not find any of this style plate on the roadside but they were primarily magnetic

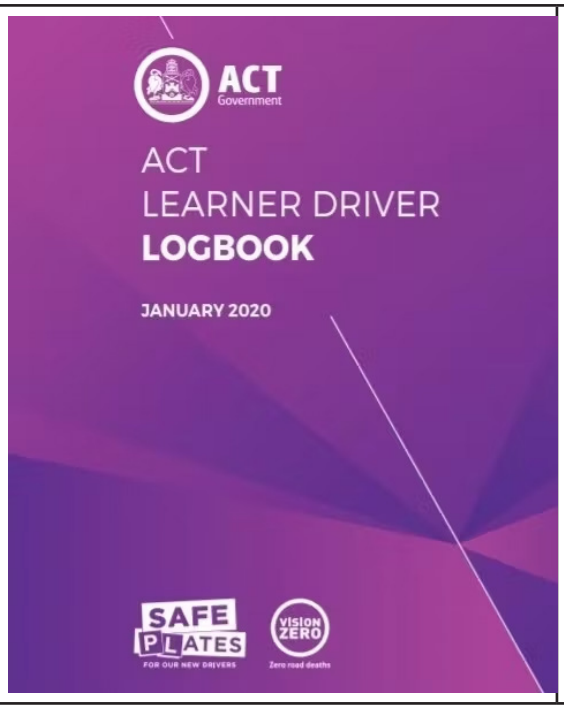
It is not very strong so if I have a large heavy plate it will come off

- It is harder to take them on and off

These clips are one of the more reliable ways of holding plates on, but they are quite annoying to take off, as the clip often grips either too tightly or too loosely.

Despite it being a very good solution, I'm unable to use it as the clips will not be strong enough to hold the torsional stresses that my heavier L plate will induce.

Current logbook



This is analogue, which means it can't suffer from bugs and server failures

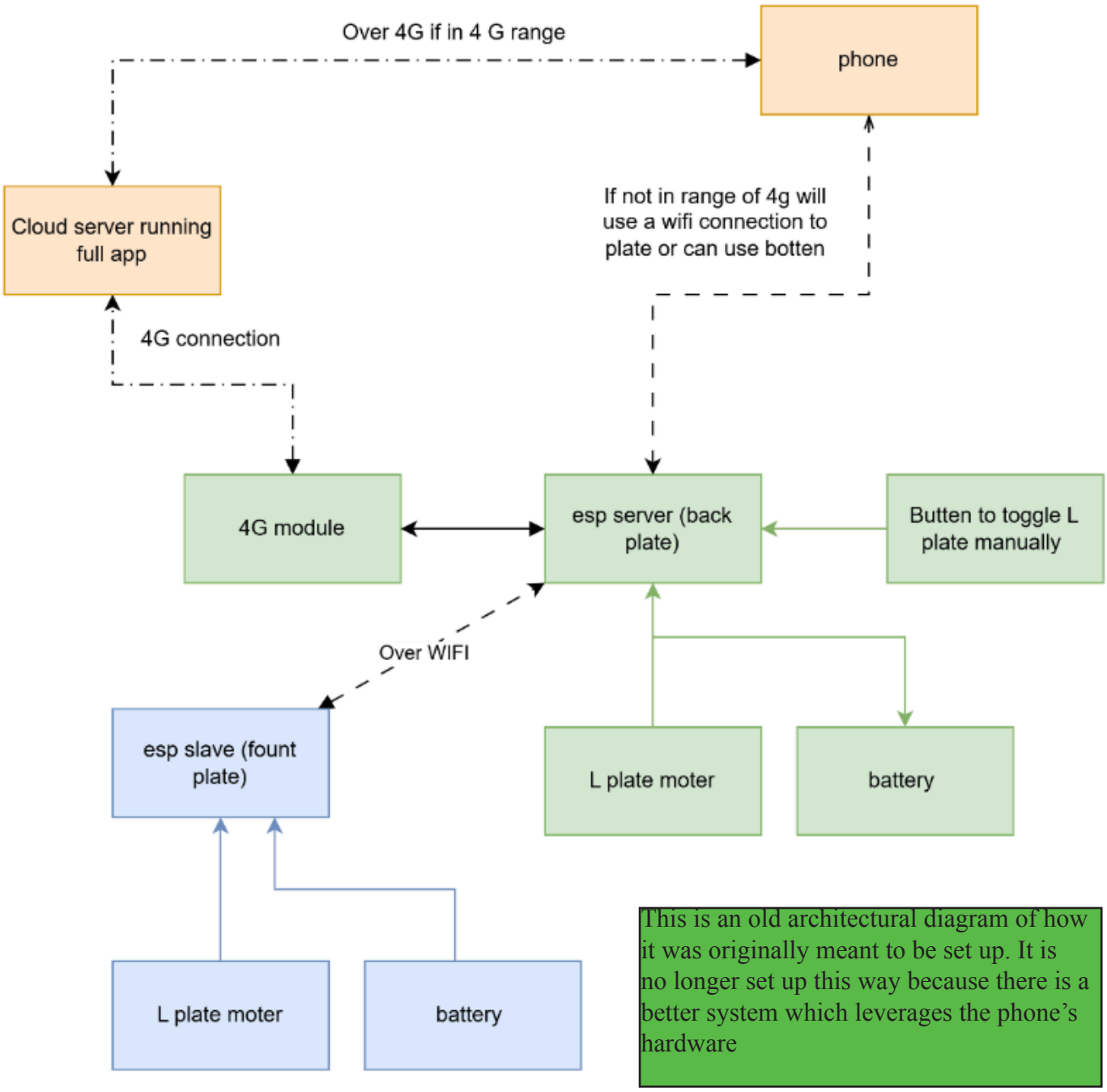
It is very easy to fake hours
Can get lost
Can have illegible handwriting

A digital logbook has become a standard in some states, but most of them still require a physical logbook

Despite the logbook being very good historically, it does not leverage the full abilities of modern tech technologies to preserve records in a way that can't be faked. There is no ability to verify that the drive actually was conducted that you were the driver. This is why I am opting for a digital logbook in my solution.

Project Development & Realisation

Idea Generation



This design was one of my first attempts to design the infrastructure for the L plate app and plate. Ultimately, I've ended up changing it quite significantly but this was a very good starting point, and through the design process and repeated evaluations I was able to come to a better design.

This design uses 4G which allows it to work off-line in isolation but it needs a mobile data connection and you also need an expensive pay per month SIM card. This results in the L plate being heavier and consuming a lot of power so battery life is reduced. This design was not very good and was quite naive but it was a starting point for further design.

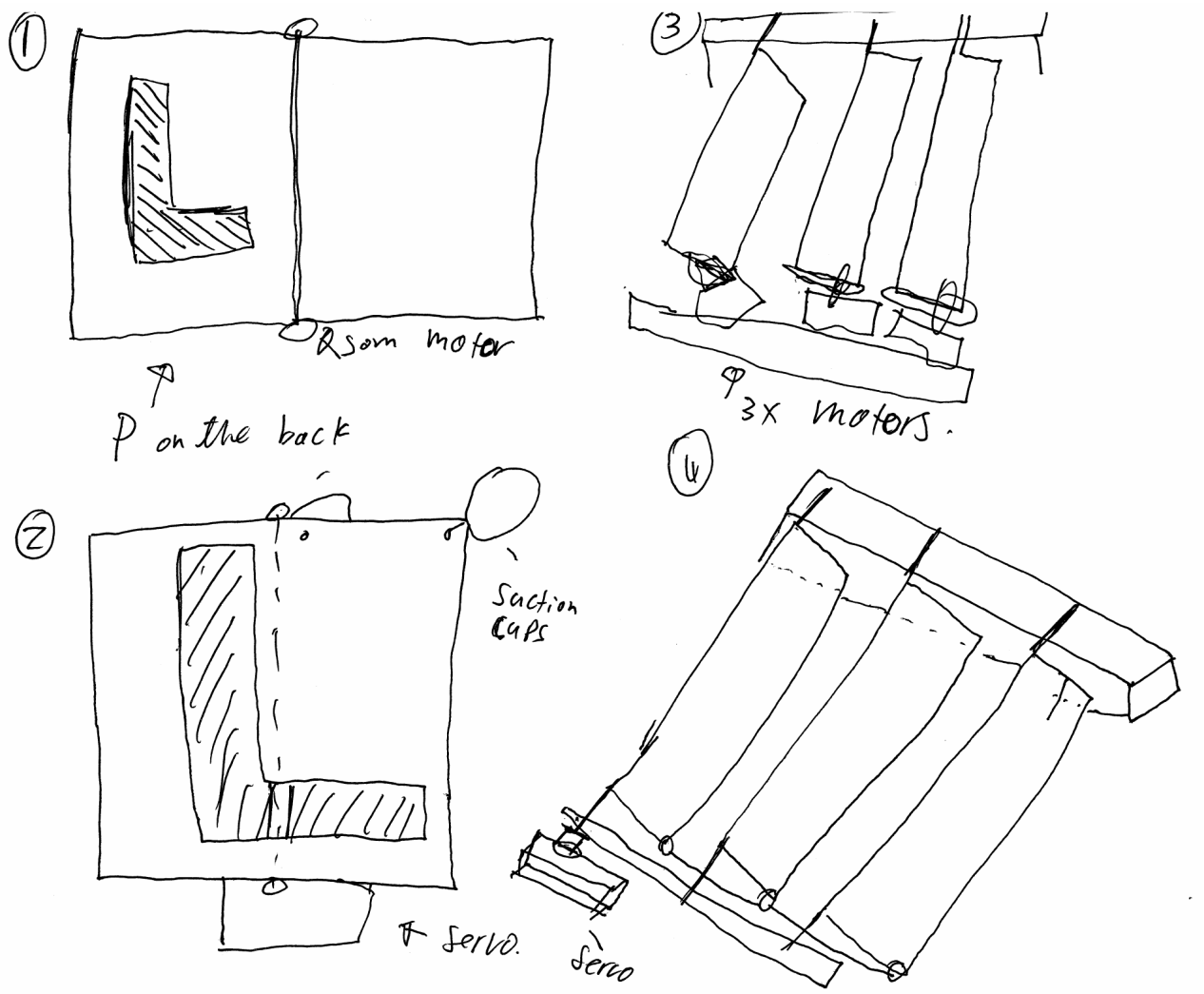
The design that I ended up going with was much more efficient as it does not require the extra component of the 4G SIM card but instead it uses Bluetooth to talk to the phone which can be easily charged. The GPS means that the per unit cost and manufacturing complexity is also decreased while having a more seamless user experience.

I could improve on the flaws of my first iteration with my second iteration and remove the need for the GPS and 4G module which would reduce the cost and enable me to move more of the processing to the phone.

Evaluating this iteration against my success criteria is what made the decision clear. A monthly SIM fee would have been an ongoing cost to the user on top of the unit price, and the larger battery needed to run 4G would have made the plate heavier and harder to mount, so this design failed both my cost and ergonomics criteria even though it worked in principle.

Project Development & Realisation

Sketches



1) This was my first design. It was a book inspired design where the L plate would fold with a motor at the crease. The advantages of this design is that it would work but it would not be very compact for space when opened and would take up twice the space of a normal L plate.

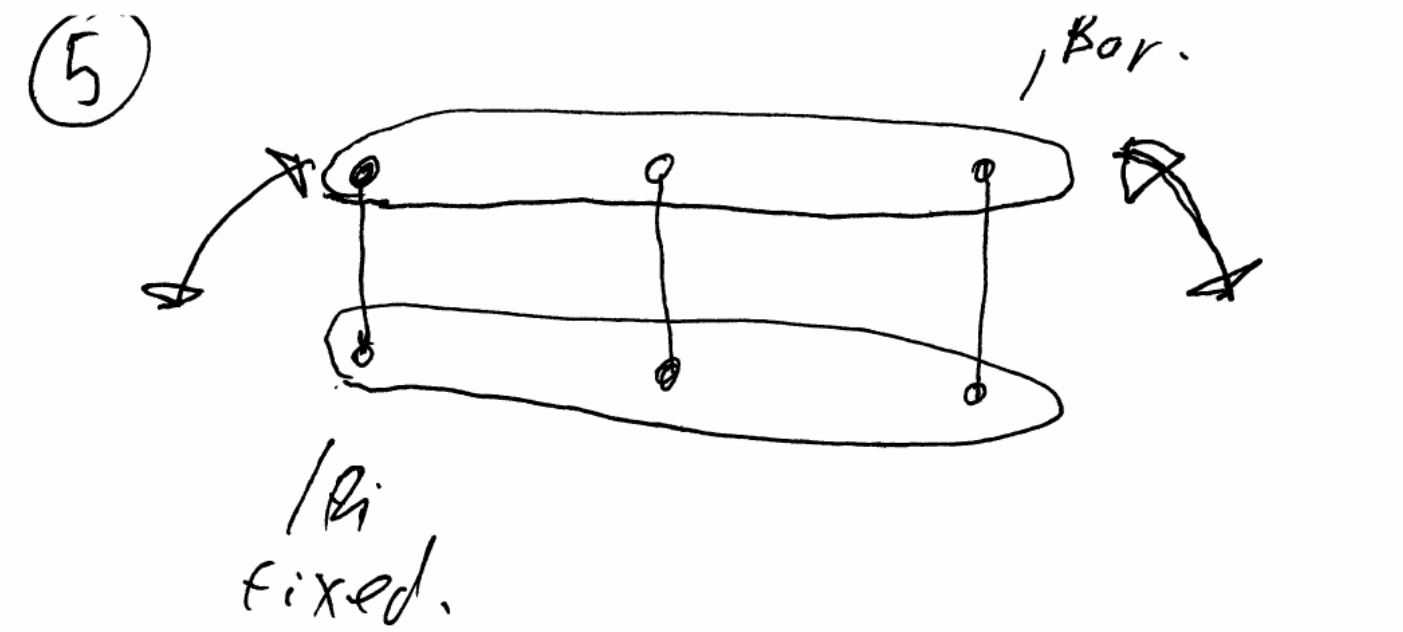
2) The second design I split the plate in half along the middle. This meant it reduced the amount of space it takes up when fully open but it does not allow you to have both a L and a P plate mode.

3) My next design involved having multiple hinges that fold and each one would be controlled by a motor. The advantage of this design is that it allowed it to take up the same amount of space as Design #2 while having the dual sided ability of Design #1. This design involved using three motors for each blade of the mechanism. This will become quite expensive, heavy and difficult to manufacture.

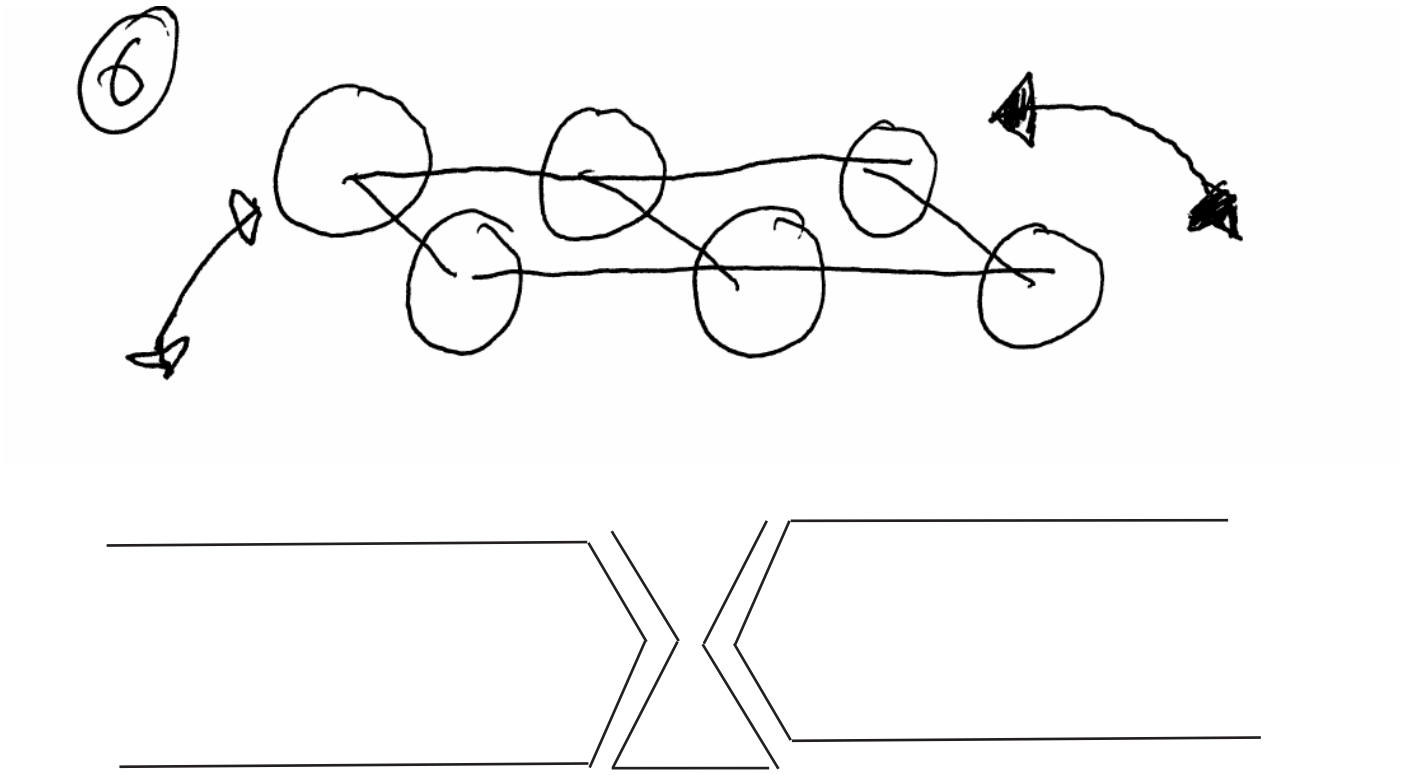
4) This one used a linkage known as the four bar linkage to couple the motion of one of the blades with the surrounding blades. This enables it to solve the issue of complexity of Design #3 while maintaining the same functionality without the cost of manufacturing complexity every single one of those linkages.

Project Development & Realisation

Sketches



5) This one uses a single bar to couple the motion of the blades. This will reduce the manufacturing cost which was a limitation of number 4. This is because its printed in place. Its one single part and not seperate parts reducing costs and making the construction easier. This design is weak because small pins are difficult to 3D print as it requires high precision



6) In this design, I have adjusted the shape of the pivot so that they can be stronger while keeping the same functionality. This will increase the printability and durability. It increases printability as printer struggled to make circles with small radiuses. So making the radius larger results in much better tolerances and more precision perhaps. A larger diameter pin has more material and more surface area. This means it will be stronger. Additionally, the dovetail pin that I will use has more surface area than a normal pin additionally increasing more durability through surface area.

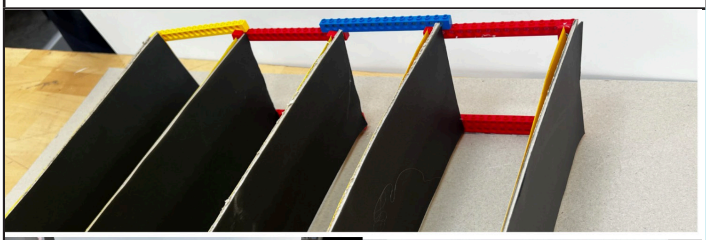
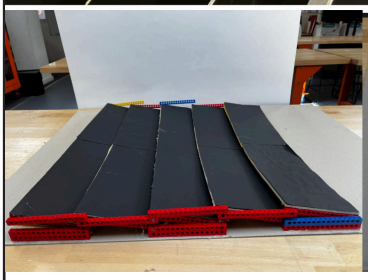
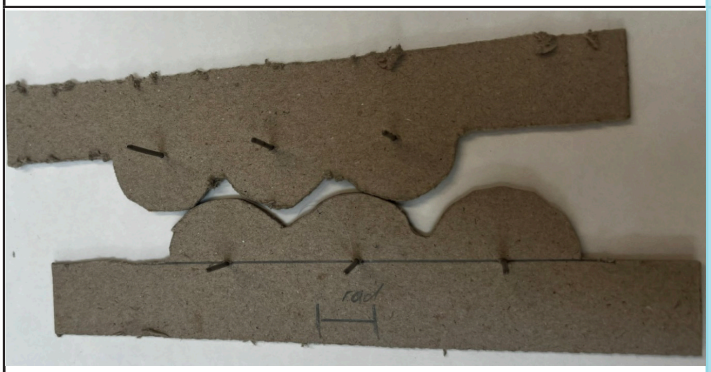
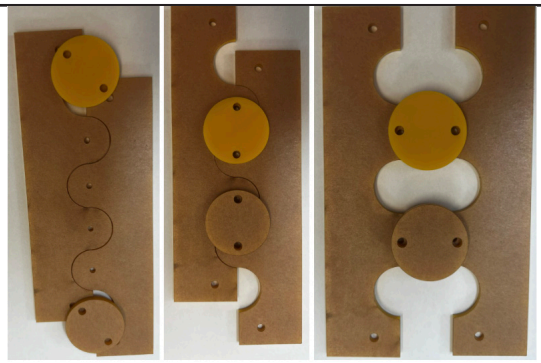
Project Development & Realisation

Factors Affecting Design

Factor	Evaluation
Work health and safety	It's important to consider workplace health and safety when making my project as I'm dealing with fumes that are carcinogenic and potentially heavy metals (lead) that are very dangerous. This means I need to wear proper protective equipment and ensure proper ventilation when manufacturing my prototypes.
Aesthetics	Aesthetics are critical when designing Lflip as I would like my plate to look professionally made and not have scuffs, dents and hot glue all over the place. Given that many people in my target market care about how their cars look it is important to make it look aesthetically pleasing and they're willing to use it.
Needs	It is essential that I consider the consumer needs when designing this licence plate. This project must solve a need for the consumer. If it doesn't it will not get bought by the consumers. I will do this by conducting surveys and also asking my classmates as they are the target market.
Environmental consequences	I must consider the environmental consequences when designing this licence plate as I am attempting to address many environmental issues such as L plates falling off causing litter and the overconsumption of plastic with more plates being bought. I must make sure my design addresses these changing needs as this is very important.
Quality	It is important to consider the quality when designing my Lflip. If my product looks high-quality, clean and is durable enough to always work in all weather and has good battery life it means that my target market will buy it as they will see it as a good investment that will not break.
Life cycle analysis	It is valuable to consider the life cycle of my product. Due to increasing social concerns about the environment, I must make sure that my product is reusable and has a second life (working as a P plate) this means that I not only extended life-cycle but it is also made up of recyclable materials.
Function	It is important that I consider the functionality of my Lflip as it primarily must work as an L plate that flips automatically. If it doesn't work reliably then customers will not buy it and the product will be a failure.
Ergonomics	Ergonomics is important for this app as it must be easy to use, intuitive with the hardware binding and pairing to the app in a way that makes sense to the user. To ensure that it is ergonomic I will test it with users that have not seen the app before and I will watch them use the app (without instruction) to see if they succeed and where they fail in navigating the app. I will use this feedback to make improvements. Whether the whole system works together is what defines whether it is a good product.

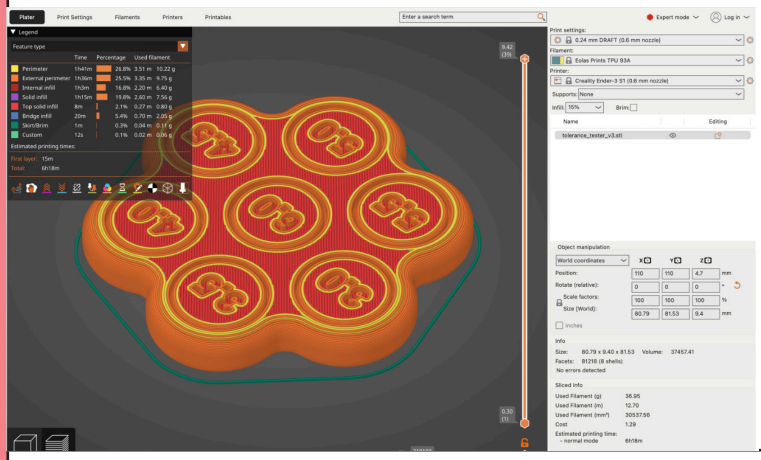
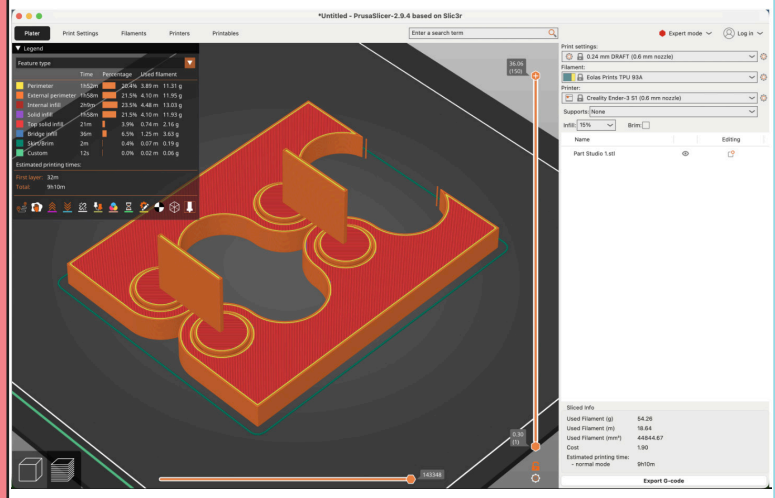
Project Development & Realisation

Prototypes

Prototype	Evaluation
 	Prototype 1: Template Steps 1. Used Lego blocks as a template to position the foundation blocks correctly 2. Hot glued one sheet of cardboard to the Lego bricks 3. Assembled the pins 4. Glued yellow and black paper to highlight how the mechanism moves and that it will work as an L plate This was the first test of the mechanism with the four bar linkage. This was an essential step to establish that this mechanism was a valid solution to the problem.
	Prototype 2: 1. I had a discussion with my mentor and we brainstormed ideas together until we came up with this idea of a mechanism 2. Next, I laid out cardboard and then cut it out 3. To simulate pins and joints I put nails through the centres of the holes 4. I then attached a plate to the nails to check if the mechanism could actuate fully
	Prototype 3: This mechanism worked well but I had a few invalid assumptions about the spacing of the two constraint pins and it helped remove those before moving onto prototype four which is identical save this pin adjustment.
	Prototype 4: I made the laser cut acrylic on Adobe illustrator and I adjusted the hole placement as per the previous one. I could reuse my same file but only making some small tweaks. I also changed from large circles to smaller ovals because this would allow me to see what was happening in the mechanism and better. This prototype taught me that my mechanism was completely valid although this particular mechanism was not optimal or optimised but it did work which was all it needed to do. There was lots of room for improvement but it was a foundation that would work.

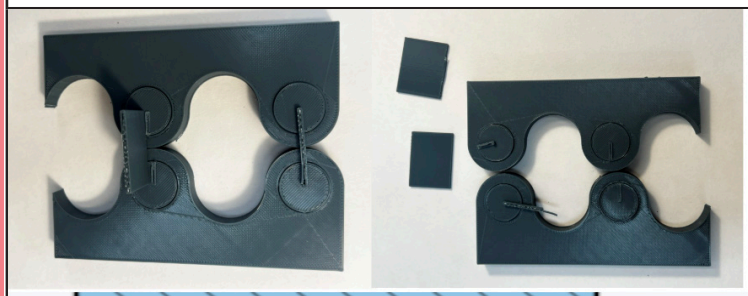
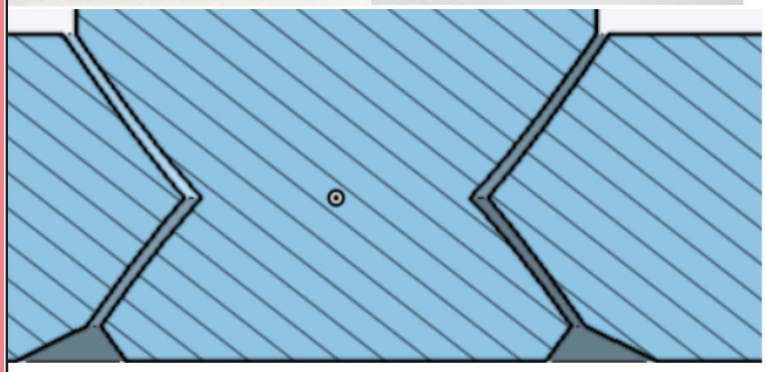
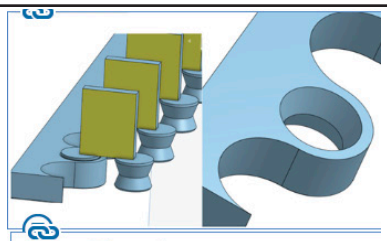
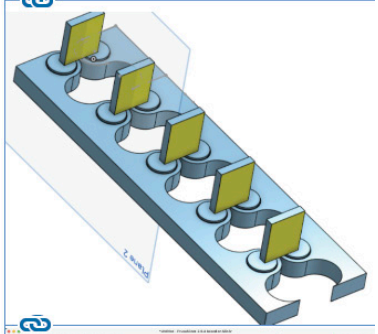
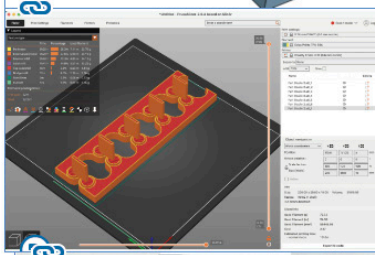
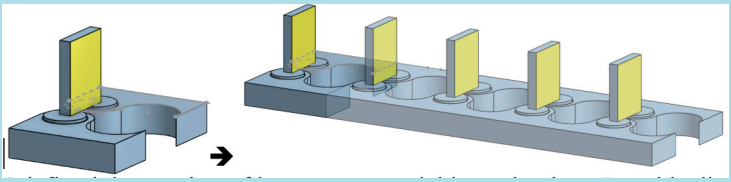
Project Development & Realisation

Prototypes

Prototype	Evaluation
   	Prototype 5: 1. Find the STL https://www.thingiverse.com/thing:3591280 it was on a site that is a repository the 3D printed 2. Next I imported into Prusa Slicer 3. I printed it on the school's 3D printers 4. The general process of putting on a 3D print is to: 5. Load the filament 6. Preheat the bed and nozzle 7. Make sure the bed is clear 8. If necessary applying glue stick to the printer bed 9. Slice it with the default settings except a brim 10. Upload the sliced G code file onto a USB stick and begin the print 11. Click start 12. Monitor the first layer does not have any failures 13. Come back and take it off the bed 14. Then I took this off the printing bed and with my hands saw which ones of these I could move if I could move it, it was a good enough tolerance; if not, it wasn't Ultimately the purpose of this was to be able to find out what is the tolerance of the 3D printer because I was doing print-in-place mechanisms and I need to know the gap that I need to leave so for my next prototype. I worked out in CAD how thick it should be reducing the amount of iterations I had to do
	Prototype 6: I designed this particular part in Onshape using a re-volve to generate the dovetail pin I made sure I assigned the particular tolerance to a vari-able in the sketch I used so that I could easily modify it without needing to edit my sketch manually. This is called parametric design and I used it throughout the entire design cycle I exported the 3D model and then I printed it using the standard process

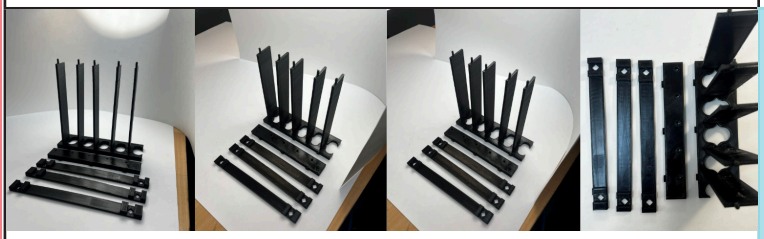
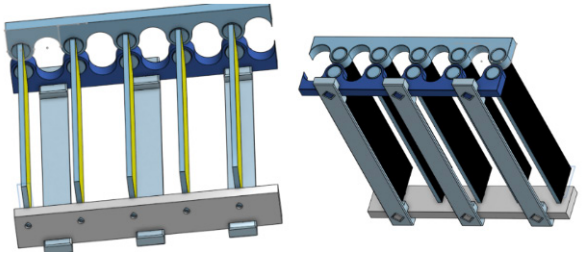
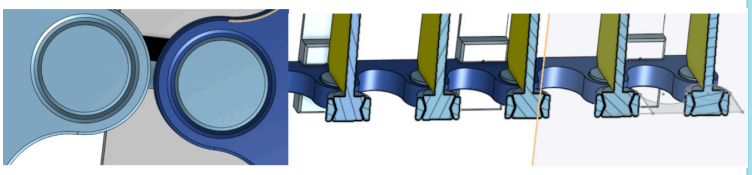
Project Development & Realisation

Prototypes

Prototype	Evaluation
 	Prototype 6 continued… This allowed me to verify that the dovetail pin mechanism worked, in addition it affirmed the design and shape work in the green acrylic 1. This is very useful because it allowed me to refine a small scale prototype knowing I have a working mechanism that only needs to be optimised from this point forward.
  	1. Given that you can quite clearly see a repeating pattern in the design that I wanted to achieve I was able to leverage the ability of CAD to do patterns so I started by making a single copy of what I wanted and then I duplicated it the number of times I wanted. 2. I defined the number of louvres as a variable so that later I could adjust the number of louvres I needed. This was very useful when testing 3, 4 and 5 different numbers of louvres.  3. I then printed it using the standard procedure Once I had printed this and verified that it worked, I could also test that a single one of these louvres could drive the entire mechanism. This allowed me to confirm that a single driving motor could work, which is the most efficient way of driving it. This is very beneficial because it confirms not only the mechanism but the way of activating it as well in future. I can start designing and optimising this mechanism for mass and area coverage as well as how to mount and drive it with a motor.

Project Development & Realisation

Prototypes

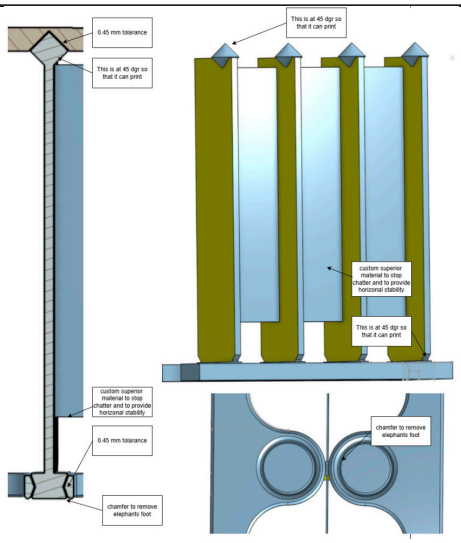
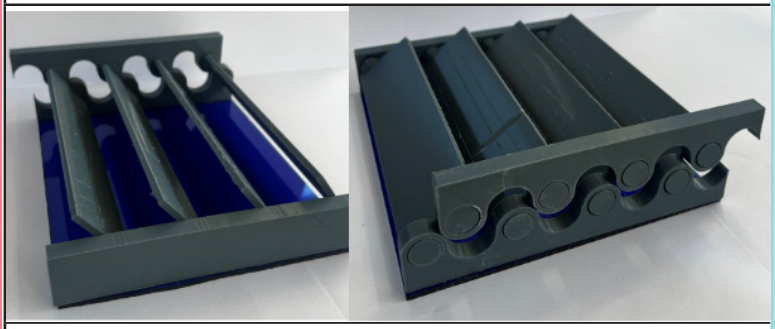
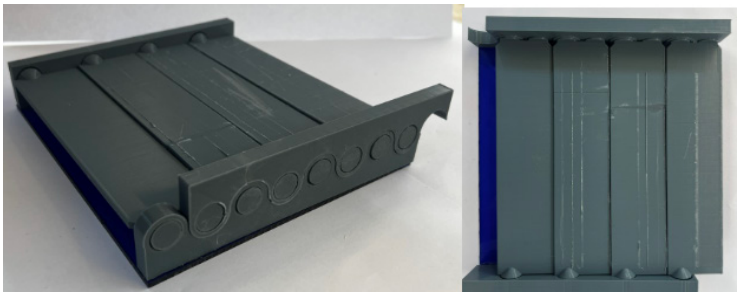
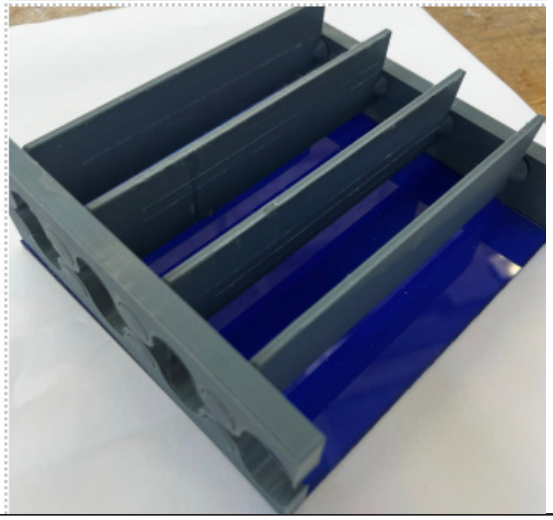
Prototype	Evaluation
	This shows the images from a high angle view laid out as if they came off the printer.
	This image highlights the limitations due to chatter of the top pin
	These 3 images show the chatter which is why I added the custom support materials
	Close-up view of the pins also shows stringing in the second photo.
	An overview of the mechanism
	Two images of the tolerances and the print in place mechanism

1. I was able to rapidly design to test whether this would work by not following parametric modelling procedures but instead just making it
2. The benefit of this is that it was faster but at the cost of having a reusable CAD model but I was able to revert at the end of this iteration back to the original model
3. I was pleased with the teardrop shaped holes which allowed me to print at a better orientation as a teardrop hole will maintain tolerance as it won't sag.
4. I also printed this using the standard procedure.

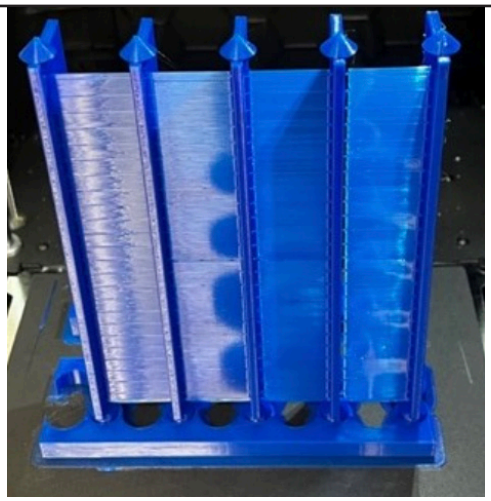
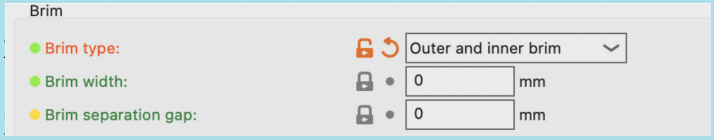
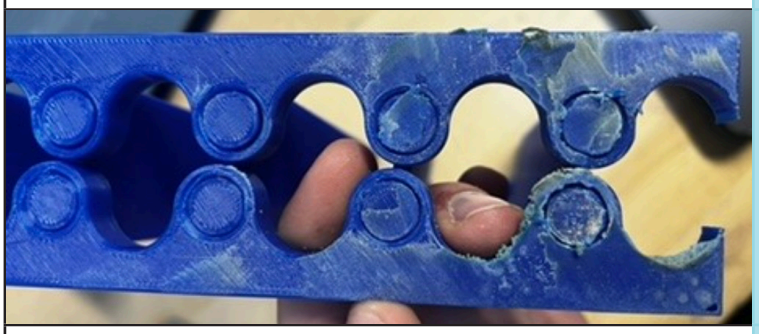
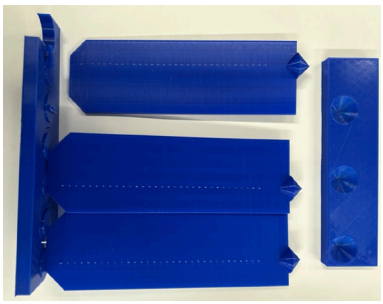
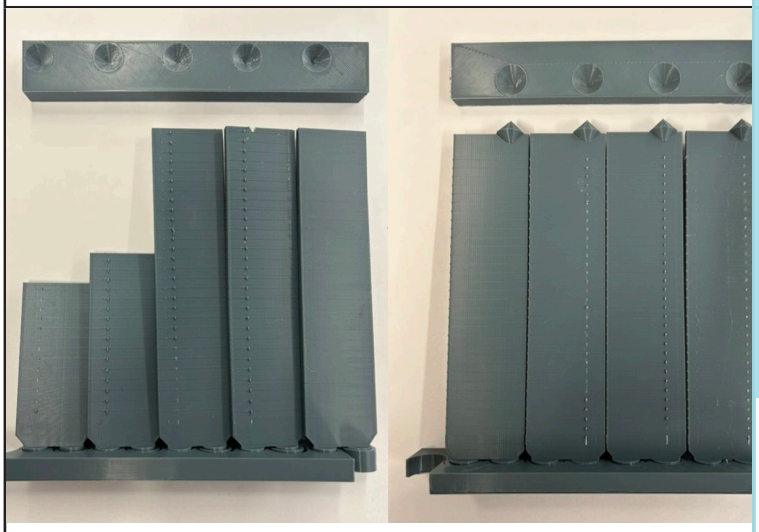
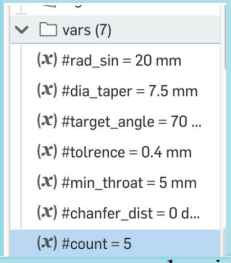
Ultimately, this iteration of the prototype was somewhat unsuccessful but it did prove that the mechanism can work with an endpoint and full length levers. The downside was that the pins are small and the bottom mounting method was not very reliable or strong as well as blocking vision so in future I need to work out some solution like a clear backing to join them as then it would not block vision.

Project Development & Realisation

Prototypes

Prototype	Evaluation
	1. After researching about 3D printing optimisation I applied lots of the things I learnt as you can see in this annotation. 2. For example, I added chamfers to remove elephant's foot. This is where the print squishes out and would cause the joint to adhere. I also added this large support material to stop them swaying. This is called design support. 3. The benefit of design support is if eventually I was to get this manufactured commercially. I wouldn't have to tune slicer settings on their end but I could simply get them to slice my model and it would work. 4. The reason I added 45° chamfers at the end is for locating as well as maximising surface finish
	Images showing it in an actuated state.
	Images showing it in its closed state.
	This print was incredibly instructive as it allowed me to determine that a lot of my optimisations were better. This mechanism was less binding than the previous iteration as well as printing cleaner. One downside is that the support material adhered to the next iteration. I need to add perforations to allow the support material to easily detach. Despite me adding a chamfer for elephant's foot it was not large enough to properly mitigate it so in the next prototype I will increase the amount of elephant's foot prevention.

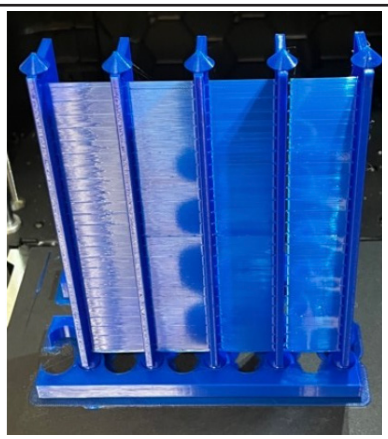
Project Development & Realisation

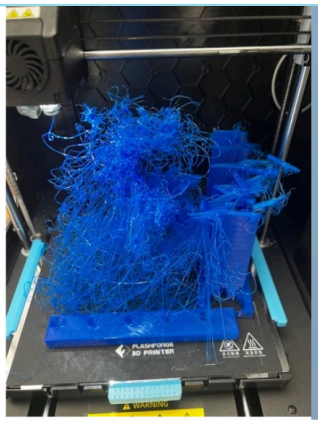
Prototype	Evaluation
	In this iteration I decided to add a brim. This would enable me to print it with better support and it did work, but it was impossible to remove without damaging the mechanism. 1. In CAD I added perforations in the support material  bound up so to fix this I used a disc sander to grind away at the mechanism where it would not come away
	This wasn't good because it left damage to the mechanism, but the mechanism did function. Going forward I will know that if I use a brim I should select an outer-only brim to ensure it is easy to remove, otherwise I will struggle removing the brim.
	Image of broken three bladed print.
	1. By adjusting the parametric variable for the count, I was able to generate three different louvres with different numbers of blades. 2. 3 blades, 4 blades and 5 blades let me test what was the optimal number of blades for my prototype. 3. I printed each of them using the standard procedure. I printed them all at the same time, though with different filaments. 

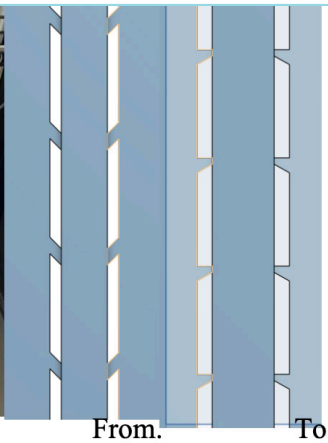
This let me determine the optimal number of blades to have on my mechanism as it was quite difficult deciding on the number. After my testing, I decided on five blades as it satisfied my requirements of having large dovetail pins while having small enough blades.

Project Development & Realisation

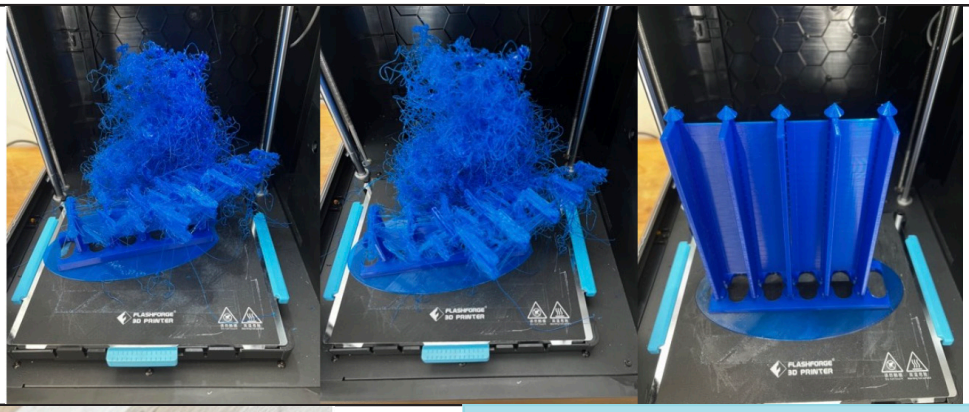
Prototype	Evaluation

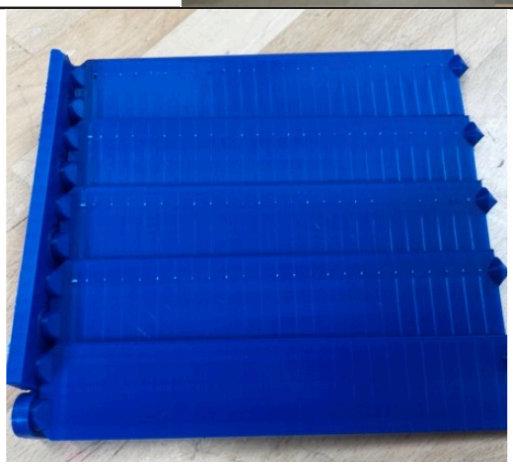






From. To



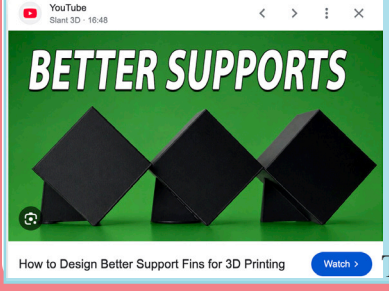


1. Onshape I changed the thickness of the designed support from 1.414 mm to 0.400 mm so that it would be easier to remove. I did this by editing a sketch.

2. Then I printed it with a brim as seen in figure 1 this made it difficult for me to work out what my changes were as I forgot that I had made the small change in the thickness of the support material.

I printed one of them and then I printed it again with a better layer height and I did lots of very good quality bed prep with a glue stick. This failed so then I printed it with a designed brim and that also didn't work. The problem was I kept on getting layer shifts and I worked out it was because the louvres were acting as a lever and they were snapping off the bed of the printer. I managed to fix this by adding support material so that they couldn't leave it and act as one homogeneous piece so it would have enough surface area to stick to the bed. I did this across multiple iterations and I also added a brim in this step. This is a print in place support material which increases adhesion.

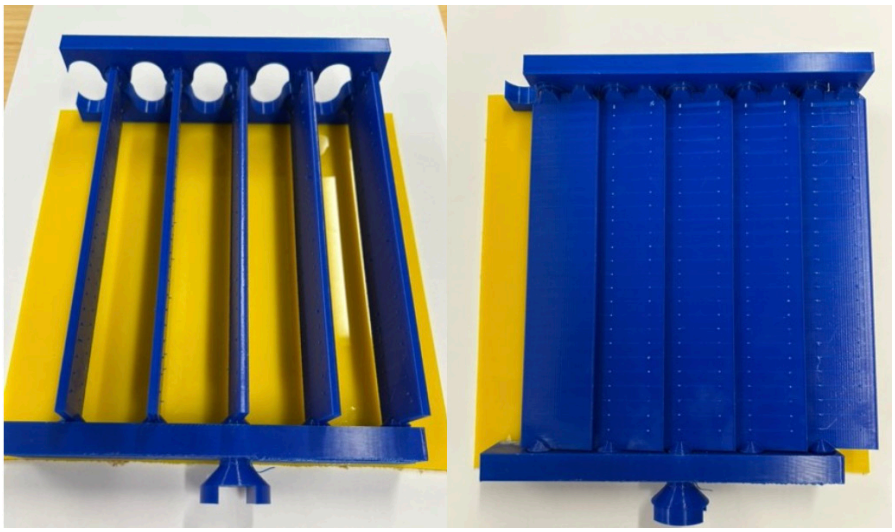
This taught me that despite having thin support material, which makes the louvres easy to remove, you have to make a compromise about strength and I made the wrong choice by making it too thin that it didn't act as support material. In the next iteration I will make it thicker.

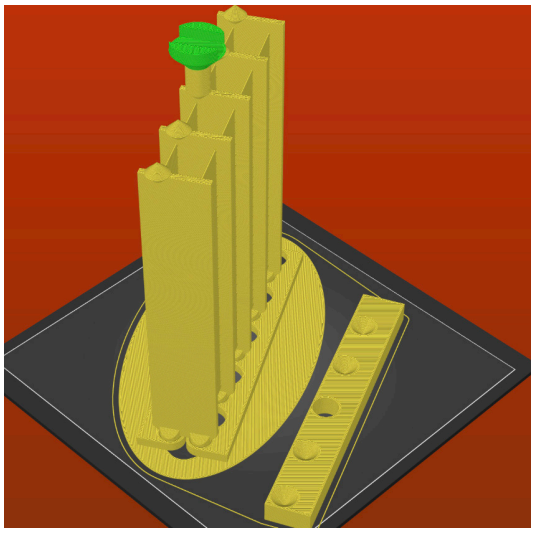


This video taught me how to design these supports (see Areas *of Investigation).*

Project Development & Realisation

Prototype	Evaluation





1. To make this I didn't have to modify much. I modelled the actuator mount itself in CAD, and the only physical change I made was drilling a 13 mm hole for it to fit through. I used a drill rather than remodelling and reprinting the whole part because I already had one printed, so this was much faster

2. Then I paused the print midway through. You can do this in the configuration in the slicer settings so that the print will automatically pause at the correct layer height. You can see this in the green and yellow slicer screenshot. The green is where it all pauses so you can change colours.

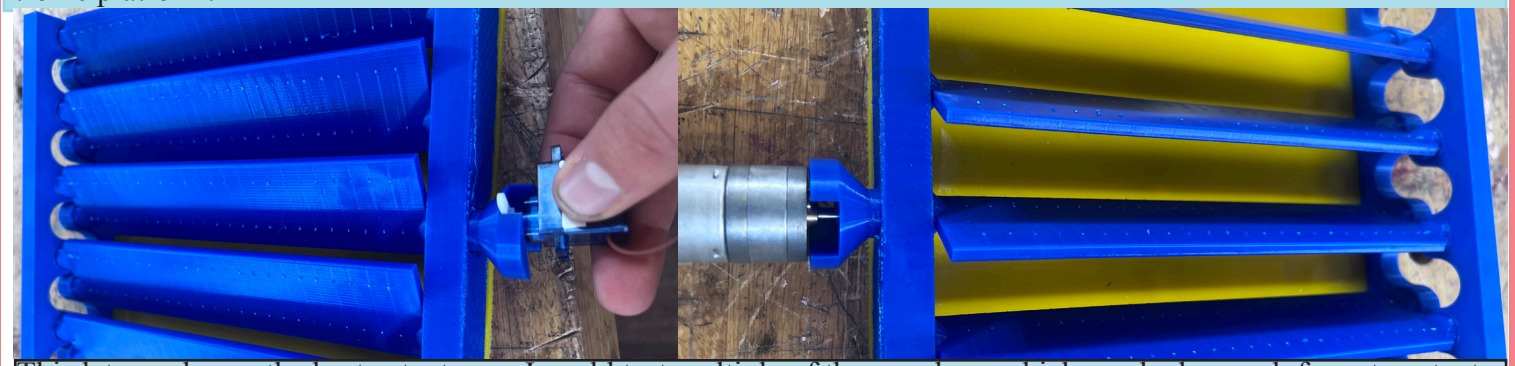
3. This allowed me to place the bar on top of the louvres in place and then I could resume it and print the drive mechanism

4. Once I finished the print, I took it off and glued it to a base and I also lubricated the hinges

This let me establish whether it was possible to drive this mechanism from one single point which is how I intended to design it and make a servo mechanism for it to go into. I could also do load testing with the servo to see if the servo was powerful enough to turn it and it was, which means I can use a small SG90 (this is a typical small servo driven by a PWM [pulse width modulation which means a series of pulses which encode digitally the position that the closed loop controller must try and drive to] signal from a microcontroller). Having a servo means that I can be more reliable and precise as it is a closed loop controller which knows position so if the gears slip or it gets held down, it will still try and move once pressure is released instead of losing position as the case with 3D printing stepper motors.

In addition, pausing the print and resuming it is how I'm going to have the multicoloured L plates and it was a good verification of this manufacturing technique.

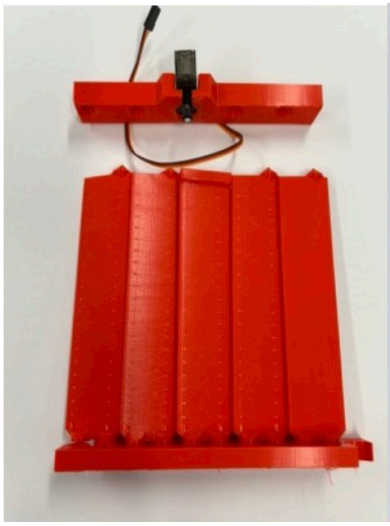
At this point, I'm happy with a louvre mechanism so far and I would like to move onto now prototyping the electronic platform.

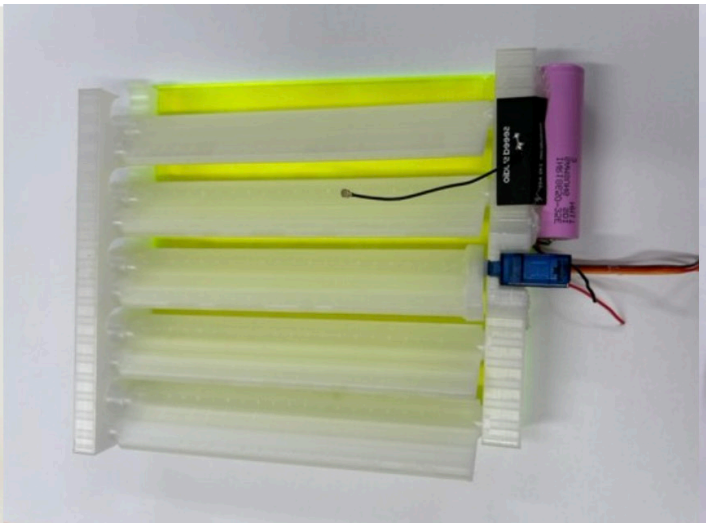


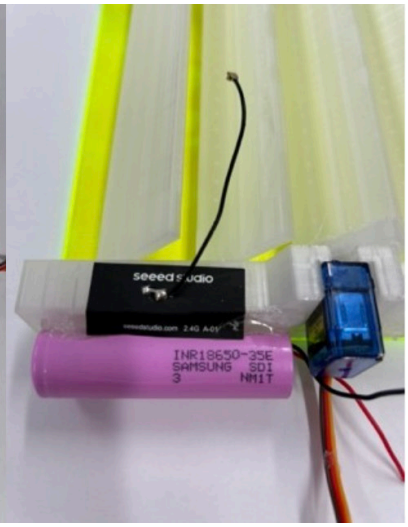
This let me choose the best actuator, as I could test multiple of them and see which one had enough force to actuate the mechanism. Of the other actuators, the DC motor on the right is far too large, whereas the light 9 g servo on the left works perfectly well.

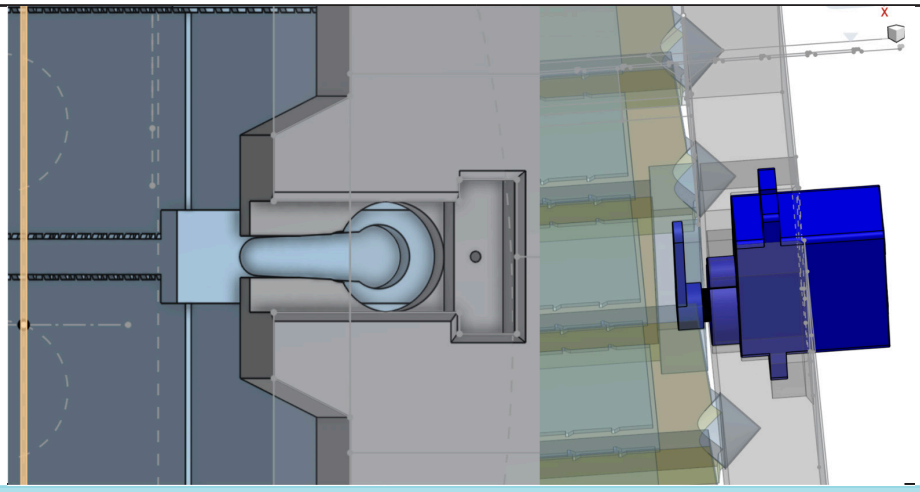
Project Development & Realisation

Prototype	Evaluation









1. The first step when modelling in CAD is to work around an existing part and import or make a model of the object you're trying to make a mount for or at least a rough approximation of the parts that you need to interface with

2. This meant that I imported a 3D model of the SG90 servo with a horn that I was planning on using as you can see in the image.

3. Then using a series of extrudes and sketches I cut away the material. In these sketches. I included an offset with my tolerance dimension. This is a globally defined dimension which is set to 0.25 mm. This is the tolerance of the 3D printer and it means that when I insert things it'll have enough clearance to physically fit.

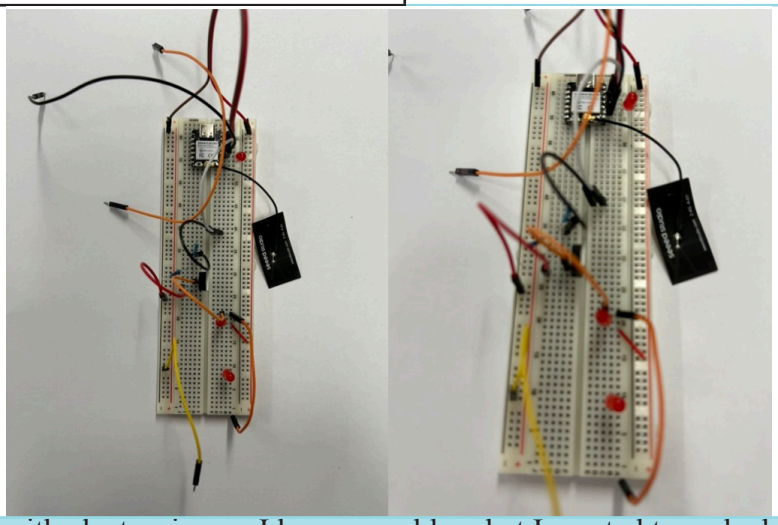
4. I decided the second servo arm mount is not necessary so I will be cutting that off so it can fit and reduce space which is one of my success criteria.

The red one is printed in PLA whereas the white one is printed in PETG. PETG is known for better UV resistance and being tougher. While PLA is much faster to print and good for rapid prototyping it does not handle UV well so that is why I printed in PETG so that I could get a feel for what that material is like. Going forward I will continue to use PETG when necessary but if there is no PETG in the DT building I will use PLA as it's fine for a mechanical prototype but does not necessarily show the final product or the same printing tolerances.

I've also learnt that my servo is strong enough to drive the mechanism. This means going forward that I can use a servo and continue designing and pursuing this design route. Next, I need to work out some way of mounting the battery without using hot glue as currently it's crude.

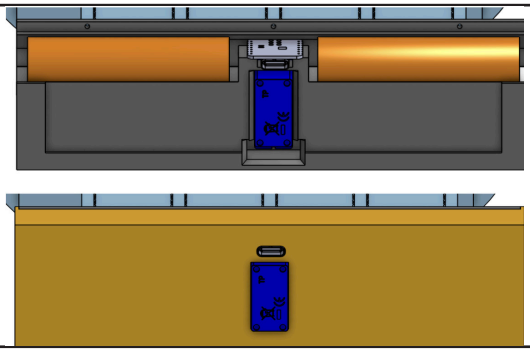
Project Development & Realisation

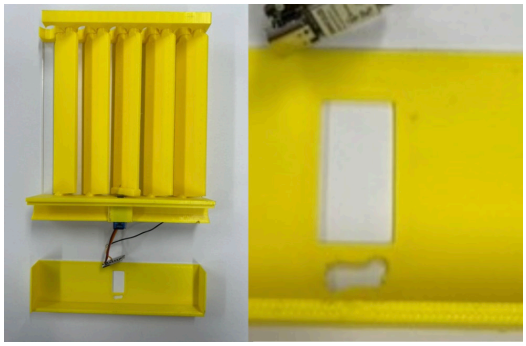
Prototype	Evaluation



I have had some experience with electronics, so I knew roughly what I wanted to make but I was in contact with my mentor and ended up having a two hour conversation where we planned out the electronics of the board. This was our first version and prototype. It worked, but I decided not to keep the MOSFET (metal-oxide-semiconductor field effect transistor) which essentially is an electronically controlled switch. This particular MOSFET defaulted to off and only turned on when a voltage was applied to it. The idea was that it would let me completely disconnect the servo so it could not draw any power at all, and I could also put the ESP32 into deep sleep, which would have given me a better battery life. In the end I decided it was not worth it, as the MOSFET added cost and complexity to the board for a fairly small gain.
Instead, I have decided to send a PWM signal of nothing to the servo which means it will not try and actively drive and finish the control loop which will have 95% of the battery savings while reducing weight and cost to the end user.

Prototype	Evaluation

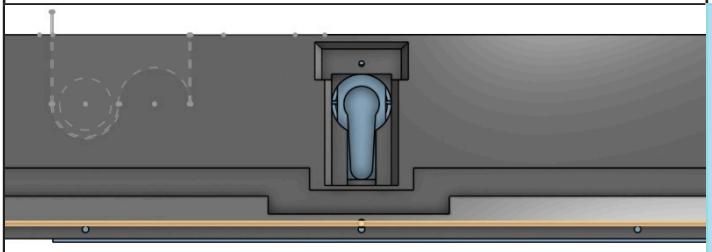
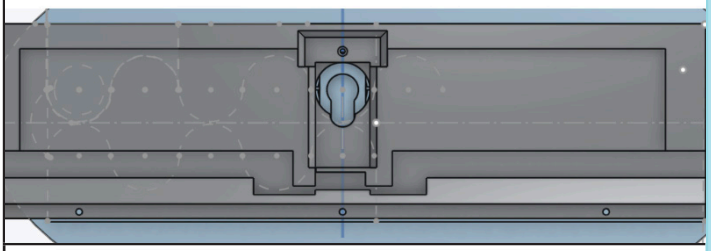
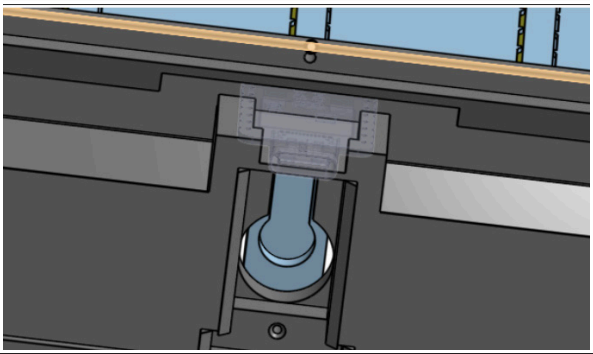
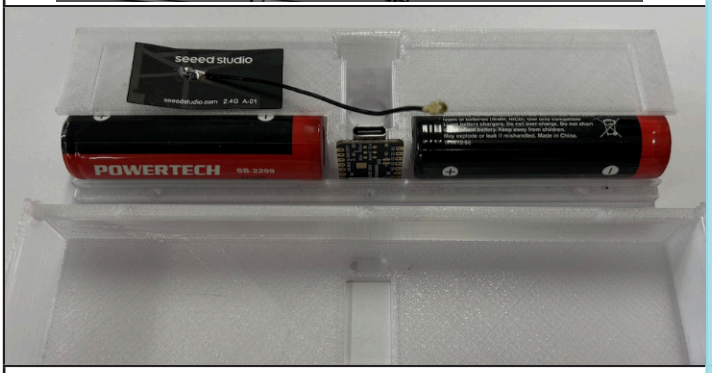
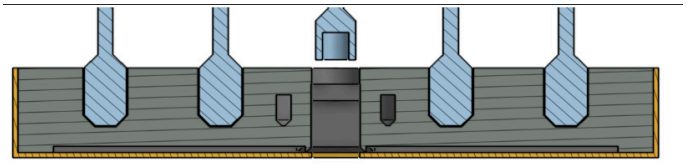
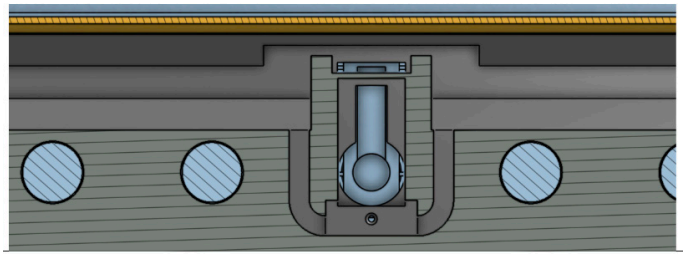




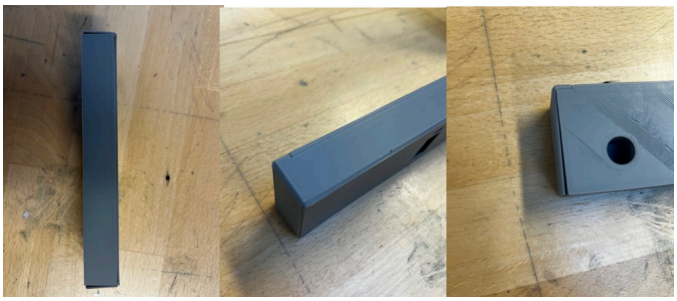
1. Following the established design pattern of placing the components first and then modelling around them I placed the batteries in a spot that I liked.
2. To mount the batteries I made a V groove. This was an easy model and there was room to route the wire underneath. It also had a snap fit so that the batteries would stay in place.
3. For mounting the ESP32 I had made a decision to mount it with the PCB labelled down. This allows the antenna exposed and the USB cable to be accessible but because the mounting was not repeatable and because of the heat generated this led to the ESP32-C3 (this is a low power micro controller with Bluetooth and Wi-Fi functionality and an integrated battery charging
module).

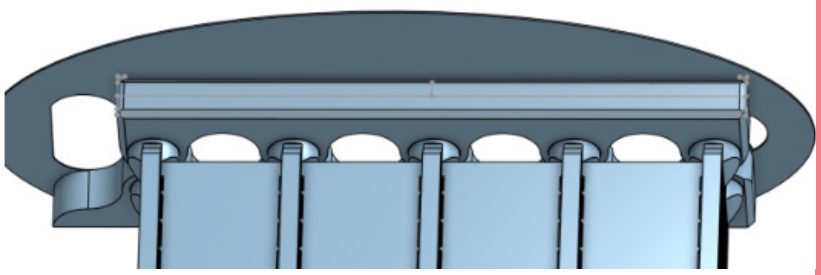
I didn't consider the wire routing as wires have a physical size and I didn't consider how they should be routed and this meant that the case would not fit on. I also did not make the hole into the correct spot for the CAD model. Since the ESP32 could move, it didn't end up being in the correct spot so I had to drill to widen it which did not work out well. I also didn't consider that there was no way to access the charging module pins underneath the ESP32 as they were hidden by the plastic. These various mistakes are expected on the first or second iteration of your electronics module due to significant number of moving parts and complexities to consider. It's good to learn from mistakes when prototyping. I needed to make sure that I included proper wire routing (this is foreshadowing) and also mount the servo correctly

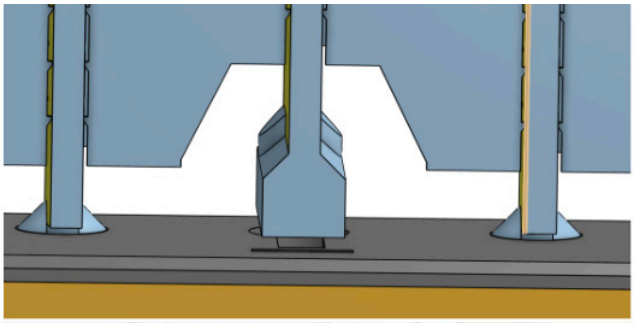
Project Development & Realisation

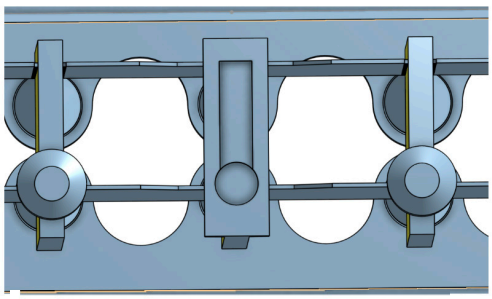
Prototype	Evaluation
   	1. As wires are very difficult to model I also made an error. I accounted for the wires on the battery holder but I didn't account for the thickness of the wires going from the servo motor to the ESP32. I also made fixed mounting on the top by turning the ESP upside down and doing a series of sketches and extrudes to make the boss that the ESP could sit on. The back of the ESP was constrained by the back plate and the sides around the CPU. The ESP and the USB port were constrained by the two walls meaning that it would be in the same spot every time. I also adjusted the location on the outside case accordingly. 2. This let me make the ESP in a repeatable spot every time which allowed me to design a correct size hole Going forward, I need to make a solution for routing the wires correctly This new layout worked really well although there was an issue with there being no way to access the boot button on the ESP32. The boot button is needed to reset the ESP32 when programming it with code. But other than that this setup was very effective despite there being no way of routing wires from the servo to the ESP32 that did not cause the case to bulge. You can see in the photo above that I have used a drill to test a different way of routing the wires. My first approach did not work very well but my second approach routing them through the body worked very well. I decided on this for the final wire routing solution from the servo to the ESP32.
 	The reason the channel for the servo wire routing is curved is that, in this print orientation, it is best to have a 45° angle. I learnt in my research that a 45° angle or a horizontal 90° angle is optimal, because if it is a horizontal, the printer can't stretch the filament from one pole to the other like a bridge and if it is 45° it is a general rule of thumb number about how steep of an angle it can have before it starts sagging. Hence, I used a combination of 45° angles and a bridge, which is the name for the 90° horizontal. This let me effectively route wires through a channel without the risk of having filament in the way as it would be difficult to clear out. I chose to make the channel on both sides instead of one because if I make it on one side and it turns out the batteries sit better if the wire goes the other way, then I'm stuck. Also, if I have it both ways, it makes it stronger as there's more material. This is because infill is not as strong as perimeters.

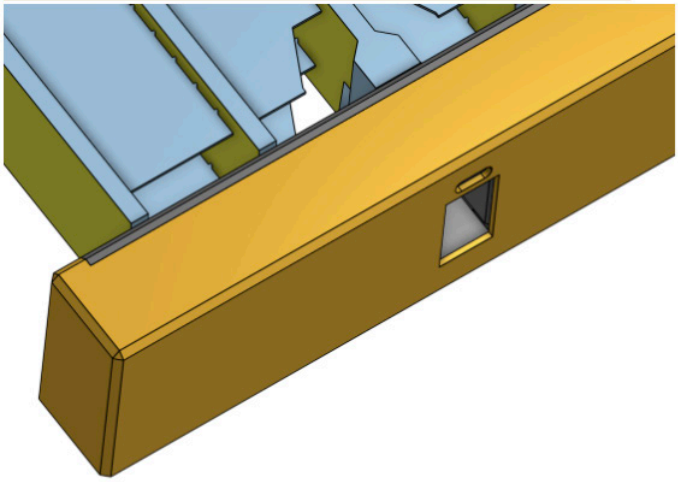
Project Development & Realisation

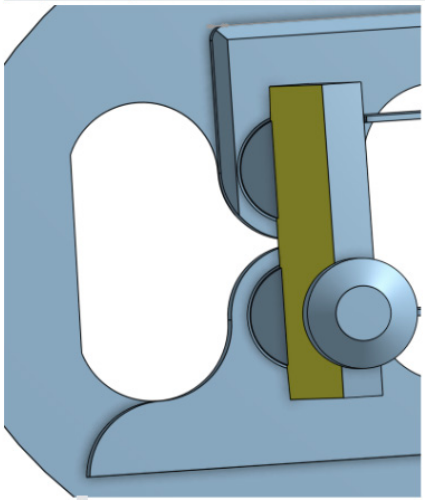


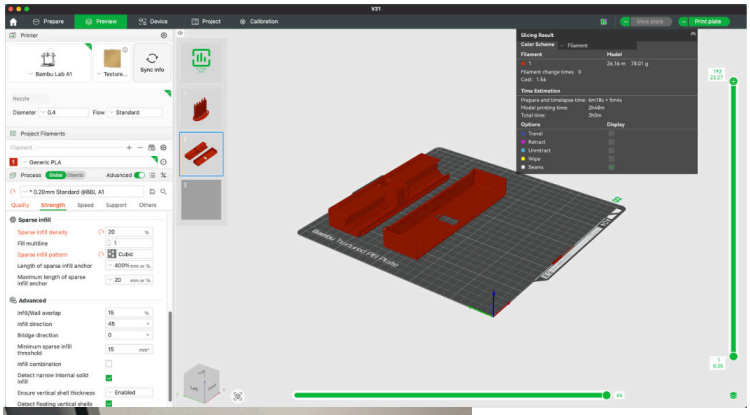


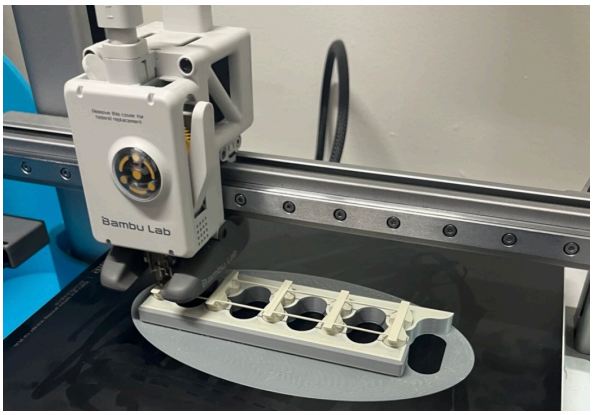


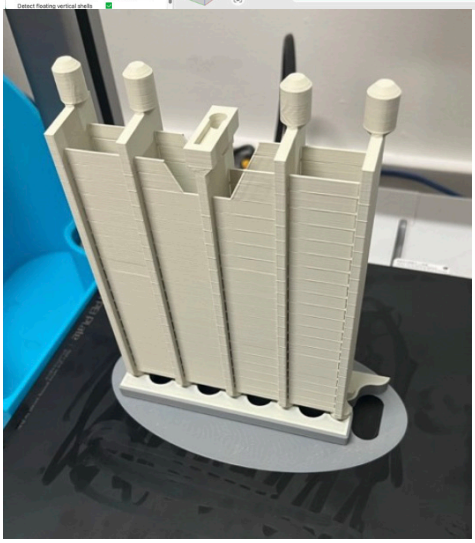


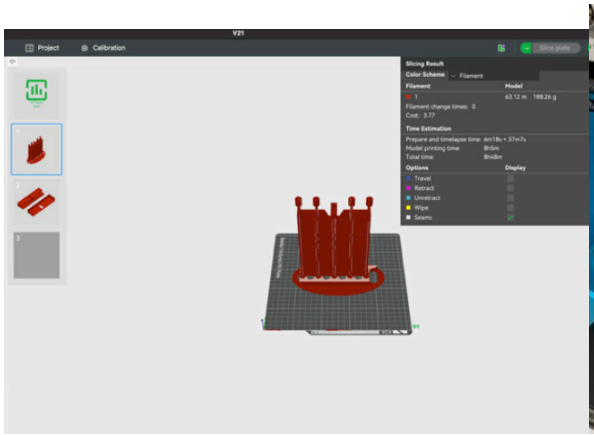


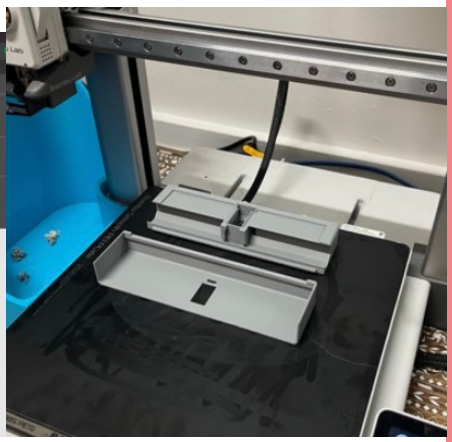






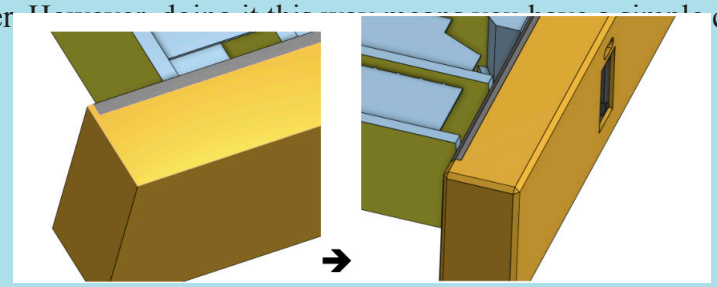




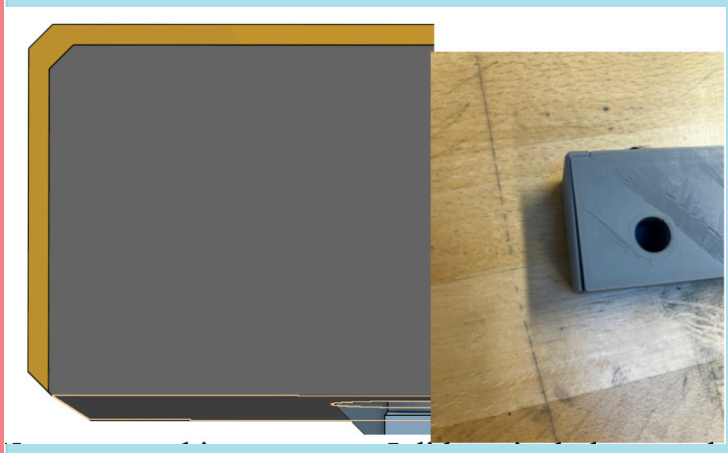


Project Development & Realisation

1. To increase the hole where the servo horn mounts as in a previous iteration I used a drill and a heat gun to mount the servo horn into the hole. I used an extrude but I also needed to chamfer the sides of the horn mount to make it stronger and not have a chance to fail when it's printing in thin air, as previous iterations did.
2. I needed to remove material from my support material so that it could accommodate the larger horn. I did this with a simple sketch and an extrude the whole way through.
3. I wanted to make the edges chamfered to improve the safety of my product. To do this I needed to often chamfer manually as I wanted to keep the brim in place.
4. You can see highlighted in figure 2 that the brim looks like it's in place and badly fitting because instead of properly modifying the parametric design I simply used an extrude to cut away and adjust the shape. This is risky but it is not worth modifying the original brim as there is a small cost in filament but a large cost wasted time fixing the parametric document so it would be easier just to cut away.
5. Because the corners were originally mounting in the edge, if you have a chamfer you need to be more precise as there is a compound chamfer. However, doing it this way means we have a simple chamfer.



6. I moved the location of the corner inwards so that there would be solid material to make the chamfer to ensure there was no gap. This would look ugly and could cut someone so I moved it inwards to have a solid piece of material. See images above.



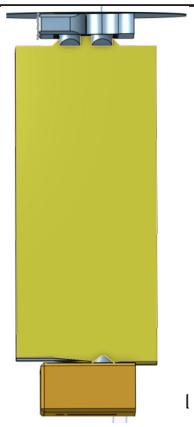
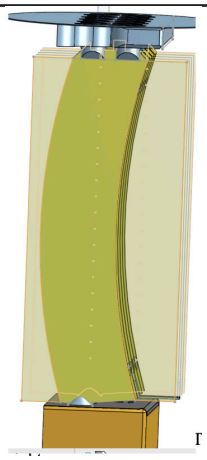
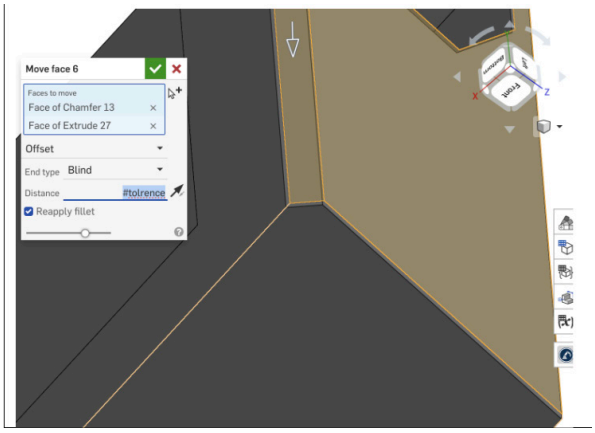
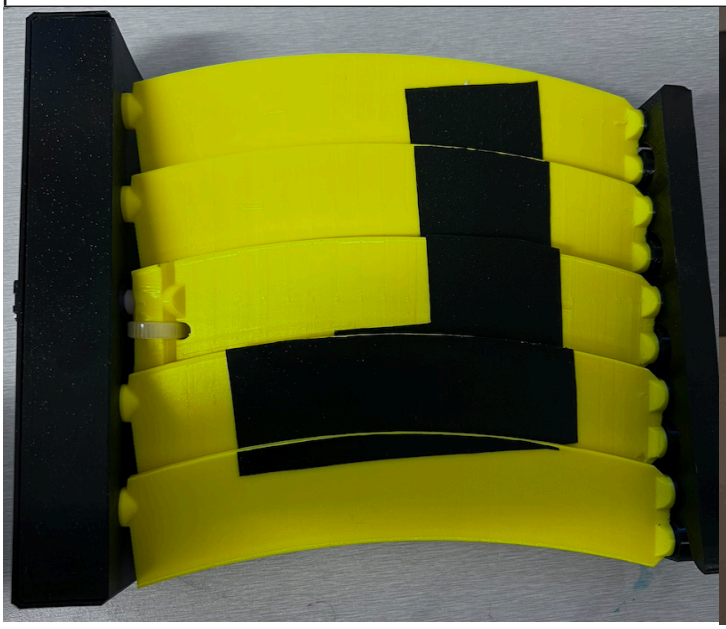
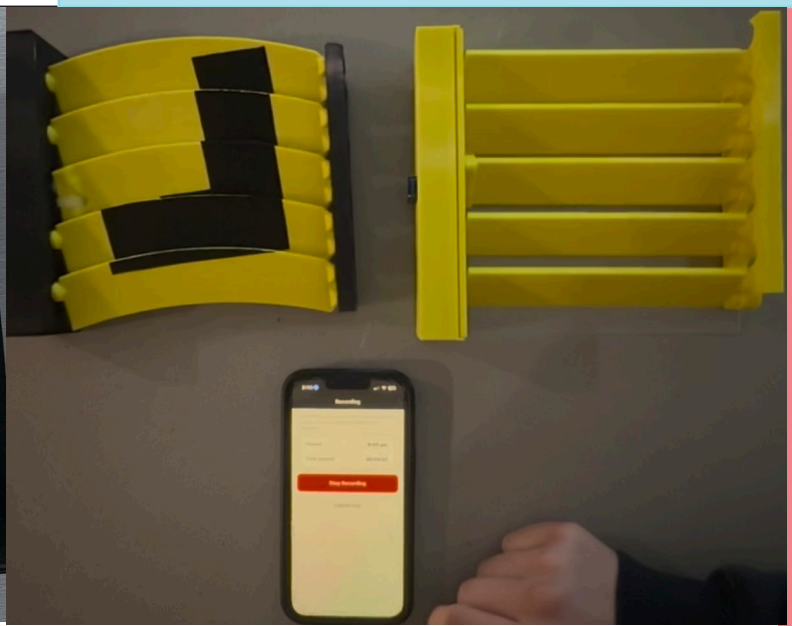
7. See top corner; I did not include any tolerance, which means that the top corner acts as a focal point and it will pry open the bottom of this bar. For the next iteration I need to increase the chamfer amount on the inside as I do on the outside so that it is a relief so that it doesn't pry open the lip.

Overall the chamfers look really nice and clean but there needs to be improvement with refining tolerance in the case but ultimately these are small refinements that make a product high-quality with good design and it's important to get this right.

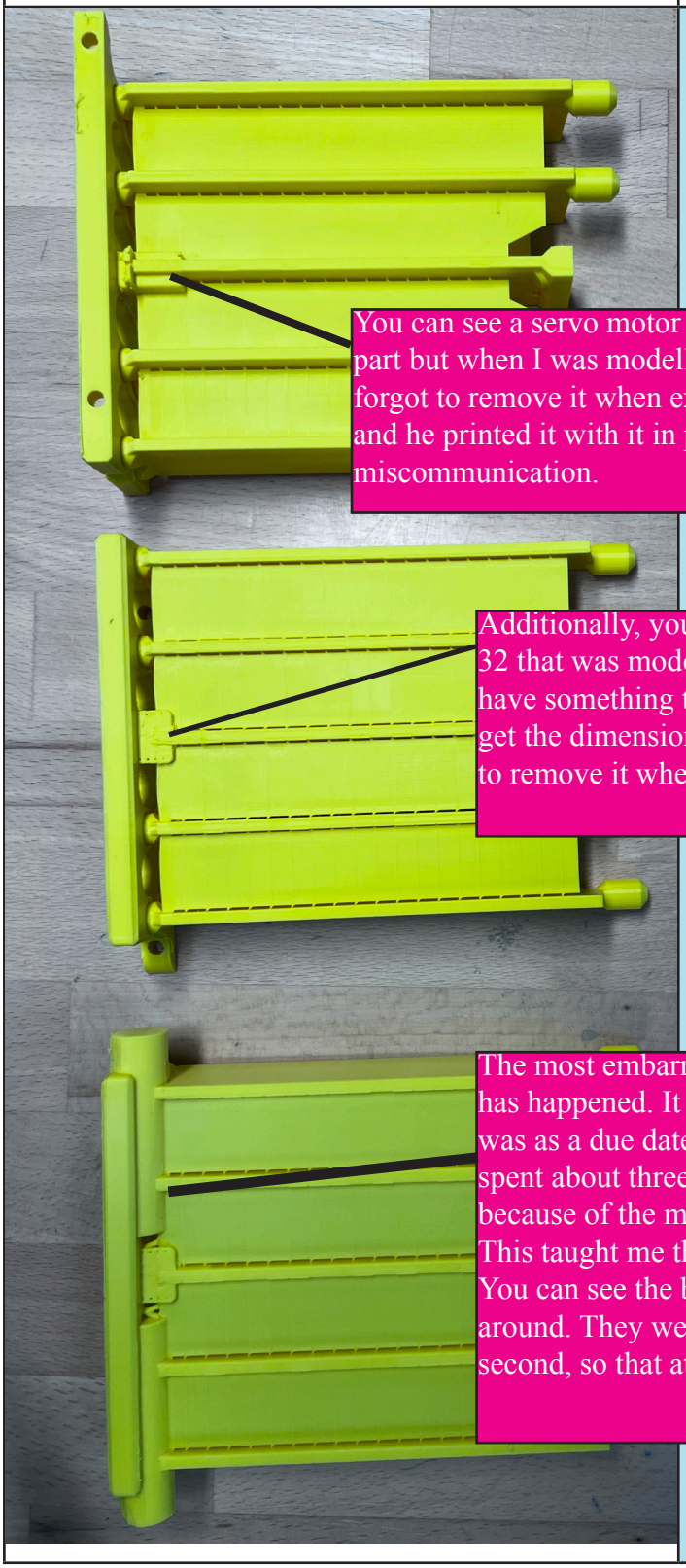
This design printed without any issues, despite the wires being slightly hard to feed through the tubes; it just takes a little patience and the results are very satisfactory. The wire routing is significantly better and easier to do, which reduces manufacturing time and therefore cost, and the reinforcement adds strength and rigidity near the motor mounting.

The chamfers at the top worked very well at reducing the overhang, so the part prints in this orientation without support and without filament entering the slot and blocking the wire.

Project Development & Realisation

Prototype	Evaluation
	I extruded this then I can cut them to make it curved
 	Then I used a sketch and an extrude to cut away the louvres in the right way Then I used a file to remove the gap I had to remove material that was both in the chamfer and on the side face so once I've done that I removed that material in CAD for the next iteration There has not been much of an improvement with the box and the flap still pries open. To solve this in my next iteration I am going to mount the screws in a position such that they will help pull the flaps in so that this is not an issue. In addition to this I am going to need to start testing and developing a method for adhering the L plate to the car so that it doesn't fall off. This design of having a curved plate does fit quite nicely with a car but it doesn't save any space because the top height is still the same, so ultimately it is no different to having straight plates. There is no gain in doing this design but there are losses in manufacturing complexity and, subjectively, worse aesthetics, which is why I am going to revert back to the other design.
	

Project Development & Realisation

Prototype images	Evaluation
	This is an example of three successive communication failures that happened with my Mentor. This is because I sent him a STL file that contained the parts that were there for moulding not the printable body so this meant that they got caught in the mechanism and would make these parts not functional. You can see a servo motor here. This is meant to be an external part but when I was modelling it I had to model the servo and I forgot to remove it when exporting my CAD file to my Mentor and he printed it with it in place instead of removing it. This is a miscommunication. Additionally, you can see the ESP 32 that was modelled so that I would have something to model around and get the dimensions right but I forgot to remove it when exporting it. The most embarrassing thing was that it is not the first time it has happened. It happened again and it was the next print. This was as a due date was coming up really closely and we had spent about three days printing these all up and they all failed because of the miscommunication. This taught me that communication is important for designers. You can see the batteries that were originally here to model around. They were also included in the first print, but not in the second, so that at least is an improvement.

Project Development & Realisation

Researching the best filaments

Image/name	Plus	Minus	Interesting
PETG	- UV stable, so it will not yellow or degrade behind glass - Glass transition around 80°C, above what a parked car reaches in summer - Tougher than PLA, though still brittle in thin sections	- Bridges and overhangs print less cleanly, so channels need 45° chamfers - Strings more, so the pins need cleaning up - Slower to print than PLA	PETG is the same polymer family as PET, the plastic used for drink bottles. The G stands for glycol, which is added to stop it crystallising so it prints clear instead of cloudy.
PLA	- Fastest and easiest to print - Best surface finish and finest layer lines - Stocked free by the DT department	- Softens near 60°C, below what a parked car reaches in summer - Degrades and goes brittle under UV	PLA is made from fermented plant starch such as corn or sugarcane rather than crude oil, and is industrially compostable.
ABS	- High heat resistance - Tough and impact resistant - Can be vapour smoothed for a premium finish - Lower cost as it is very common in injection moulding	- Warps badly without the heated enclosure the school does not have - Styrene fumes need extraction - Yellows and chalks under UV	ABS is one of the most common injection moulding materials and is what Lego bricks are moulded from.
ASA	- Best heat and UV resistance of the four - As tough as ABS	- Same warping and enclosure requirement as ABS - Fumes, higher cost, and not stocked at school	ASA was developed as a weather resistant version of ABS and is widely used for car wing mirrors and exterior trim.

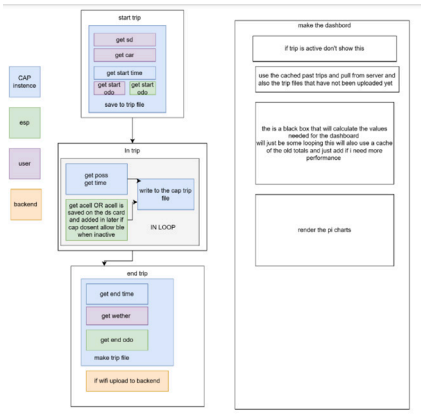
Laying out each filament in a PMI table allowed me to easily work out which one was the best and most appropriate for my design. This ultimately led me to the conclusion that PETG was the best filament to choose as it has adequate heat resistance while being much easier and safer to print at school. It is a little bit brittle, so I had to make compensations in my design to ensure that it would be strong enough.

If I were to move this to larger scale manufacturing, I would consider moving to ASA or ABS because of their better UV resistance, and I could take advantage of the lower cost of commercial manufacturing, as both are commonly used in injection moulding.

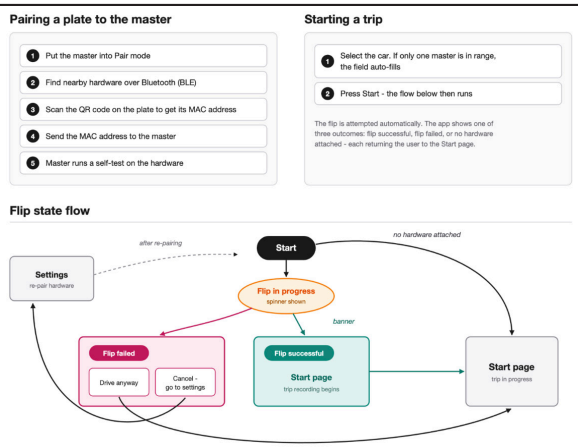
Project Development & Realisation

App Development

Now I'm documenting the development of prototyping that I conducted to develop my app and firmware for L plate. I'm a student of software engineering so I was able to use the resources such as my software teacher to help me develop this app. We also had a family friend who was professional software developer who also helped me quite considerably when developing this app. This was the first time I built an app so it was a new experience with lots of learning lots of new skills.

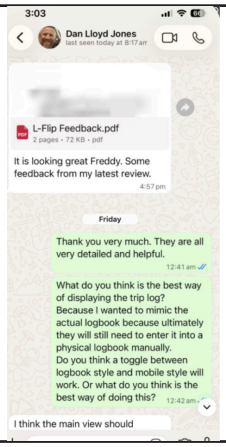


This is my initial design for the UI but ultimately I decided not to go with it as I spent quite a considerable amount of time user testing to try and optimise the UI for efficiency and intuitiveness.
Due to my testing, I was able to provide improvement to this design because the new designs were more intuitive to use and more aesthetically pleasing.



This flow chart is efficient at conveying how the pairing process works in edge cases. For example, when the plate will not connect to the phone and how it handles this gracefully

I conducted user testing on my app. I gave a development version of my app to a couple of students in my year and asked them to navigate the app and I watch them do this. This let me help her find my UI design as I could identify where they got stuck on and therefore address those issues.
This led me to change to make some of the buttons larger as some of the students had very large hands with large fingers and struggle to press them accurately.



Communicating with a mentor who is a professional software developer of mobile apps

Project Development & Realisation

App Development

Intended result	Actual result	Result	Action taken
Taken to dashboard	Taken to dashboard, you were routed through the login screen but this is so that I don't have to duplicate logic	PASS	NA
Taken to login screen without a JWT or cookie	I got taken to the onboarding screen	FAIL	Remove the cookie
Rejected by a generic but helpful error message	I got good error messages that did not give away the back-end infrastructure but were clear enough to understand	PASS	NA
Returns personalised information about your dives	I got a "Sorry, I couldn't produce an answer for that. Please try again." I got this message	PASS	The reason this did not work the first time is because I made the chat and then the tool separately
Verify that the last AI chats are gone	They were not removed	FAIL	Execute a force reload; this is an easy fix. Open further investigation — there is another cause
Verify that there is some sort of error message that doesn't allow this	Got a helpful error message	PASS	NA
You log in successfully	It worked but I got the onboarding pages. This will be annoying	PASS	Make the onboarding only come up if flag exists, not if flag doesn't exist. Fix in progress
I would expect that all the text is readable and the themes match up with no excessive amounts of light	I found a few text boxes and themes that were not properly in dark mode	FAIL	Update my CSS and theming as well as my colour palette to work with dark mode
The expected result is in the first case to get trip information and other specific information about the user	My server hardening worked very well, but the AI tool fabricated data (see 1) for the output	PASS	The fix is to harden the system prompt
I'll be able to see it in the GPS map. GPS points will still be logged despite the screen being off		not run	
Get a pop-up or other notification from the LM that you're out of tokens	This was really bad design on my behalf. All you got was a pop-up saying please check your connection	PASS	I was HTML-escaping the content twice which meant that I was getting things like " in the output
The master and slave will connect, then they will seamlessly toggle with an intuitive UI	This works as expected	PASS	NA
The app icon should be a bright yellow with a black L	The app was an olive green-yellow. This looked bad	FAIL	So I changed it to a bright yellow

I conducted something called acceptance testing. This is where you will test the functionality of your app and after testing the functionality of your app you need to verify that it did what you expected it to do. If it doesn't you have a problem and you need to fix it but if it does work it passes and you know you have verified the app works. This is different to something called unit testing which is where you test each piece of code
individually.

Acceptance testing is where you verify the app. Does what you wanted to do instead of testing the code you test whether the functionality works. You do this by first writing down what it needs to do then you check and verify that it does what you expected to do. This is a very beneficial type of testing as it catches a large amount of user interface bugs although it does not catch some more insidious code bugs.

This sort of testing allowed me to verify that my app worked so that I could be confident such that my users would be able to navigate it without any issues. This testing was conducted at the end of the development of the app.

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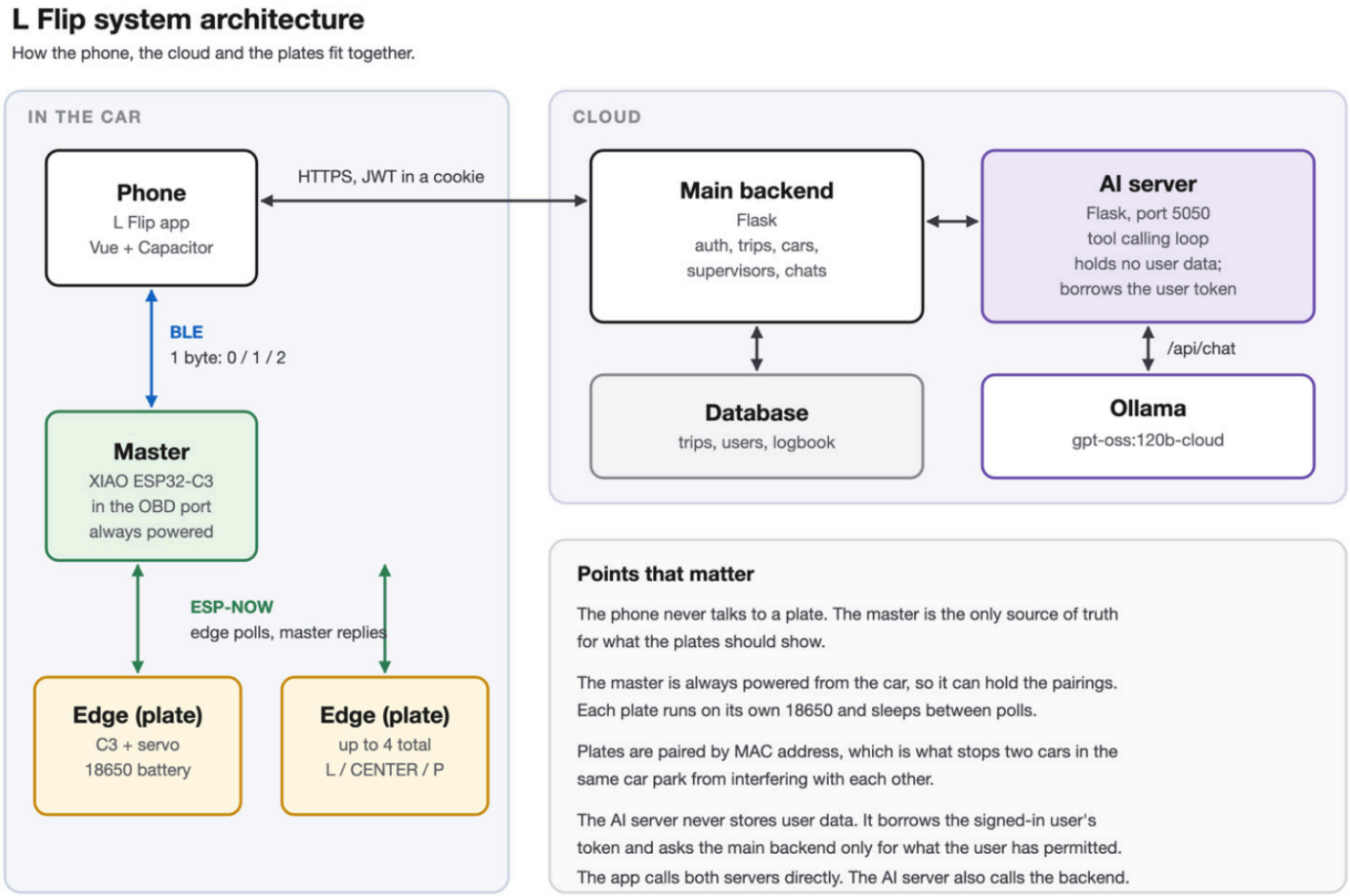
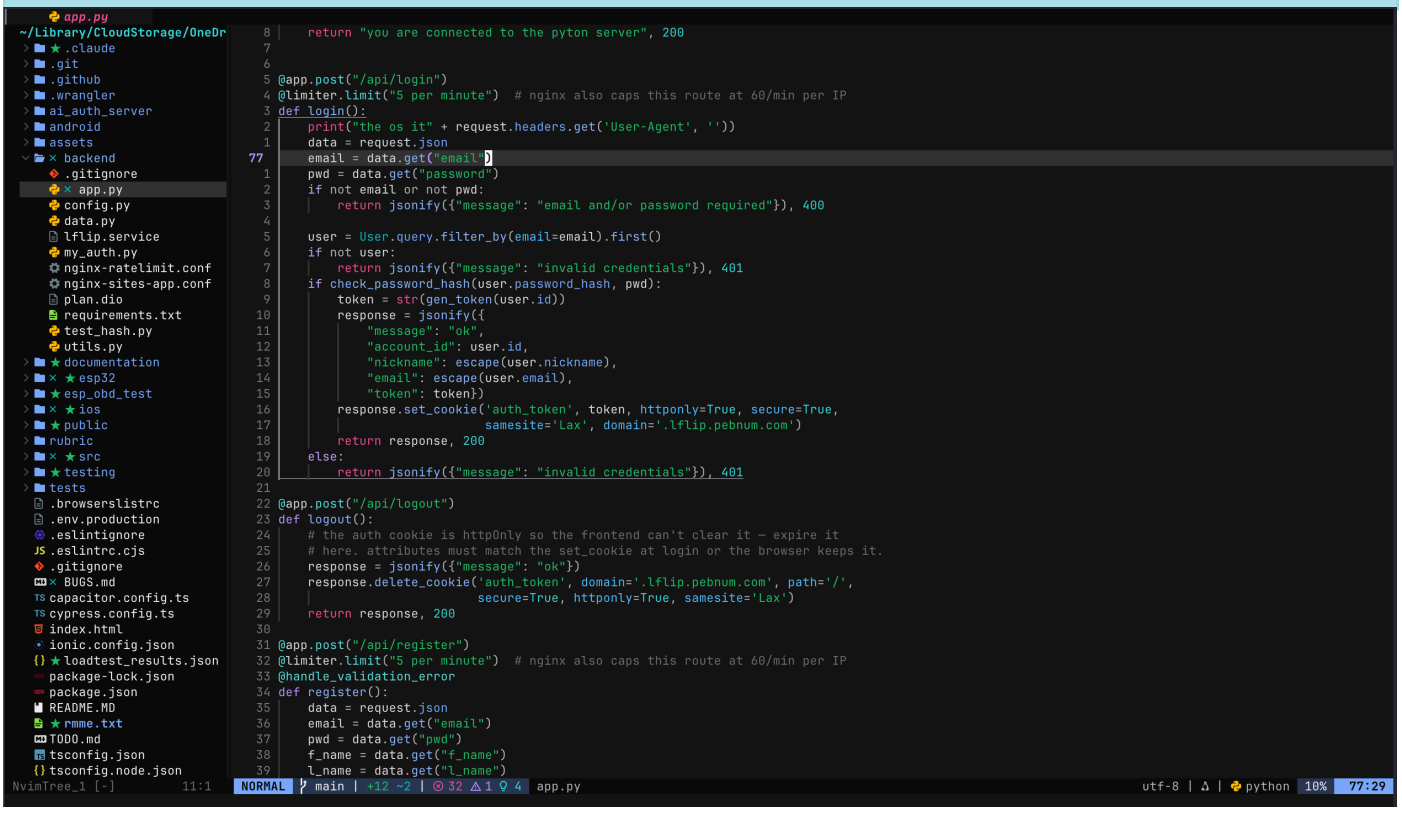


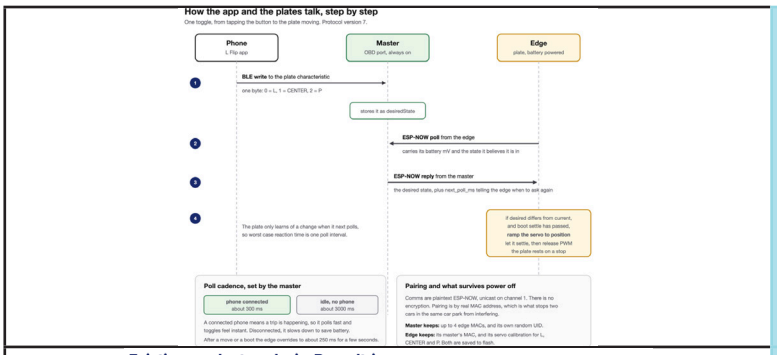
This is how my Software works
There are lots of components that need to interact with each other which is complex, so these visualization were really helpful to understand how my own system works so that I could find bugs in my code.

This architecture has been validated by testing and this gives me confidence to move further onwards. It follows industry best practices for security as well as having isolation between the AI server and the main back end as AI products can often deal or lead to security vulnerabilities in the app. That is why I have it isolated.

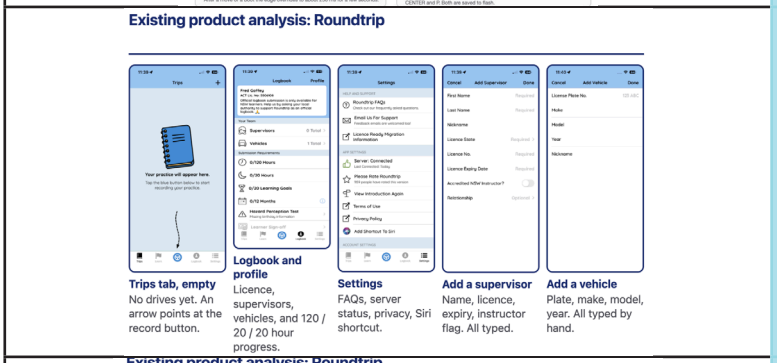


Project Development & Realisation

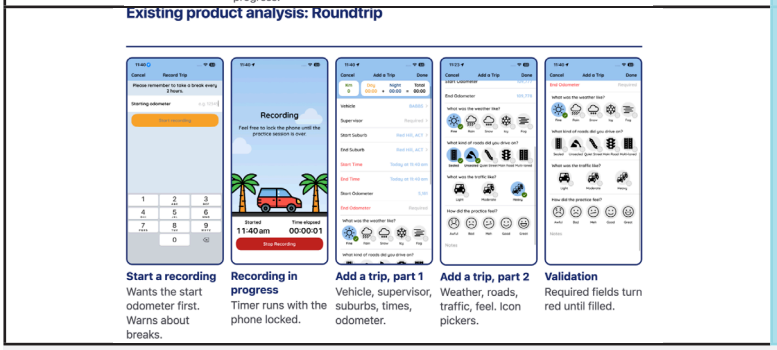
App Development



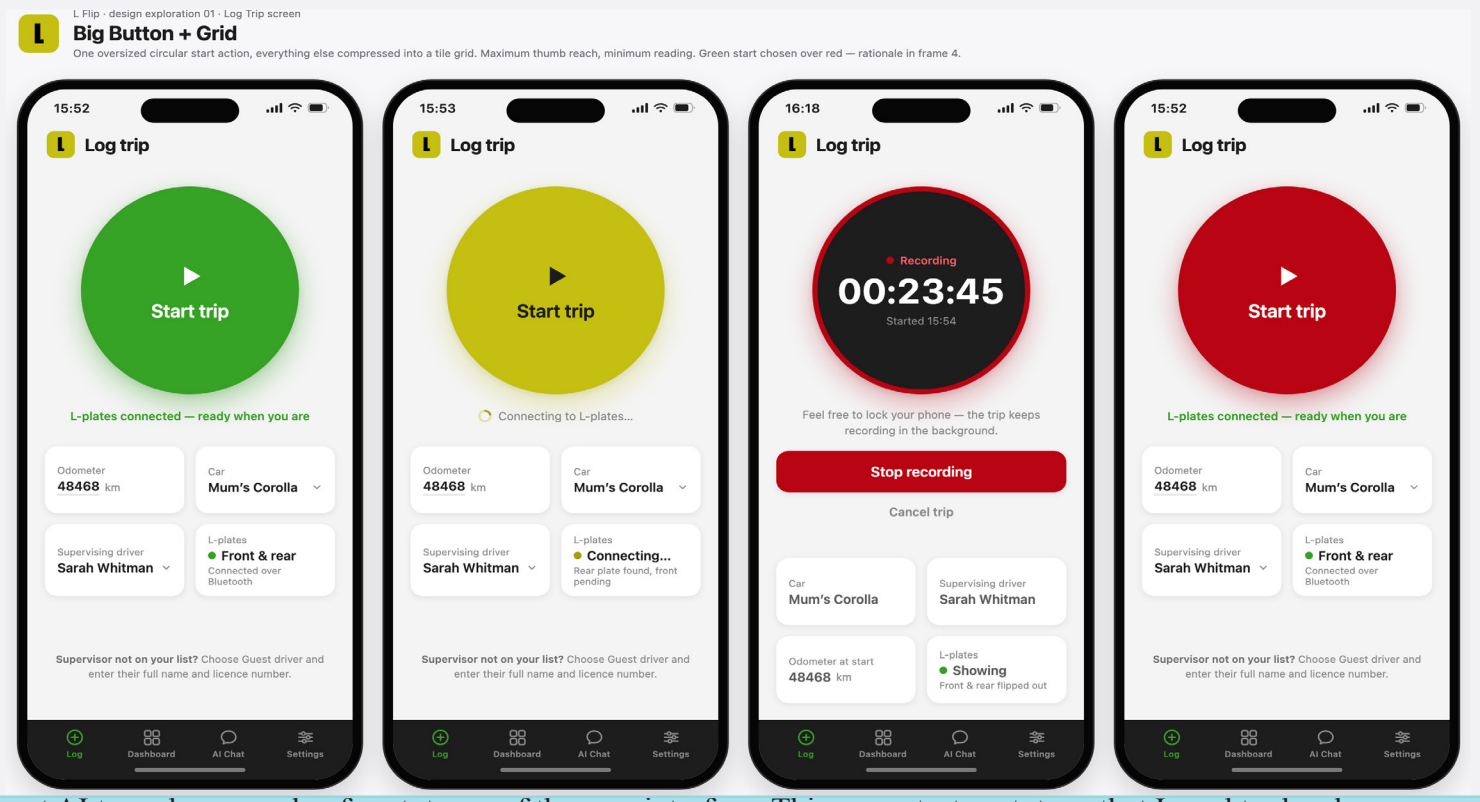
This is a diagram outlining how I am intending to have my network protocols to work. I design this so that the user would not have to think about how it works, but it is intuitive.
Also, so that it can reset itself with no maintenance. This proved to work quite well.



This is me analysing my competitor's app and their app works quite well. It lacks a lot of the advanced features that mine is designed to be integrated with hardware and log hours very fast with minimal friction.
I was able to gain meaningful insight from what they have and it helped me work out how I was going to design mine and differentiate my app.



The way that they implemented weather selection UI helped me choose to use a drop-down instead of a text box. Because a text box means that it is hard to search in categories whereas clicking a drop-down or button is much more efficient.



I got AI to make a couple of prototypes of the user interface. This was a test prototype that I used to develop my app. It let me test out the colour scheme and the placement of buttons in relation to the user experience. Doing this allowed me to design an optimal user experience that was grounded in real world test data and not just my own preferences. I was pretty happy with how this prototype went and therefore, decided to keep most of the components such as the large stop button. But I copied from the log book in terms of the design so it was familiar.

Project Development & Realisation

Identification and Justification of Ideas and Resources

Tools Actual

Name	Image	Example uses	Evaluation
1. Soldering iron TS101		- Soldering batteries in place - Soldering wires to ESP32 - Inserting threaded inserts	This is a very high-quality soldering iron despite its inexpensive price. It efficiently and easily soldered all I needed.
2. Stanley knife		- Cutting off support material - Removing brim - Removing tab in servo	This tool is quite dangerous, but it is also very useful. When I was using this, I was always careful to make sure that I could not be cut.
3. Side cutters		- Cutting the wire to length - Cutting off the mounting tab on the servo	This is a very nice tool as it allows you to cut precisely and also cut flush with surfaces. Although the school's ones were quite blunt, so I brought my own from home.
4. Phillips head screwdriver		- Screwing the screws into the case	A screwdriver allows me to efficiently put screws into PETG material
5. Hot Glue Gun		- Used to mount the ESP32 in place	This tool was very useful, but it was important not to overuse it as large amounts of hot glue are a clear sign which implies your product is inferior quality.
6. Wire strippers		- To strip all the wire that need to be soldered - To crimp the antenna on the ESP32 in place	These tools let me remove precisely the amount of wire that I wanted allowing me to easily trim and solder.

Project Development & Realisation

Identification and Justification of Ideas and Resources

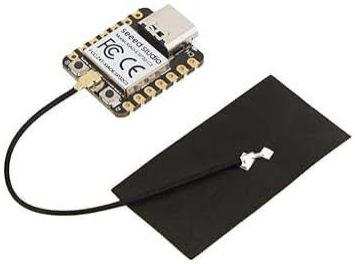
Materials Actual

Name of material	Image	Where it will be used	Evaluation
7. Poly-ethylene Tere-phthal-ate Glycol (PETG)		The PETG printing filament will be used to make the main body of the L plate.	Yellow and black PETG contrast well making the piece both aesthetically pleas-ing and easily identifiable as the colour contrast make the plates identifiable on the road. This filament also produces a smooth finish with small layer lines while having superior thermal properties meaning that it will not melt or deform when placed on a hot car.
8. Suc-tion cups with threads		These suction cups will be used to attach the L plate to the inside of the car	This solution was successful as the thread-ed inserts allowed me to attach the suction cups to the inside of the car. These suc-tion cups are also successful because they do not damage the interior or exterior of the car like magnetic L plates and do not require prior attachment or being continu-ously affixed to the car, like clip on plates.
9. Thread-ed insert		These threaded inserts will be used to attach the suction cups to main 3D printed body of the car	These can be quite easily inserted into 3D printed filament as the filament melts with the temperature of a handheld soldering iron. They are also successful because, though they are easy to attach they provide a strong multi-use threaded joint. Finally threaded inserts are superior to screws as they do not wear out as a result of continuous use, while resulting in a refined product.
10. Silicone wire		This silicone wire will be used to connect the batteries and the servo to each other	Silicone wire is very thin and is very nice to use because it's flexible. Although silicone wiring is expensive I believe that it was a worthwhile investment as it was flexible enough to allow me to do complex routing while remaining thin enough to not get in the way of the case and fit in the space constrained spaces.
11. Servo Motor		This is a small cheap inexpensive servo motor which allows me to effectively actuate the louvres (the horizontal-ly angled, and movable blades which display the "L").	Due to this servo being so common there are lots of tutorials online meaning that it is simple to program. This is what makes this servo motor a great choice for my Major Design Project (MDP).

Project Development & Realisation

Identification and Justification of Ideas and Resources

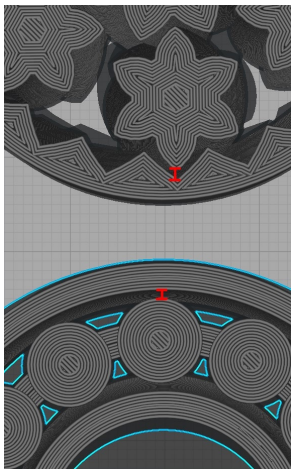
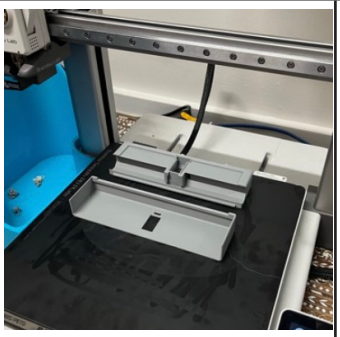
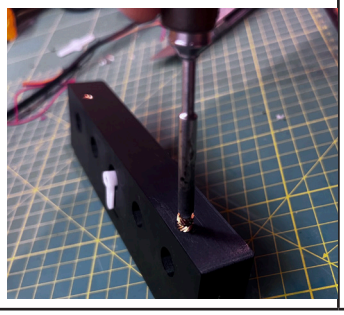
Materials Actual

Name of material	Image	Where it will be used	Evaluation
12. XIAO		The XIAO ESP32-C3 will be the main micro controller for my project	This is a very low-cost efficient micro controller that is easy to program as it has extensive English documentation meaning that it is easy to find things out about it. This means that it was an excellent choice of micro controller for my project. In addition, the micro controller has an on-board battery management system that allows it to charge the 18650 cell battery seen in Fig. 13, below.
13. Protected 18650 cell		The cell that I chose has a large battery capacity as it is the primary power supply for the electronics in my design.	Over discharging is a primary issue because if the battery is over discharged, it will lose its capacity and potentially cause it to catch fire. To combat this I have used protected cells. Therefore, over discharging is an issue that this product solves because the battery will automatically cut the flow of current once it reaches below a certain level.
14. Screw		This screw will be used to attach the case to the battery mount	These screws are self tapping, which means that they make a thread for themselves. This means I do not need threaded inserts. This will reduce the part count decreasing the cost of the product.
15. Cable Tie		This cable tie was used to fix the louvre mechanism to the servo housing box	This cable tie was instrumental in removing one part as before I had an acrylic base plate but using a cable tie to fix it to the servo horn allow me to remove one part decrease in manufacturing complexity thus reducing manufacturing time complexity.

Project Development & Realisation

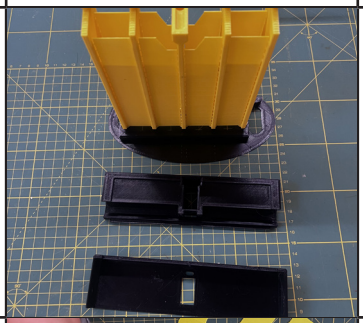
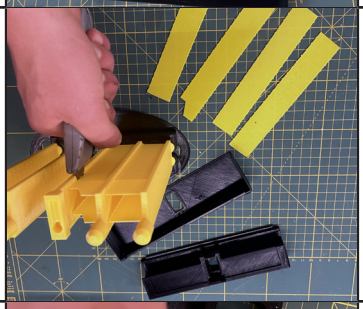
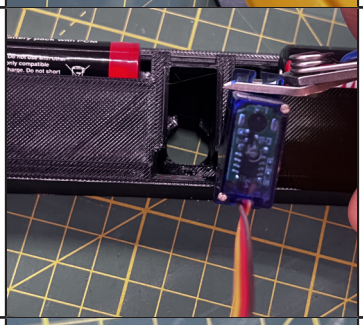
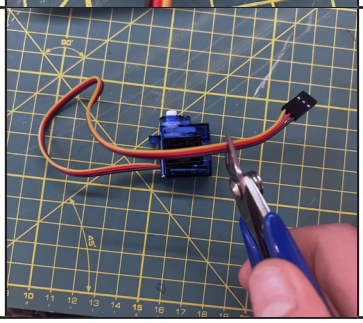
Identification and Justification of Ideas and Resources

Techniques

Skill	Photos	Application	Evaluation
3D modelling printing place		- This skill was key for designing the louvres and the mechanism that makes them work	This skill was fundamental to the success of my prototype and project.
Programming		- To program the whole app - To Write the code which goes on the ESP 32	As I take the subject of software engineering, I already have some experience writing code but this project tested my skills in different ways as writing code for microcontrollers is different from writing code for larger machines like laptops and servers.
3D printing		- To print all the prototypes	I used 3D printing extensively throughout the entire production process and learning how to operate a printer was instrumental to the success of my project.
Soldering		- Soldering was used throughout the process of assembling the prototypes	Getting good at soldering was instrumental for the success of my project as a lot of these solder joints were quite delicate, and being skilled at it meant I did not damage electronic components.
Threaded inserts		- Threaded inserts were used to make the threads for the suction cups to attach to.	This is a relatively easy skill, but it was instrumental for the success as it allowed me to create reusable threads in plastic parts increasing their repairability of my product.

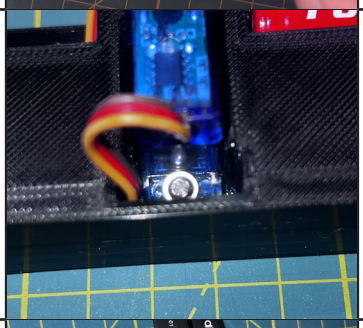
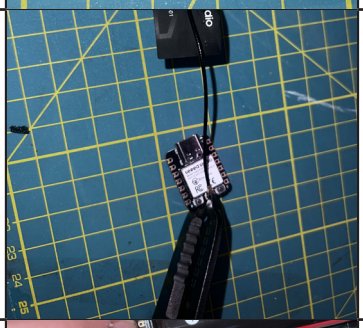
Project Development & Realisation

Production process

#	Photo	Description
1		3D print the following parts using default printing settings but you need to add a pause and change the colour of filament at the spot on the L plate so that you can have multicolour 3D printing. Once you've done that, you need to rinse the glue stick residue from the base.
2		Using a push cut (this is a type of cut with a knife that is designed for high-powered cutting tasks in a straight line without needing fine detail control) when you are doing this, you need to make sure the blade of the knife follows closely with the surface of the louvre to remove any protruding support material.
3		Using a draw cut, where you cut with the blade facing towards you gives fine control and precision. You need to be careful not to cut yourself so your hands need to be out of the line that the knife is moving in. Use this to carefully slice the 3D printed brim away from the part. You need to take extra care that there is no overhanging material. If there is overhanging material use a scrape cut (this is where the knife is perpendicular to the material you're removing).
4		Using a pair of side snips carefully remove one of the mounting pads. The use of side snips allows you to precisely remove the pads without any excess material. The reason I chose to remove one of them is because two was unnecessarily strong while increasing the size. If I could remove one pad, I could use the enclosure to keep the servo mounted correctly and have one less screwed in place. This results in minimal force being applied in that direction.
5		The servo has a moulded strain relief tab where the lead exits the case, which improves water resistance and prevents the wires being pulled from their solder joints. Neither applies here, as the leads are soldered to the ESP32 and routed so they can't be tugged. Removing the tab shortens the enclosure by 5 mm in the X axis.
6		Using side snips, remove the pins as there is no use for them and they will inhibit the wire from feeding through the routing

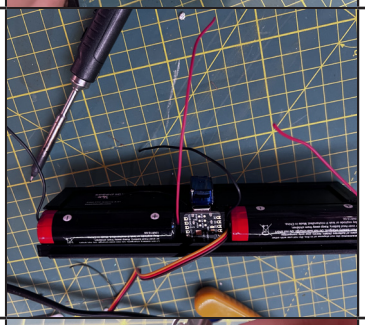
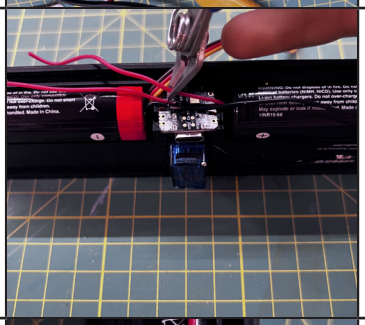
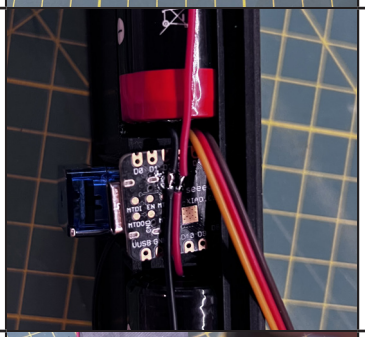
Project Development & Realisation

Production process

#	Photo	Description
7		I fed the wire through the hole carefully which was a little bit difficult so I took my time and made sure the angles were optimal so that it would feed correctly.
8		Screw the screw to attach the servo in place. There is a hole already printed in there with a chamfer to make the screw self locate. Use the self tapping screws provided with the servo.
9		Attach the antenna to the ESP32 and move and crimp it in place as shown in the image.
10		Remove the backing paper off the antenna exposing the contact adhesive layer underneath.
11		Remove the contact adhesive backing to mount the antenna in the far corner of the cut out that is being provided. You will need to route the wire so that it pokes out of a hole near where the USB C plug is.
12		To solder something, especially a battery, you need to tin it (this is where you apply a small layer of solder to it so that it is easier to melt when you want to attach the wire).

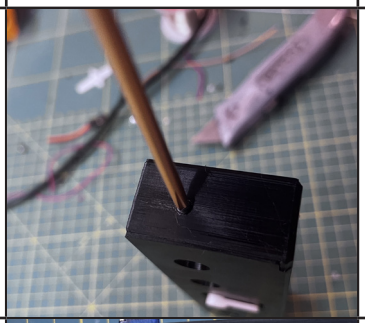
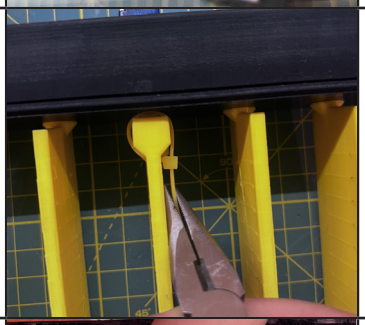
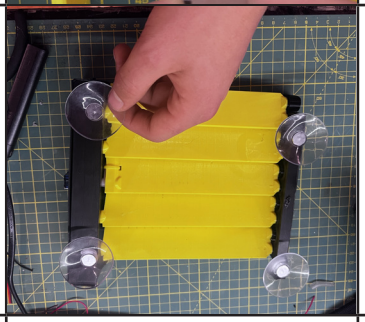
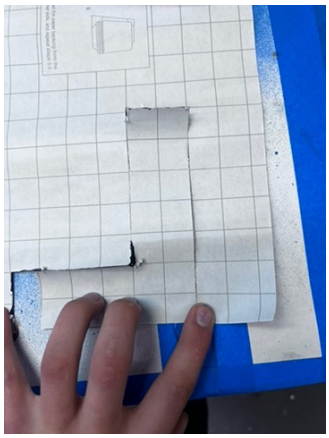
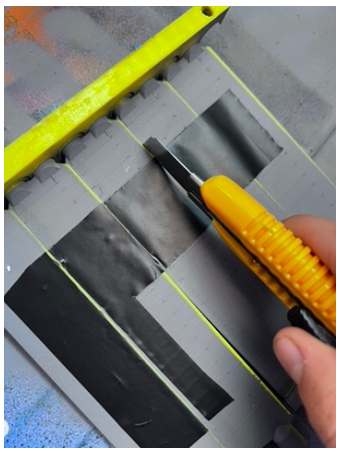
Project Development & Realisation

Production process

#	Photo	Description
13		Cut and solder a wire to the battery. Make sure you use black for the minus side and red for the plus side.
14		Place the batteries into their clips noting there is excess wire, but this is much easier than having to take the battery out and resolder it.
15		Using side snips, trim the length of the wire so that it reaches the charging pads on the ESP32.
16		Solder the wires in place as seen in the image
17		Using the self tapping screw, provided with a servo, screw the horn in place.
18		Set the soldering iron to 300°C, then insert M3 threaded inserts into the holes careful to keep the soldering iron perpendicular to the 3D printed part.

Project Development & Realisation

Production process

#	Photo	Description
19		Clip on the outer part of the case and screw in self tapping screws to keep the case in place.
20		Insert the louvres into the holes where they belong and use a small cable tie to keep the horn in the hole. I considered designing some sort of clip but using a cable tie removes mechanical complexity and improves repairability.
21		Screw in suction cups into the threaded inserts.
22		Use a Stanley knife to cut out the contact in the shape of an L Measure it to ensure it is the official size
23		Using a Stanley knife to cut the L plate where these slots are for the louvres. This will ensure that the louvres can rotate while the L plate remains the official size. * The images from a prototyping stage, but it is still valid

Project Development & Realisation

Use of Communication and Presentation Techniques

	On shape is a web-based 3D CAD platform and it allowed me to rapidly prototype my designs. Another feature of shape is that it has version histories so every edit I make has a saved checkpoint and you can revert back to previous versions. In software we have something called git. This is a version control system that uses branching. This makes it very familiar for me. Onshape also let me generate cross-sectional views of my 3D models which were very useful in conveying how the internal geometry worked
	InDesign helped me re-organise and format the folio so it is in line with the marking criteria for it to be succinct and well organised. I was able to colour code each section and make it more aesthetically pleasing
	Microsoft Word let me draft and spellcheck my words in writing before I submitted. It also gave me the ability to share with my teacher to gain feedback. This was useful for improving my portfolio and communicating articulately and succinctly.
	I used Excel to easily construct a well formatted time action plan, that I could print out and easily write down the steps and how long it took me to complete. I could also easily evaluate these steps.
	Adobe illustrator let me design and test my very first iteration on the laser cutter.
	draw.io let me generate beautiful diagrams that could clearly communicate the flow of complex unintuitive concepts in programming. This let me communicate visually instead of through code examples or pseudocode which despite being easier to read than real code, it is still difficult.

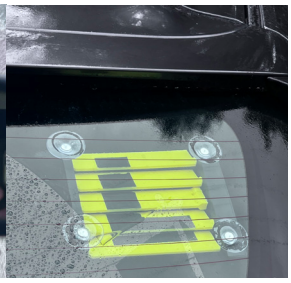
These communication tools let me effectively convey my ideas. Without these I would have to write dense paragraphs of wordy complicated technical text and this would struggle to communicate effectively what I wanted to say. Instead, I was able to use tools like Onshape which was especially useful to show things like internal geometry without having to physically cut open a model and take a photo of it.

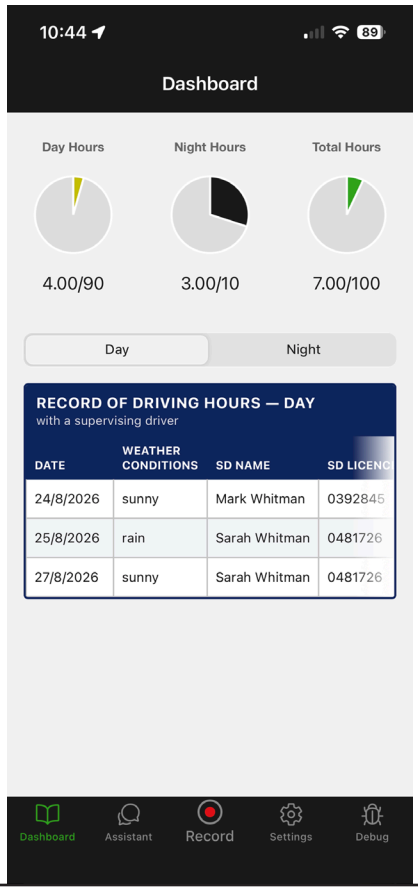
Project Evaluation

Evaluation of Functional and Aesthetic Aspects of Design

Functional evaluation

The functionality of Lflip played a pivotal role in the success of my project.





Fast and reliable actuation of my product as well as the auto filling the log book. Lflip successfully and effectively does that. This is because of the correctly sized actuator as well as a highly functional integrated app which allows it to work reliably. This product works very well as it saves a large amount of time for starting a drive.

The total time that it actuates/moves in is 1.8 seconds total time. My initial criterion was 3 seconds, so this comfortably beats the target. I am, therefore, happy with the actuation rate.

Maintaining a **clear line of sight** through the back windscreen as this improves the safety. As is visible in the images, there is an unobstructed view through the back windscreen. This makes it very effective in ensuring the overall safety of the driver and the people around them.

Fast logging of trips and minimal friction

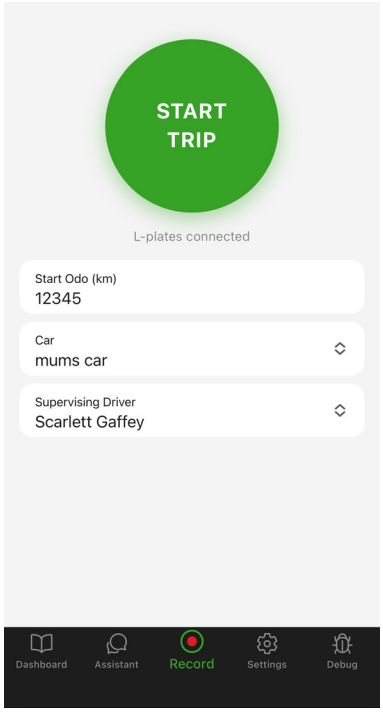
The main goal of the project was to be able to quickly log drives so that there is less friction to start a trip. This then, would encourage people to always put on their plates, as the added convenience of opening an app removes much of the hassle with traditional L plate designs. I was highly successful in this aspect as it takes roughly three seconds to start a drive which is a significant improvement from the previous 50 seconds.

Project Evaluation

Evaluation of Functional and Aesthetic Aspects of Design

Aesthetic evaluation

The aesthetics of my product significantly contributed to the success of my project



3D Printing

The surface finish of my project looks very good as I used a high quality 3D printer. This makes my product appear as a high-quality premium product, whereas it is actually relatively low-cost manufacture at scale.

Because of its 3D printed nature it allows me to reduce the amount of parts used. This means that you do not have numerous ugly screws and bolts protruding everywhere.

The advantage of 3D printing allows me to make an aesthetically pleasing product

Colour scheme

The colour scheme of Lflip contributed considerably to the success of the product. The outer casing of Lflip was printed in black. This was so that it would look like the L plate meaning that it was a single aesthetic design language.

In addition to the design of the plate mechanism, the app design was carefully crafted to mimic what the official ACT government apps should look like as I found a PDF online and followed that. Hence the use of purple.

Overall the colour scheme of my app made it feel like much more of an official and proper product.

App design

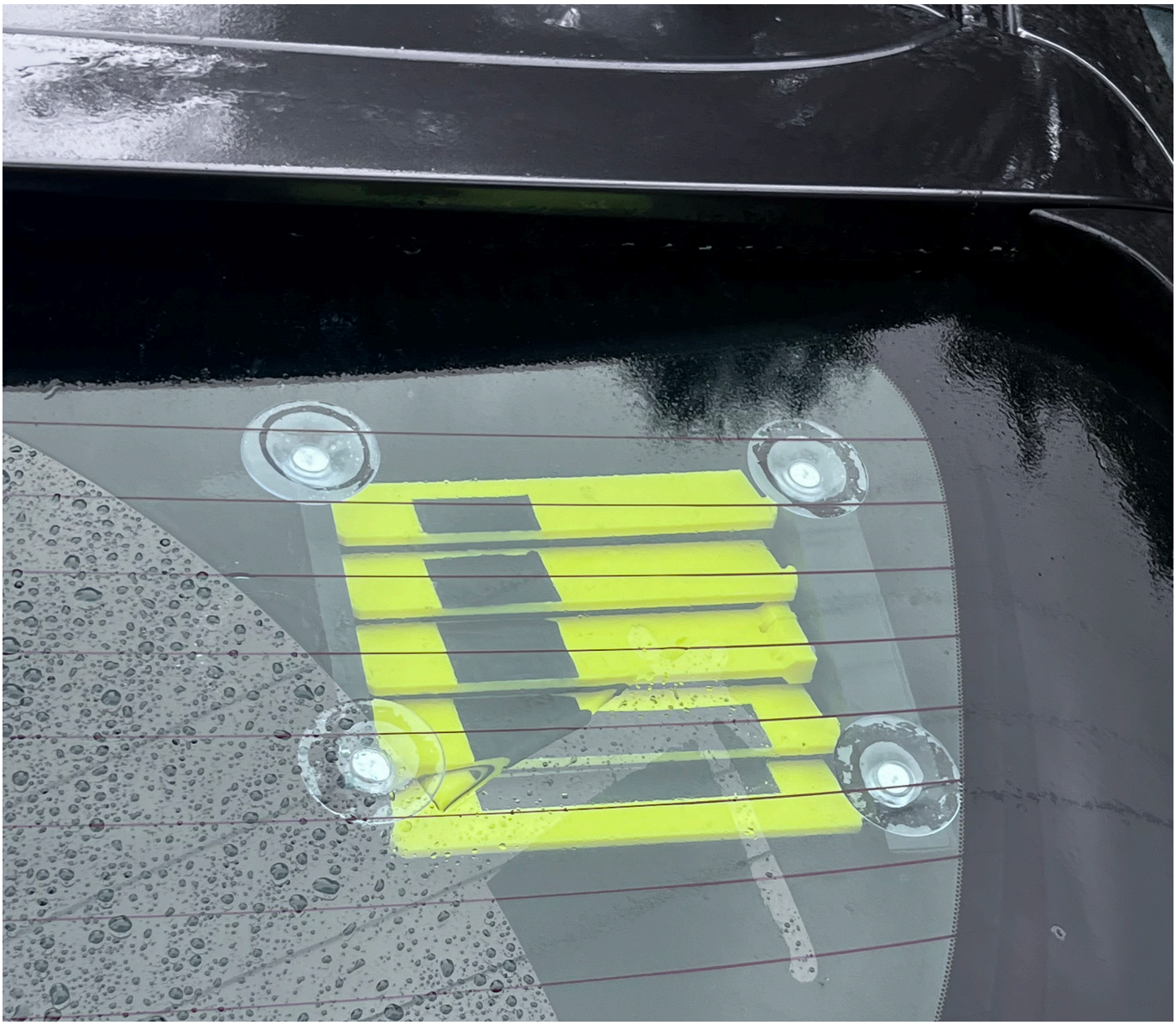
The app design contributed to the professionalism of the product, as each screen looked very aesthetically pleasing. In terms of the app being aesthetically pleasing it's got a minimal aesthetic, minimum official colour palette, which closely aligns to the official government recommendations for the colours. The navigation is clear and intuitive as the fonts and colours follow the UI and UX design principle of consistency. This allows the language the be clear, and users know exactly how to navigate without needing additional instructions.

The programing language I used was modern and I followed design best practices to make it intuitive, modern and easy to use. This contributes to the success of the project.

Project Evaluation

Evaluation of the Project's Impact on the Individual

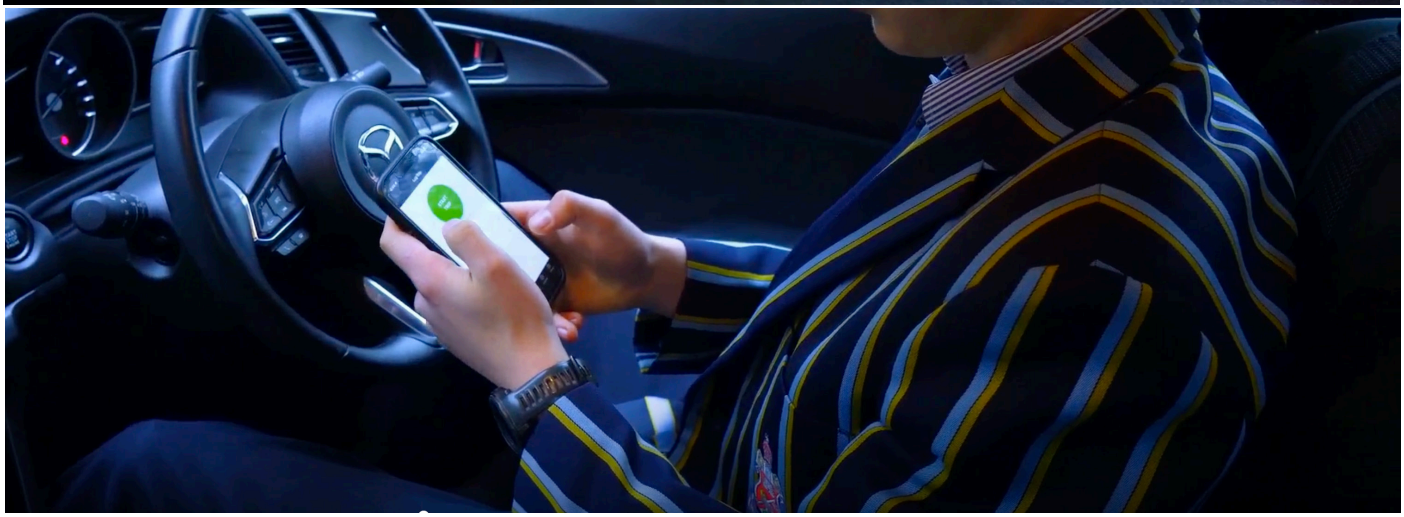
	Impact
Individual	This project directly affects the benefits of the individual as per planned. The 'individual' in this context is my referred demographic of people who have newly gotten their L's and need to log their hours. This means that they are less likely to drive illegally without plates on as this is something that's happened to individuals numerous times when driving without L plates on. Therefore, individuals are able to easily access their logged hours, as it's automated, making the whole process more accurate and efficient. Because of the combination of a very well designed user interface and a sleek and professional looking product it means that this will not impact the aesthetics of their car. For example, if they have a luxury car, this should not deter it from being used. Plates also regularly fall off the vehicle causing inconveniences, and often L platers end up driving without their plates on, without their knowledge. As this design is located on the inside of the vehicle, this will ensure individuals are not driving illegally without their plate nor having to purchase new plates, resulting in cost efficiency.



Project Evaluation

Evaluation of the Project's Impact on Society

	Impact
Society	This design is impactful overall within society as overall roads become a safer place. When drivers have the correct plates on it ensures the surrounding drivers are aware of their capabilities in driving, and hence, can be more cautious around them. This in the long run reduces the chances of an accident and therefore creates a safer environment for learners. As this product does not currently exist, it also acknowledges the gap in the market within a society, potentially increasing awareness of the necessity and appropriation of this product. This product was a resounding success as it is so easy and efficient to activate the L/P plates that the user will do it almost automatically. This ensures the safety of all road users and the general public. So ultimately this project was a success for society.



Project Evaluation

Evaluation of the Project's Impact on the Environment

	Impact
Environment	My project has a significant impact on the environment. Due to L plates causing a large amount of roadside littering mounting them internally does solve this issue. As this is located on the inside of the vehicle, even if it does detach, it falls within the car, creating less waste. This is very beneficial for the environment as when L plates currently in use break down they produce microplastics, which are harmful for the environment. When evaluated in terms of the environment, my product is very successful as it significantly reduces the amount of roadside litter and plastic waste. Life cycle analysis: A plain plate is mono-material and at least theoretically recyclable; the product is mixed e-waste with a battery that's a landfill fire risk. The only caveat in the environmental aspect is that the product uses PETG - which is a petroleum driven creation. Next time, perhaps a more environmentally friendly filament can be explored.

Project Evaluation

Relationship of the final product, system or environment to the project proposal

Criteria	Evaluation	Rating
Safety	My product is very safe as it has a large number of strong suction cups to hold it in place and attach to the car. In addition to this it encourages safer driving by making it easier to be taken on and off the car so that there's less friction, therefore people are more likely to drive with it on and less likely to get in a car crash, keeping them and those around them safer. The main thing limiting this from becoming a 10/10 is the lack of complete reliability and assurance that all the electronic components will last and survive in a hot car in addition to the suction cups' reliability not being tested thoroughly. Moreover, this product limits the visibility out of the rear windscreen. This is because this product does not currently have the capabilities to be displayed externally on the car, and is larger than standard L and P plates, causing less visibility than other products on the market.	8/10 I rated an 8 out of 10 because I believe it is still very safe, but there is a danger that the batteries might catch fire in the high temperatures of a car, or if they are punctured in a crash. This is what stops it being rated higher.

Criteria	Evaluation	Rating
Ergonomics	This product is very ergonomic. Both the actual physical L plate and the App work together seamlessly in a way that's very comfortable for the user to use. This is because I applied user-centred design when designing both the app and the hardware to make it as intuitive as possible. A slight limitation is that the way the L plates pair with one another is unique. This means it can be slightly difficult for the users to debug issues, but ultimately this is not a significant problem as I have handled this as seamlessly as possible with the code. To improve the ergonomics, I could add a tutorial video and improve the accessibility of my documentation so that it is accessible for people with vision impairment.	9/10 I chose to give this a 9 out of 10 because it was very ergonomic and even users who had never used it before could use it, but I only tested it on people without accessibility needs.

Project Evaluation

Relationship of the final product, system or environment to the project proposal

Criteria	Evaluation	Rating
Legality	My project is legal. I have carefully read the relevant road rules and determined that for an ACT learner driver this product is legal. It displays an official-size plate in the back of the car, and it can't be moved while driving because the software in the app prohibits that. Unfortunately, this is not legal for a New South Wales driver because they must have plates displayed externally, but it is legal for most drivers in most of the other states.	7/10 The reason I give it a 7 out of 10 is because it is almost legal in all states, but that remaining state of New South Wales means that it is not fully legal in all parts of Australia. In addition to this there is a perceived illegality as people feel like it should be illegal so this is a barrier to overcome with marketing.

Criteria	Evaluation	Rating
Aesthetics	The aesthetic of my product was a large factor contributing to the success of my project. The striking black and yellow of the L plate was continued in the housing and the dovetail pins. This meant that it looked like a single cohesive design language maintaining the black and yellow colour scheme throughout. For the app design I followed the colour scheme available on the ACT government website. This let me make it look like an official app. Due to the use of high-quality PETG and low layer height in printing it made the surface look smooth and matte, which is very aesthetically pleasing as it does not increase glare when driving. The black border around the Lflip highlights for drivers that the individual driving is on their Ls. This helps safety while being aesthetically pleasing to look at for other drivers. Overall, the aesthetics of my project contribute greatly to its success.	8/10 I give this rating because it is very aesthetically pleasing because of the colour scheme being quite visibly distinct. In addition to this I also think the app is quite well designed as it follows modern design best practices meaning that it looks professional.

Project Evaluation

Relationship of the final product, system or environment to the project proposal

Criteria	Evaluation	Rating
Quality	My product quality contributed significantly to the success of the project. The reason it is such high-quality is because I chose high-quality materials and manufacturing processes which produce a high-quality result. Processes such as 3D printing often have a reputation for low quality, but this is untrue because it allows you to leverage geometries which are not possible with traditional manufacturing allowing you to produce stronger and lower part count items. This lower number of parts decreases the chance of failure. In addition to this I chose high-quality electronics and good quality batteries and implemented both physical and software stops to avoid damaging the batteries. This means that it will last for a long time. Overall, because of the manufacturing process and design decisions, this has led to a high-quality product.	8/10 I gave it an 8 out of 10 as it's very high-quality, but it's still made of plastic which is less durable than metal or carbon fibre. Both of these would've produced a high-quality product but would've been a failure in terms of my other criteria of success.

Criteria	Evaluation	Rating
Cost	My product cost is one of the major limitations of the product. This is because of the electronic components I chose. I chose to use an expensive microcontroller, the ESP32-C3, from a reputable company. This is a board designed for developers, which includes lots of features that I'm not using. If I was to improve it in future, I would change to a production board and make my own PCB (Printed Circuit Board). This will reduce the cost of the microcontroller and electronics from $30 to $5. In addition to this the cost of the batteries is about $50. I can reduce this by using unprotected cells and adding the protection in the PCB. This will reduce the cost of the cell from approximately $50 to $10. The cost of the printing filament is largely unchanged and approximately $5. So this goes from a unit cost of $85 each to $20. This is a significant reduction.	5/10 7/10* * With the proposed future changes. I give it a 5 out of 10 because these changes have not been implemented yet, but it would be quite easy to make this a 7 or 8 out of 10.

Project Evaluation

Final SWOT analysis

Strengths	Using the design process to iterate and produced a good prototype through countless prototypes. Attention to detail allowed this to happen and also a software background allowed me to see cross disciplinary opportunities to iterate an app. The product looks scalable, professional and may be manufactured at a scale. It is a ready product, that can be immediately used or mass manufactured at a reasonable cost. The assembly process has been optimised for efficiency reducing the overall labour cost. This is because of design support material being significantly removed or to generate support material.
Weaknesses	The 'L' plate looks not as sturdy and looks fragile. As it is made of contact tape. A better approach would have been multi material 3d printing or multi extruder machines. We just simply did not have access to that at school or elsewhere. Time constraints also played a role in this as I have 11 units that I also needed to focus on. The way that the two plates pair, this has not been fully refined in terms of design. I'd prefer instead a master and a slave two peers. Which means one plate is connected to the phone and it guides the other plate. This peer to peer design would allow two plates to create separately making the troubleshooting of issues to be an easier process. The app also draws a lot of battery at the moment.
Opportunities	**Dual sided L to P Plates:** Would like to explore opportunities of having L plates on one side and P on the other further maximising the cost and sustainability aspect of the design. **More integrated app:** The L plate does not allow multiple users to connect to it currently. For example if there were two L platers in a family, it can only connect to one, so they'd have to buy two of the same products. **Dynamic Sleeping:** The microcontrol in the L plate could perhaps go to sleep saving battery **Potential Charging Dock:** A charging dock perhaps within the car or elsewhere compatible with the product will allow you to charge it. **Design enhancements:** The L plate itself design wise could be more refined maybe with spray paint, or vinyl, sleeves over the panels etc. This, however, due to time constraints could not be investigated as it was not quite a functional priority.
Threats	**Battery died the day before this was due:** This caused this introduced unexpected functionality and I thought I had broken the product, but it was just a battery that needed to be charged with the integrated charger. **The prototype that was meant to be the final broke:** I dropped my prototype for two days before the submission date and it caused me to have to reprint it. **Miscommunications with my mentor:** This wasted three days worth of time when I was a week out from the deadline. **Countless software bugs:** The nature of software development is that bugs do happen and this was an ongoing challenge I had to face **3D Printer Failures:** During the initial prototyping phases the printer failed this called me to doubt my design it was actually both the printer and my designs fault that happened simultaneously this made it hard to work out the root cause.

Project Evaluation

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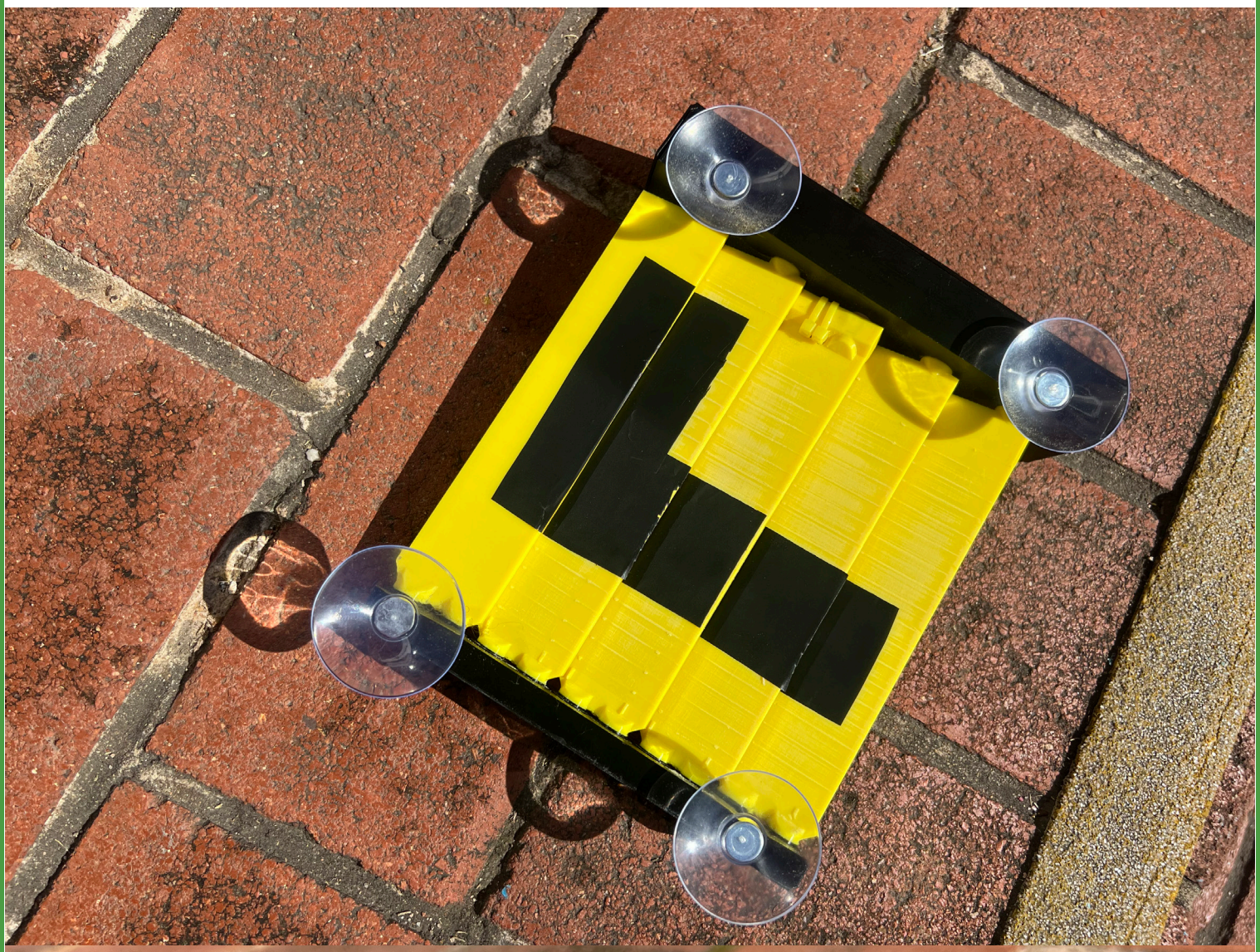
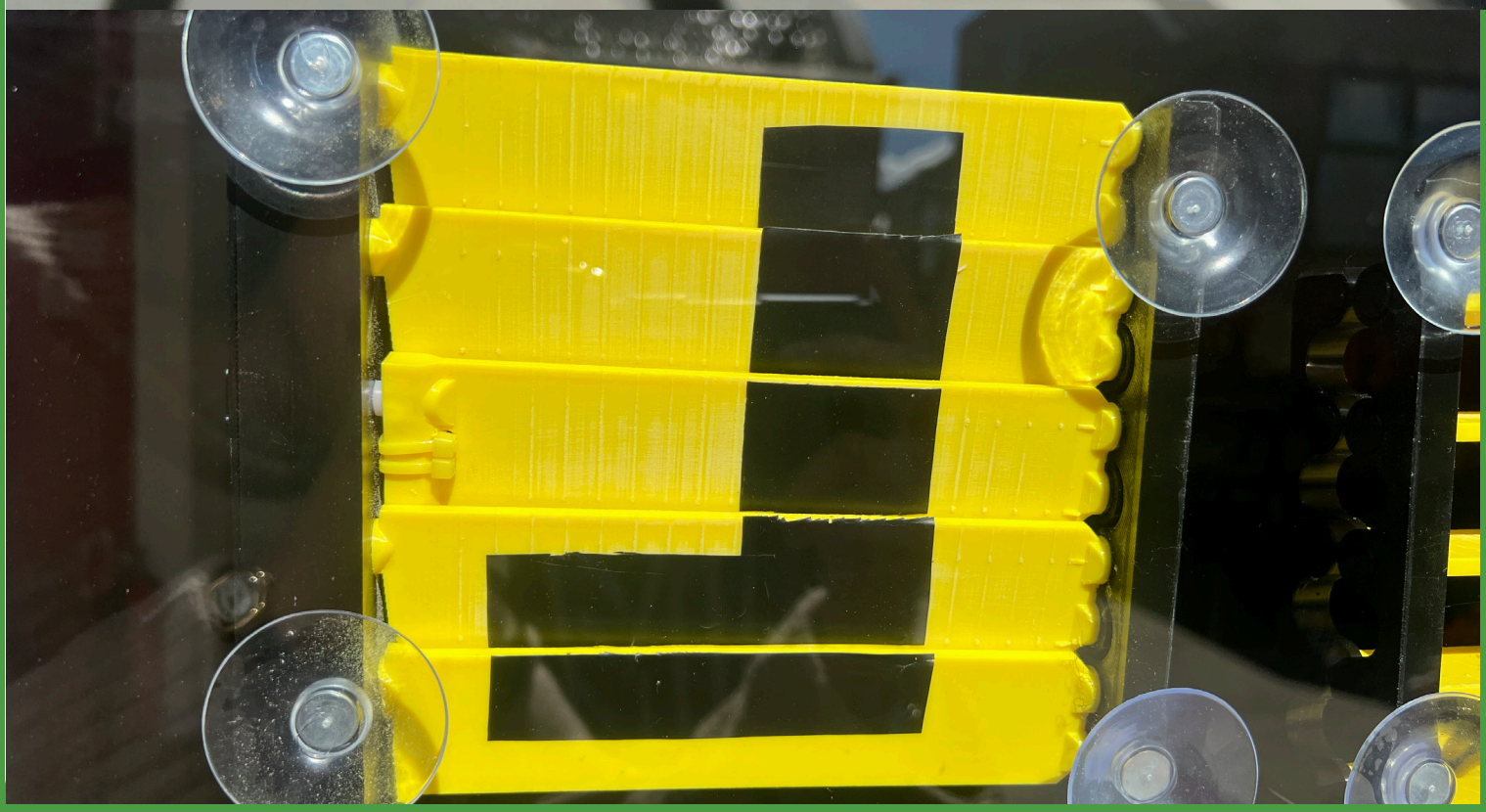


Project Evaluation



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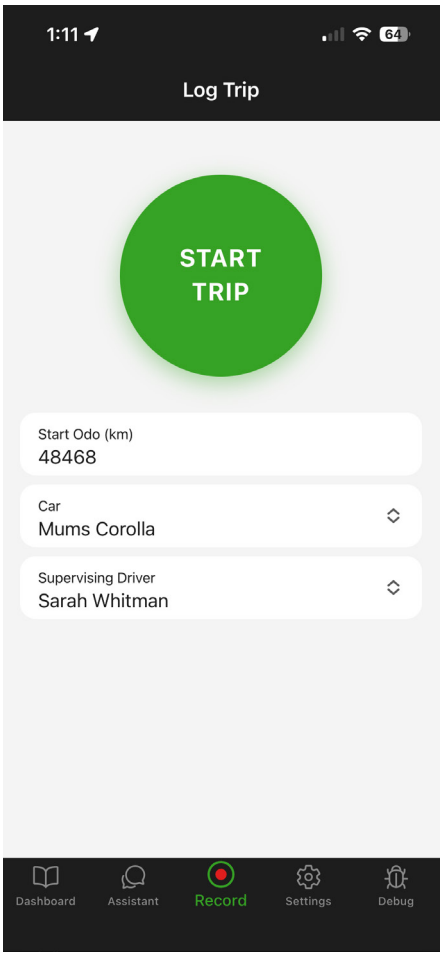

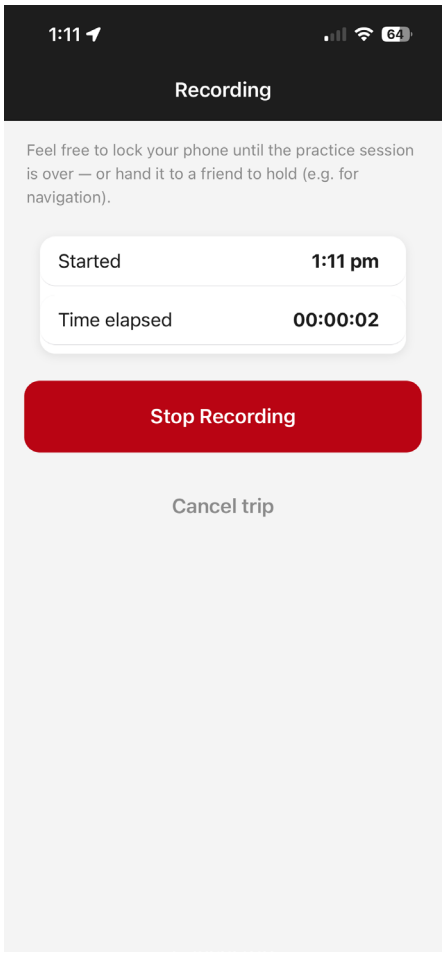

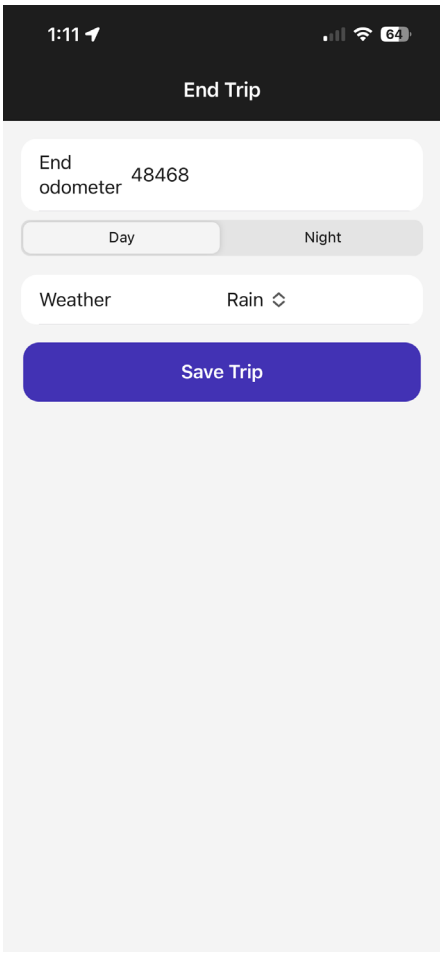

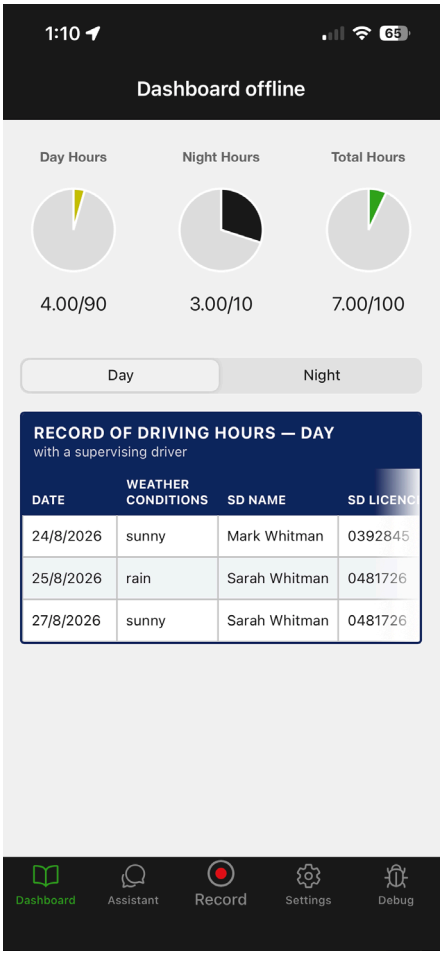

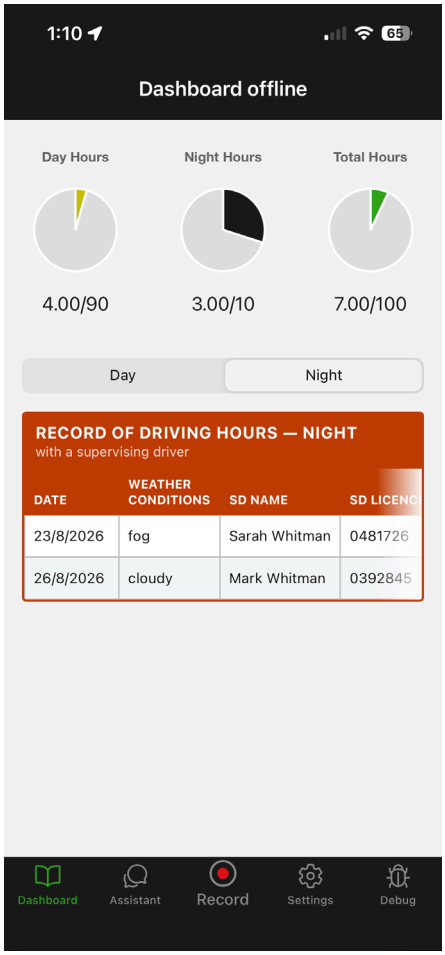

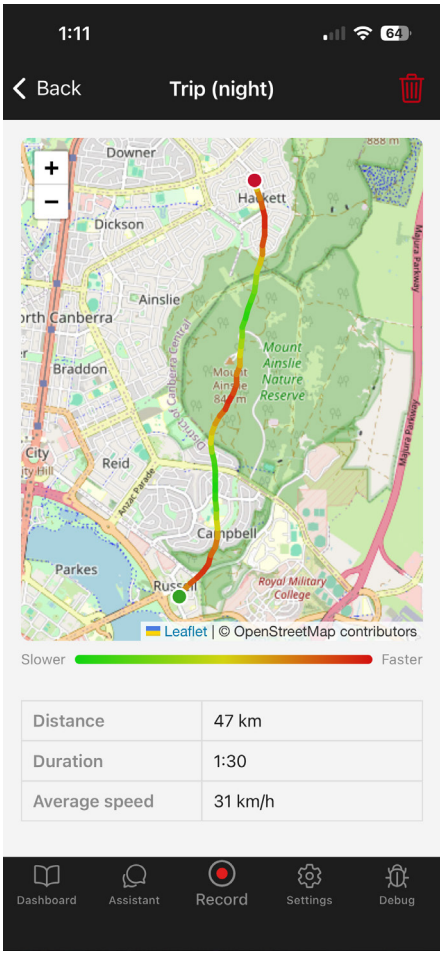

Project Evaluation